

kohlhase-omdoc — formula report

459 inline formulas (section omitted; 4 did not render), 6 display equations, 103 tables, 271 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
kohlhase-omdoc_ EQ0004 (26.1) [conf 0.035]	297	■ 0.035 ● U +44	<code>\backslash \text { CSum\{k\}1\infty\{\Cexp\{x\}k\}</code> instead of <code>\backslash \text { sum_{k = 1}^{} }{ }_{1}</code> <code>\text { infty x^k }</code>	<i>(not rendered)</i>	<code>\CSum{k}1\infty{\Cexp{x}k}</code> instead of <code>\sum_{k = 1}^{\infty} x^k</code>
kohlhase-omdoc_ EQ0005 [conf 0.092]	309	■ 0.092 ● U +105	<code>\begin{figure} \[\begin{aligned} \left.\mathcal{R}\right. \\ \frac{\mathbf{A}:\mathbf{B}}{\mathbf{A}\mathbf{B}:\mathbf{C}} \rightarrow \mathbf{B} \\ \quad \mathbf{B}:\mathbf{B} \rightarrow \mathbf{A} \mathbf{B}:\mathbf{C} \\ \right) \Rightarrow \frac{\frac{\frac{\forall X.\mathbf{B}(X) \Rightarrow \mathbf{C}(\mathbf{A}X)}{\mathbf{B}(\mathbf{B}) \Rightarrow \mathbf{C}(\mathbf{A}\mathbf{B})} \mathbf{B}(\mathbf{B})}{\mathbf{C}(\mathbf{A}\mathbf{B})} \\ \mathcal{R} \frac{\frac{\forall X.\mathbf{B}(X) \Rightarrow \mathbf{C}(\mathbf{A}X)}{\mathbf{B}(\mathbf{B}) \Rightarrow \mathbf{C}(\mathbf{A}\mathbf{B})} \mathbf{B}(\mathbf{B})}{\mathbf{C}(\mathbf{A}\mathbf{B})} \\ \left) \Rightarrow \frac{\frac{\frac{\forall X.\mathbf{B}(X) \Rightarrow \mathbf{C}(\mathbf{A}X)}{\mathbf{B}(\mathbf{B}) \Rightarrow \mathbf{C}(\mathbf{A}\mathbf{B})} \mathbf{B}(\mathbf{B})}{\mathbf{C}(\mathbf{A}\mathbf{B})} \right)</code>	<i>(not rendered)</i>	$\mathcal{R} \frac{\mathbf{A}:\mathbf{B} \rightarrow \mathbf{C} \quad \mathbf{B}:\mathbf{B}}{\mathbf{A}\mathbf{B}:\mathbf{C}} \Bigg) = \frac{\frac{\forall X.\mathbf{B}(X) \Rightarrow \mathbf{C}(\mathbf{A}X)}{\mathbf{B}(\mathbf{B}) \Rightarrow \mathbf{C}(\mathbf{A}\mathbf{B})} \mathbf{B}(\mathbf{B})}{\mathbf{C}(\mathbf{A}\mathbf{B})}$ $\mathcal{R} \frac{\frac{\forall X.\mathbf{B}(X) \Rightarrow \mathbf{C}(\mathbf{A}X)}{\mathbf{B}(\mathbf{B}) \Rightarrow \mathbf{C}(\mathbf{A}\mathbf{B})} \mathbf{B}(\mathbf{B})}{\mathbf{C}(\mathbf{A}\mathbf{B})} \Bigg) = \frac{\frac{\forall X.\mathbf{B}(X) \Rightarrow \mathbf{C}(\mathbf{A}X)}{\mathbf{B}(\mathbf{B}) \Rightarrow \mathbf{C}(\mathbf{A}\mathbf{B})} \mathbf{B}(\mathbf{B})}{\mathbf{C}(\mathbf{A}\mathbf{B})}$
kohlhase-omdoc_ EQ0003	188	■ 0.131 ● C +3	<code>\begin{aligned} & \mathcal{T}_2 \xrightarrow{\sigma} \mathcal{C}_2 \quad \mathcal{B}_1 \xrightarrow{\beta} \mathcal{B}_2 \xrightarrow{\sigma} \mathcal{C}_2 \\ & \mathcal{T}_1 \xrightarrow{\sigma} \mathcal{A}_2 \xrightarrow{\sigma} \mathcal{C}_2 \\ & \mathcal{A}_1 \xrightarrow{\alpha} \mathcal{A}_2 \xrightarrow{\sigma} \mathcal{C}_2 \end{aligned}</code>	$\mathcal{T}_2 \xrightarrow{\sigma} \mathcal{C}_2 \quad \mathcal{B}_1 \xrightarrow{\beta} \mathcal{B}_2 \xrightarrow{\sigma} \mathcal{C}_2$ $\mathcal{T}_1 \xrightarrow{\sigma} \mathcal{A}_2 \xrightarrow{\sigma} \mathcal{C}_2$ $\mathcal{A}_1 \xrightarrow{\alpha} \mathcal{A}_2 \xrightarrow{\sigma} \mathcal{C}_2$	$\mathcal{T}_2 \xrightarrow{\sigma} \mathcal{C}_2 \quad \mathcal{B}_1 \xrightarrow{\beta} \mathcal{B}_2 \xrightarrow{\sigma} \mathcal{C}_2$ $\mathcal{T}_1 \xrightarrow{\sigma} \mathcal{A}_2 \xrightarrow{\sigma} \mathcal{C}_2 \quad \mathcal{A}_1 \xrightarrow{\alpha} \mathcal{A}_2 \xrightarrow{\sigma} \mathcal{C}_2$
kohlhase-omdoc_ EQ0006	332	■ 0.672 ● S +1	<code>\begin{aligned} & ([0-9] , .) \mid (* \mid \mathrm{e}([0-9] , .) *) ? \\ & \mathrm{e}([0-9] , .) * ? \end{aligned}</code>	$([0-9] , .)$ $(* \mid \mathrm{e}([0-9] , .) *) ?$	$([0-9] , .)$ $(* \mid \mathrm{e}([0-9] , .) *) ?$
kohlhase-omdoc_ EQ0001	060	■ 0.863 ● U +38	<code>\forall M . \operatorname{magma}(M) \Leftrightarrow \exists E, F . M=(E, F) \wedge \text { law_of_composition }(E, F)</code>	<i>(not rendered)</i>	$\forall M. \operatorname{magma}(M) \Leftrightarrow \exists E, F. M = (E, F) \wedge \text{law_of_composition}(E, F)$
kohlhase-omdoc_ EQ0002	121	■ 1.000 ● N +0	<code>\frac{1}{1+\frac{1}{1+\dots}}</code>	$\frac{1}{1+\frac{1}{1+\dots}}$	$\frac{1}{1+\frac{1}{1+\dots}}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Tables

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																										
kohlhase-omdoc_TAB_001	028	—	(no LaTeX source; 1084 × 302 px region)	—	<table><tr><td>by</td><td>W3C Math WG</td><td>OPENMATH society</td></tr><tr><td>origin</td><td>math for HTML</td><td>integration of CAS</td></tr><tr><td>coverage</td><td>content + presentation; K-14</td><td>content; extensible</td></tr><tr><td>status</td><td>Version 2.2e (VI 2003)</td><td>Version 2 (VI 2004)</td></tr><tr><td>activity</td><td>maintenance</td><td>maintenance</td></tr><tr><td>Info</td><td>http://w3c.org/Math/</td><td>http://www.openmath.org/</td></tr></table> Fig. 2.1. The status of markup standardization for mathematical formulae OPENMATH was originally a development driven mainly by the Cammutor DOC uses three levels of modeling corresponding to the concerns raised previously. We have visualized this architecture in Figure 3.1. <table><tr><th>Level of Representation</th><th>OMDOC Example</th></tr><tr><td><i>Theory Level:</i> Development Graph – Inheritance via symbol-mapping – Theory inclusion via proof-obligations – Local (one-step) vs. global links </td><td></td></tr><tr><td><i>Statement Level:</i> – Axiom, definition, theorem, proof, example,... – Structure explicit in statement forms and references </td><td><pre><definition for="#plus" type="recursive"> <CMP>Addition is defined by recursion on the second argument </CMP> <FMP>X + 0 = 0</FMP> <FMP>X + s(Y) = s(X + Y)</FMP> </definition></pre></td></tr><tr><td><i>Object Level:</i> OPENMATH/MATHML – Objects as logical formulae – Semantics by pointing to theory level </td><td><pre><OMA> <OMS cd="arith1" name="plus" /> <OMV name="X" /> <OMS cd="nat" name="zero" /> </OMA></pre></td></tr></table> Fig. 3.1. OMDOC in a nutshell (the three levels of modeling)	by	W3C Math WG	OPENMATH society	origin	math for HTML	integration of CAS	coverage	content + presentation; K-14	content; extensible	status	Version 2.2e (VI 2003)	Version 2 (VI 2004)	activity	maintenance	maintenance	Info	http://w3c.org/Math/	http://www.openmath.org/	Level of Representation	OMDOC Example	<i>Theory Level:</i> Development Graph – Inheritance via symbol-mapping – Theory inclusion via proof-obligations – Local (one-step) vs. global links		<i>Statement Level:</i> – Axiom, definition, theorem, proof, example,... – Structure explicit in statement forms and references	<pre><definition for="#plus" type="recursive"> <CMP>Addition is defined by recursion on the second argument </CMP> <FMP>X + 0 = 0</FMP> <FMP>X + s(Y) = s(X + Y)</FMP> </definition></pre>	<i>Object Level:</i> OPENMATH/MATHML – Objects as logical formulae – Semantics by pointing to theory level	<pre><OMA> <OMS cd="arith1" name="plus" /> <OMV name="X" /> <OMS cd="nat" name="zero" /> </OMA></pre>
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kohlhase-omdoc_TAB_002	041	—	(no LaTeX source; 1135 × 637 px region)	—																											
kohlhase-omdoc_TAB_003	050	—	(no LaTeX source; 1142 × 120 px region)	—																											

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Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																
kohlhase-omdoc_TAB_004	051	—	(no LaTeX source; 1141 × 1740 px region)	—	384 Textbooks and Articles <table><tr><td>2,3</td><td>The parser is told to use a document type definition for validation. The string <code>omdoc</code> specifies the name of the root element, the identifier <code>PUBLIC</code> specifies that the DTD (we use the “OMDoc basic” DTD; see Subsection 22.3.1), which can be identified by the public identifier in the first string and looked up in an XML catalog or (if that fails) can be found at the URL specified in the second string. A DTD declaration is not strictly needed for an OMDoc document, but is recommended, since the DTD supplies default values for some attributes.</td><td>24.1 p. 228</td></tr><tr><td>4</td><td>In general, XML files can contain as much whitespace as they want between elements, here we have used it for structuring the document.</td><td></td></tr><tr><td>5</td><td>Start tag of the root element of the document. It declares the version (OMDoc 1.2) via the <code>version</code>, and an identifier of the document using the <code>xml:id</code> attribute. The optional <code>modules</code> specifies the sub-language used in this document. This is used when no DTD is present (see Subsection 22.3.1).</td><td>11.1 p. 90</td></tr><tr><td>6,7</td><td>the namespace prefix declarations for the Dublin Core, Creative Commons, and OPENMATH namespaces. They declare the prefixes <code>dc:</code>, <code>cc:</code>, and <code>om:</code>, and bind them to the specified URLs. We will need the OPENMATH namespace only in the third markup step described in Section 4.3, but spurious namespace prefix declarations are not a problem in the XML world.</td><td>10 p. 83</td></tr><tr><td>8</td><td>the namespace declaration for the document; if not prefixed, all elements live in the OMDoc namespace.</td><td>10.1 p. 83</td></tr><tr><td>9–29</td><td>The metadata for the whole document in Dublin Core format</td><td>11.2 p. 92</td></tr><tr><td>10</td><td>The title of the document</td><td>12.1 p. 98</td></tr><tr><td>11</td><td>The document creator, here in the role of a translator</td><td>12.2 p. 101</td></tr><tr><td>12</td><td>The date and time of first creation of the document in ISO 8601 norm format.</td><td>12.1 p. 99</td></tr><tr><td>13</td><td>The date and time of the last update to the document in ISO 8601 norm format.</td><td>12.1 p. 99</td></tr><tr><td>14–16</td><td>A short description of the contents of the document</td><td>12.1 p. 99</td></tr><tr><td>17–19</td><td>Here we acknowledge that the OMDoc document is just a translation from an earlier work.</td><td>12.1 p. 100</td></tr><tr><td>20</td><td>The type of the document, this can be <code>Dataset</code> (un-ordered mathematical knowledge) or <code>Text</code> (arranged for human consumption).</td><td>12.1 p. 100</td></tr><tr><td>21</td><td>The format/MIME type [FB96] of the document, for OMDoc, this is <code>application/omdoc+xml</code>.</td><td>12.1 p. 100</td></tr><tr><td>22</td><td>The copyright resides with the creator of the OMDoc document</td><td>12.1 p. 100</td></tr><tr><td>23–28</td><td>The creator licenses the document to the world under certain conditions as specified in the Creative Commons license specified in this element.</td><td>12.3 p. 102</td></tr></table>	2,3	The parser is told to use a document type definition for validation. 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kohlhase-omdoc_TAB_005	052	—	(no LaTeX source; 1141 × 890 px region)	—	4.2 Structure and Statements39 <table><tr><td>24,25</td><td>The <code>cc:permissions</code> element gives the world the permission to reproduce and distribute it freely. Furthermore the license grants the public the right to make derivative works under certain conditions.</td><td>12.3 p. 103</td></tr><tr><td>26</td><td>The <code>cc:prohibitions</code> can be used to prohibit certain uses of the document, but this one is unencumbered.</td><td>12.3 p. 103</td></tr><tr><td>27</td><td>The <code>cc:requirements</code> states conditions under which the license is granted. In our case the licensee is required to keep the copyright notice and license notices intact during distribution, to give credit to the copyright holder, and that any derivative works derived from this document must be licensed under the same terms as this document (the copyleft clause).</td><td>12.3 p. 103</td></tr><tr><td>31-37</td><td>The <code>omtext</code> element is used to mark up text fragments. Here, we have simply used a single <code>omtext</code> to classify the whole text in the fragment as <code>unspecific</code> “text”.</td><td>14.3 p. 124</td></tr><tr><td>32-36</td><td>The <code>CMP</code> element holds the actual text in a multilingual group. Its <code>xml:lang</code> specifies the language. If the document is used with a DTD or an XML schema (as we are) this attribute is redundant, since the default value given by the DTD or schema is <code>en</code>. More keywords in other languages can be given by adding more <code>CMP</code> elements.</td><td>14.1 p. 122</td></tr><tr><td>33-35</td><td>The text of the $\LaTeX$ fragment we are migrating. For simplicity we do not change the text, and leave that to later stages of the migration.</td><td></td></tr><tr><td>38</td><td>The closing tag of the root <code>omdoc</code> element. There may not be text after this in the file.</td><td>11.1 p. 90</td></tr></table>	24,25	The <code>cc:permissions</code> element gives the world the permission to reproduce and distribute it freely. Furthermore the license grants the public the right to make derivative works under certain conditions.	12.3 p. 103	26	The <code>cc:prohibitions</code> can be used to prohibit certain uses of the document, but this one is unencumbered.	12.3 p. 103	27	The <code>cc:requirements</code> states conditions under which the license is granted. In our case the licensee is required to keep the copyright notice and license notices intact during distribution, to give credit to the copyright holder, and that any derivative works derived from this document must be licensed under the same terms as this document (the copyleft clause).	12.3 p. 103	31-37	The <code>omtext</code> element is used to mark up text fragments. Here, we have simply used a single <code>omtext</code> to classify the whole text in the fragment as <code>unspecific</code> “text”.	14.3 p. 124	32-36	The <code>CMP</code> element holds the actual text in a multilingual group. Its <code>xml:lang</code> specifies the language. If the document is used with a DTD or an XML schema (as we are) this attribute is redundant, since the default value given by the DTD or schema is <code>en</code> . More keywords in other languages can be given by adding more <code>CMP</code> elements.	14.1 p. 122	33-35	The text of the $\LaTeX$ fragment we are migrating. For simplicity we do not change the text, and leave that to later stages of the migration.		38	The closing tag of the root <code>omdoc</code> element. There may not be text after this in the file.	11.1 p. 90
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kohlhase-omdoc_TAB_006	053	—	(no LaTeX source; 1125 × 931 px region)	—	<table><tr><td>1</td><td>The <code>omtext</code> element classifies the text fragment as a definition, other types for mathematical statements include axiom, example, theorem, and lemma. Note that the numbering of the original text is lost, but can be re-created in the text presentation process. The optional <code>xml:id</code> attribute specifies a document-unique identifier that can be used for reference later.</td><td>14.3 p. 124</td></tr><tr><td>2</td><td>A multilingual group of <code>CMP</code> elements that hold the text (in our case, there is only the English default). Here the $\TeX$ formulae have been marked up with <code>legacy</code> elements characterizing them as such. This might simplify a later automatic transformation to <code>OPENMATH</code> or <code>Content-MATHML</code>.</td><td>13.5 p. 120</td></tr><tr><td>4-13</td><td>We have classified every paragraph in the original as a separate <code>omtext</code> element, which does not carry a <code>type</code> since it does not fit any other mathematical category at the moment.</td><td>14.3 p. 124</td></tr><tr><td>15</td><td>The three examples in the original in Figure 4.1 are grouped into an enumeration. We use the <code>OMDOC omgroup</code> element for this. The optional attribute <code>xml:id</code> can be used for referencing later. We have chosen <code>enumeration</code> for the <code>type</code> attribute to specify the numbering of the examples in the original.</td><td>11.4 p. 93</td></tr><tr><td>16</td><td>We can use the <code>metadata</code> of the <code>omgroup</code> element to accommodate the title “Examples” in the original. We could enter more metadata at this level.</td><td>14.1 p. 122</td></tr><tr><td>18</td><td>The <code>type</code> attribute of this <code>omtext</code> element classifies this text fragment as an example.</td><td>14.3 p. 124</td></tr></table>	1	The <code>omtext</code> element classifies the text fragment as a definition , other types for mathematical statements include axiom , example , theorem , and lemma . Note that the numbering of the original text is lost, but can be re-created in the text presentation process. The optional <code>xml:id</code> attribute specifies a document-unique identifier that can be used for reference later.	14.3 p. 124	2	A multilingual group of <code>CMP</code> elements that hold the text (in our case, there is only the English default). Here the $\TeX$ formulae have been marked up with <code>legacy</code> elements characterizing them as such. This might simplify a later automatic transformation to <code>OPENMATH</code> or <code>Content-MATHML</code> .	13.5 p. 120	4-13	We have classified every paragraph in the original as a separate <code>omtext</code> element, which does not carry a <code>type</code> since it does not fit any other mathematical category at the moment.	14.3 p. 124	15	The three examples in the original in Figure 4.1 are grouped into an enumeration. We use the <code>OMDOC omgroup</code> element for this. The optional attribute <code>xml:id</code> can be used for referencing later. We have chosen <code>enumeration</code> for the <code>type</code> attribute to specify the numbering of the examples in the original.	11.4 p. 93	16	We can use the <code>metadata</code> of the <code>omgroup</code> element to accommodate the title “Examples” in the original. We could enter more metadata at this level.	14.1 p. 122	18	The <code>type</code> attribute of this <code>omtext</code> element classifies this text fragment as an example.	14.3 p. 124			
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																								
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kohlhase-omdoc_TAB_010	094	—	(no LaTeX source; 1096 × 190 px region)	—	OMDOC also uses the following namespaces¹: <table><tr><td>Format</td><td colspan="2">namespace URI</td><td>see</td></tr><tr><td>Dublin Core</td><td colspan="2">http://purl.org/dc/elements/1.1/</td><td>Sections 12.1 and 12.2</td></tr><tr><td>Creative Commons</td><td colspan="2">http://creativecommons.org/ns</td><td>Section 12.3</td></tr><tr><td>MATHML</td><td colspan="2">http://www.w3.org/1998/Math/MathML</td><td>Section 13.2</td></tr><tr><td>OPENMATH</td><td colspan="2">http://www.openmath.org/OpenMath</td><td>Section 13.1</td></tr><tr><td>XSLT</td><td colspan="2">http://www.w3.org/1999/XSL/Transform</td><td>Chapter 19</td></tr></table>	Format	namespace URI		see	Dublin Core	http://purl.org/dc/elements/1.1/		Sections 12.1 and 12.2	Creative Commons	http://creativecommons.org/ns		Section 12.3	MATHML	http://www.w3.org/1998/Math/MathML		Section 13.2	OPENMATH	http://www.openmath.org/OpenMath		Section 13.1	XSLT	http://www.w3.org/1999/XSL/Transform		Chapter 19																																																																																												
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MATHML	http://www.w3.org/1998/Math/MathML		Section 13.2																																																																																																																						
OPENMATH	http://www.openmath.org/OpenMath		Section 13.1																																																																																																																						
XSLT	http://www.w3.org/1999/XSL/Transform		Chapter 19																																																																																																																						
kohlhase-omdoc_TAB_011	099	—	(no LaTeX source; 1142 × 331 px region)	—	<table><tr><td>omdoc</td><td>version, xmlns</td><td>xml:id, type, class, style, version, modules</td><td>+</td><td>⟨⟨top-level⟩⟩*</td></tr><tr><td>omgroup</td><td></td><td>xml:id, modules, type, class, style</td><td>+</td><td>⟨⟨top-level⟩⟩*</td></tr><tr><td>metadata</td><td></td><td>xml:id, inherits, class, style</td><td>−</td><td>⟨⟨MDelt⟩⟩*</td></tr><tr><td>ref</td><td>xref</td><td>type, class, style</td><td>−</td><td></td></tr><tr><td>ignore</td><td></td><td>type, comment</td><td>−</td><td>ANY</td></tr></table> where ⟨top-level⟩ stands for top-level OMDoc elements, and ⟨MDelt⟩ for those introduced in Chapter 12 Fig. 11.1. OMDoc Elements for Specifying Document Structure.	omdoc	version, xmlns	xml:id, type, class, style, version, modules	+	⟨⟨top-level⟩⟩*	omgroup		xml:id, modules, type, class, style	+	⟨⟨top-level⟩⟩*	metadata		xml:id, inherits, class, style	−	⟨⟨MDelt⟩⟩*	ref	xref	type, class, style	−		ignore		type, comment	−	ANY																																																																																											
omdoc	version, xmlns	xml:id, type, class, style, version, modules	+	⟨⟨top-level⟩⟩*																																																																																																																					
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Conf. MathPix confidence:

■

 ≥0.9

■

 0.5–0.9

■

 <0.5

— no value

Residual render vs scan (inkdrill):

●

 C component

●

 W weak

●

 S stable

●

 N noise

●

 K clean

●

 not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																														
kohlhase-omdoc_TAB_012	105	—	(no LaTeX source; 1121 × 516 px region)	—	<table><tr><td>dc:subject</td><td></td><td>xml:lang</td><td>⟨math vernacular⟩</td></tr><tr><td>dc:description</td><td></td><td>xml:lang</td><td>⟨math vernacular⟩</td></tr><tr><td>dc:publisher</td><td></td><td>xml:id, class, style</td><td>ANY</td></tr><tr><td>dc:date</td><td></td><td>action, who</td><td>ISO 8601</td></tr><tr><td>dc:type</td><td></td><td></td><td>fixed: "Dataset" or "Text"</td></tr><tr><td>dc:format</td><td></td><td></td><td>fixed: "application/omdoc+xml"</td></tr><tr><td>dc:identifier</td><td></td><td>scheme</td><td>ANY</td></tr><tr><td>dc:source</td><td></td><td></td><td>ANY</td></tr><tr><td>dc:language</td><td></td><td></td><td>ISO 639</td></tr><tr><td>dc:relation</td><td></td><td></td><td>ANY</td></tr><tr><td>dc:rights</td><td></td><td></td><td>ANY</td></tr></table> for ⟨math vernacular⟩ see Section 14.1 Fig. 12.1. Dublin core metadata in OMDoc	dc:subject		xml:lang	⟨math vernacular⟩	dc:description		xml:lang	⟨math vernacular⟩	dc:publisher		xml:id, class, style	ANY	dc:date		action, who	ISO 8601	dc:type			fixed: "Dataset" or "Text"	dc:format			fixed: "application/omdoc+xml"	dc:identifier		scheme	ANY	dc:source			ANY	dc:language			ISO 639	dc:relation			ANY	dc:rights			ANY																		
dc:subject		xml:lang	⟨math vernacular⟩																																																																
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dc:date		action, who	ISO 8601																																																																
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dc:rights			ANY																																																																
kohlhase-omdoc_TAB_013	111	—	(no LaTeX source; 1144 × 299 px region)	—	12.3 Managing Rights103 <table><tr><th rowspan="2">Element</th><th colspan="2">Attributes</th><th rowspan="2">Content</th></tr><tr><th>Req.</th><th>Optional</th></tr><tr><td>cc:license</td><td></td><td>jurisdiction</td><td>permissions, prohibitions, requirements</td></tr><tr><td>cc:permissions</td><td></td><td>reproduction, distribution,</td><td>EMPTY</td></tr></table>	Element	Attributes		Content	Req.	Optional	cc:license		jurisdiction	permissions, prohibitions, requirements	cc:permissions		reproduction, distribution,	EMPTY																																																
Element	Attributes		Content																																																																
	Req.	Optional																																																																	
cc:license		jurisdiction	permissions, prohibitions, requirements																																																																
cc:permissions		reproduction, distribution,	EMPTY																																																																
kohlhase-omdoc_TAB_014	111	—	(no LaTeX source; 1044 × 159 px region)	—	- cc:prohibitions are the things the license prohibits. <table><tr><th>Attribute</th><th>Prohibition</th><th>Default</th></tr><tr><td>commercial.use</td><td>stating that rights may be exercised for commercial purposes.</td><td>permitted</td></tr></table> - cc:requirements are restrictions imposed by the license.	Attribute	Prohibition	Default	commercial.use	stating that rights may be exercised for commercial purposes.	permitted																																																								
Attribute	Prohibition	Default																																																																	
commercial.use	stating that rights may be exercised for commercial purposes.	permitted																																																																	
kohlhase-omdoc_TAB_015	111	—	(no LaTeX source; 1017 × 100 px region)	—	<table><tr><td>copyleft</td><td>thor</td><td>derivative works, if authorized, must be licensed under the same terms as the work</td><td>required</td></tr></table> The Creative Commons Initiative is committed to the success of education that	copyleft	thor	derivative works, if authorized, must be licensed under the same terms as the work	required																																																										
copyleft	thor	derivative works, if authorized, must be licensed under the same terms as the work	required																																																																
kohlhase-omdoc_TAB_016	111	—	(no LaTeX source; 1019 × 186 px region)	—																																																															
kohlhase-omdoc_TAB_017	114	—	(no LaTeX source; 829 × 156 px region)	—	<table><tr><th>legacy</th><th>format</th><th>xml:id, formalism</th><th>#PCDATA</th></tr></table> . Mathematical objects in OMDoc pitulation in the next two sections is not normative, please consult it is most relevant to the OMDoc format. All elements of the XML encoding live in the namespace http://www.openmath.org/OpenMath, for which we traditionally use the namespace prefix om:.	legacy	format	xml:id, formalism	#PCDATA																																																										
legacy	format	xml:id, formalism	#PCDATA																																																																
kohlhase-omdoc_TAB_018	115	—	(no LaTeX source; 1087 × 500 px region)	—	<table><tr><th rowspan="2">Element</th><th colspan="2">Attributes</th><th rowspan="2">Content</th></tr><tr><th>Required</th><th>Optional</th></tr><tr><td>OMOBJ</td><td></td><td>id, cdbase, class, style</td><td>⟨OMel⟩?</td></tr><tr><td>OMS</td><td>cd, name</td><td>id, cdbase, class, style</td><td>EMPTY</td></tr><tr><td>OMV</td><td>name</td><td>id, class, style</td><td>EMPTY</td></tr><tr><td>OMA</td><td></td><td>id, cdbase, class, style</td><td>⟨OMel⟩*</td></tr><tr><td>OMBIND</td><td></td><td>id, cdbase, class, style</td><td>⟨OMel⟩, OMBVAR, ⟨OMel⟩</td></tr><tr><td>OMBVAR</td><td></td><td>id, class, style</td><td>(OMV OMATTR)+</td></tr><tr><td>OMFOREIGN</td><td></td><td>id, cdbase, class, style</td><td>ANY</td></tr><tr><td>OMATTR</td><td></td><td>id, cdbase, class, style</td><td>⟨OMel⟩</td></tr><tr><td>OMATP</td><td></td><td>id, cdbase, class, style</td><td>(OMS, ⟨⟨OMel⟩ OMFOREIGN⟩)+</td></tr><tr><td>OMI</td><td></td><td>id, class, style</td><td>[0–9]*</td></tr><tr><td>OMB</td><td></td><td>id, class, style</td><td>#PCDATA</td></tr><tr><td>OMF</td><td></td><td>id, class, style, dec, hex</td><td>#PCDATA</td></tr><tr><td>OME</td><td></td><td>id, class, style</td><td>⟨OMel⟩?</td></tr><tr><td>OMR</td><td>href</td><td></td><td>⟨OMel⟩?</td></tr></table> where ⟨OMel⟩ is (OMS OMV OMI OMB OMSTR OMF OMA OMBIND OME OMATTR)	Element	Attributes		Content	Required	Optional	OMOBJ		id, cdbase, class, style	⟨OMel⟩?	OMS	cd, name	id, cdbase, class, style	EMPTY	OMV	name	id, class, style	EMPTY	OMA		id, cdbase, class, style	⟨OMel⟩*	OMBIND		id, cdbase, class, style	⟨OMel⟩, OMBVAR, ⟨OMel⟩	OMBVAR		id, class, style	(OMV OMATTR)+	OMFOREIGN		id, cdbase, class, style	ANY	OMATTR		id, cdbase, class, style	⟨OMel⟩	OMATP		id, cdbase, class, style	(OMS, ⟨⟨OMel⟩ OMFOREIGN⟩)+	OMI		id, class, style	[0–9]*	OMB		id, class, style	#PCDATA	OMF		id, class, style, dec, hex	#PCDATA	OME		id, class, style	⟨OMel⟩?	OMR	href		⟨OMel⟩?
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OMA		id, cdbase, class, style	⟨OMel⟩*																																																																
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OMR	href		⟨OMel⟩?																																																																
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured																																																																			

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																								
kohlhase-omdoc_TAB_019	120	—	(no LaTeX source; 1117 × 935 px region)	—	13.1 OpenMath 113 <table><tr><th>Shared</th><th>Exploded</th></tr><tr><td><OMOBJ> <OMA> <OMS cd="nat" name="plus"/> <OMA id="t1"> <OMS cd="nat" name="plus"/> <OMA id="t11"> <OMS cd="nat" name="plus"/> <OMI>1</OMI> <OMI>1</OMI> </OMA> <OMR href="#t11"/> </OMA> <OMR href="#t1"/> </OMA> </OMOBJ></td><td><OMOBJ> <OMA> <OMS cd="nat" name="plus"/> <OMA> <OMS cd="nat" name="plus"/> <OMA> <OMS cd="nat" name="plus"/> <OMI>1</OMI> <OMI>1</OMI> <OMI>1</OMI> </OMA> <OMA> <OMS cd="nat" name="plus"/> <OMI>1</OMI> <OMI>1</OMI> </OMA> </OMA> <OMA> <OMS cd="nat" name="plus"/> <OMA> <OMS cd="nat" name="plus"/> <OMI>1</OMI> <OMI>1</OMI> </OMA> <OMA> <OMS cd="nat" name="plus"/> <OMI>1</OMI> <OMI>1</OMI> </OMA> </OMA> </OMA> </OMOBJ></td></tr></table> Fig. 13.5. The OPENMATH objects from Figure 13.4 in XML encoding	Shared	Exploded	<OMOBJ> <OMA> <OMS cd="nat" name="plus"/> <OMA id="t1"> <OMS cd="nat" name="plus"/> <OMA id="t11"> <OMS cd="nat" name="plus"/> <OMI>1</OMI> <OMI>1</OMI> </OMA> <OMR href="#t11"/> </OMA> <OMR href="#t1"/> </OMA> </OMOBJ>	<OMOBJ> <OMA> <OMS cd="nat" name="plus"/> <OMA> <OMS cd="nat" name="plus"/> <OMA> <OMS cd="nat" name="plus"/> <OMI>1</OMI> <OMI>1</OMI> <OMI>1</OMI> </OMA> <OMA> <OMS cd="nat" name="plus"/> <OMI>1</OMI> <OMI>1</OMI> </OMA> </OMA> <OMA> <OMS cd="nat" name="plus"/> <OMA> <OMS cd="nat" name="plus"/> <OMI>1</OMI> <OMI>1</OMI> </OMA> <OMA> <OMS cd="nat" name="plus"/> <OMI>1</OMI> <OMI>1</OMI> </OMA> </OMA> </OMA> </OMOBJ>																																				
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kohlhase-omdoc_TAB_020	121	—	(no LaTeX source; 1160 × 218 px region)	—	<table><tr><td><OMOBJ> <OMA id="bar"> <OMS cd="nat" name="plus"/> <OMI>1</OMI> <OMR href="#baz"/> </OMA> </OMOBJ></td><td><OMOBJ> <OMA id="baz"> <OMS cd="nat" name="plus"/> <OMI>1</OMI> <OMR href="#bar"/> </OMA> </OMOBJ></td></tr></table>	<OMOBJ> <OMA id="bar"> <OMS cd="nat" name="plus"/> <OMI>1</OMI> <OMR href="#baz"/> </OMA> </OMOBJ>	<OMOBJ> <OMA id="baz"> <OMS cd="nat" name="plus"/> <OMI>1</OMI> <OMR href="#bar"/> </OMA> </OMOBJ>																																						
<OMOBJ> <OMA id="bar"> <OMS cd="nat" name="plus"/> <OMI>1</OMI> <OMR href="#baz"/> </OMA> </OMOBJ>	<OMOBJ> <OMA id="baz"> <OMS cd="nat" name="plus"/> <OMI>1</OMI> <OMR href="#bar"/> </OMA> </OMOBJ>																																												
kohlhase-omdoc_TAB_021	122	—	(no LaTeX source; 1142 × 439 px region)	—	13.2 Content MathML 115 <table><tr><th rowspan="2">Element</th><th colspan="2">Attributes</th><th rowspan="2">Content</th></tr><tr><th>Required</th><th>Optional</th></tr><tr><td>m:math</td><td></td><td>id, xlink:href</td><td>⟨⟨CMeℓ⟩⟩*</td></tr><tr><td>m:apply</td><td></td><td>id, xlink:href</td><td>m:bvar?,⟨⟨CMeℓ⟩⟩*</td></tr><tr><td>m:csymbol</td><td>definitionURL</td><td>id, xlink:href</td><td>m:EMPTY</td></tr><tr><td>m:ci</td><td></td><td>id, xlink:href</td><td>#PCDATA</td></tr><tr><td>m:cn</td><td></td><td>id, xlink:href</td><td>([0-9] 1 1.)([*]([0-9] 1 1.)*)?</td></tr><tr><td>m:bvar</td><td></td><td>id, xlink:href</td><td>m:ci m:semantics</td></tr><tr><td>m:semantics</td><td></td><td>id, xlink:href, definitionURL</td><td>⟨⟨CMeℓ⟩⟩,(m:annotation m:annotation-xml)*</td></tr><tr><td>m:annotation</td><td></td><td>definitionURL, encoding</td><td>#PCDATA</td></tr></table>	Element	Attributes		Content	Required	Optional	m:math		id, xlink:href	⟨⟨CMeℓ⟩⟩*	m:apply		id, xlink:href	m:bvar?,⟨⟨CMeℓ⟩⟩*	m:csymbol	definitionURL	id, xlink:href	m:EMPTY	m:ci		id, xlink:href	#PCDATA	m:cn		id, xlink:href	([0-9] 1 1.)([*]([0-9] 1 1.)*)?	m:bvar		id, xlink:href	m:ci m:semantics	m:semantics		id, xlink:href, definitionURL	⟨⟨CMeℓ⟩⟩,(m:annotation m:annotation-xml)*	m:annotation		definitionURL, encoding	#PCDATA		
Element	Attributes		Content																																										
	Required	Optional																																											
m:math		id, xlink:href	⟨⟨CMeℓ⟩⟩*																																										
m:apply		id, xlink:href	m:bvar?,⟨⟨CMeℓ⟩⟩*																																										
m:csymbol	definitionURL	id, xlink:href	m:EMPTY																																										
m:ci		id, xlink:href	#PCDATA																																										
m:cn		id, xlink:href	([0-9] 1 1.)([*]([0-9] 1 1.)*)?																																										
m:bvar		id, xlink:href	m:ci m:semantics																																										
m:semantics		id, xlink:href, definitionURL	⟨⟨CMeℓ⟩⟩,(m:annotation m:annotation-xml)*																																										
m:annotation		definitionURL, encoding	#PCDATA																																										
kohlhase-omdoc_TAB_022	128	—	(no LaTeX source; 1146 × 469 px region)	—	<table><tr><th></th><th>Required</th><th>Optional</th><th>U</th><th></th></tr><tr><td>CMP</td><td></td><td>xml:lang, id</td><td>—</td><td>⟨⟨math vernacular⟩⟩</td></tr><tr><td>FMP</td><td></td><td>xml:id, logic</td><td>—</td><td>(assumption*, conclusion*) OMOBJ [m:math legacy</td></tr><tr><td>assumption</td><td></td><td>xml:id, inductive, class, style</td><td>+</td><td>(OMOBJ [m:math legacy)</td></tr><tr><td>conclusion</td><td></td><td>xml:id, class, style</td><td>+</td><td>(OMOBJ [m:math legacy)</td></tr><tr><td>phrase</td><td></td><td>xml:id, class, style, index, verbalizes, type</td><td>—</td><td>⟨⟨math vernacular⟩⟩</td></tr><tr><td>term</td><td>cd, name</td><td>role, xml:id, class, style</td><td>—</td><td>⟨⟨math vernacular⟩⟩</td></tr><tr><td>omtext</td><td></td><td>xml:id, type, for, class, style, verbalizes</td><td>+</td><td>CMP+, FMP+</td></tr></table> Fig. 14.1. The OMDoc elements for specifying mathematical properties		Required	Optional	U		CMP		xml:lang, id	—	⟨⟨math vernacular⟩⟩	FMP		xml:id, logic	—	(assumption*, conclusion*) OMOBJ [m:math legacy	assumption		xml:id, inductive, class, style	+	(OMOBJ [m:math legacy)	conclusion		xml:id, class, style	+	(OMOBJ [m:math legacy)	phrase		xml:id, class, style, index, verbalizes, type	—	⟨⟨math vernacular⟩⟩	term	cd, name	role, xml:id, class, style	—	⟨⟨math vernacular⟩⟩	omtext		xml:id, type, for, class, style, verbalizes	+	CMP+, FMP+
	Required	Optional	U																																										
CMP		xml:lang, id	—	⟨⟨math vernacular⟩⟩																																									
FMP		xml:id, logic	—	(assumption*, conclusion*) OMOBJ [m:math legacy																																									
assumption		xml:id, inductive, class, style	+	(OMOBJ [m:math legacy)																																									
conclusion		xml:id, class, style	+	(OMOBJ [m:math legacy)																																									
phrase		xml:id, class, style, index, verbalizes, type	—	⟨⟨math vernacular⟩⟩																																									
term	cd, name	role, xml:id, class, style	—	⟨⟨math vernacular⟩⟩																																									
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kohlhase-omdoc_TAB_023	129	—	(no LaTeX source; 1107 × 265 px region)	—	Language [CML00]). <table><tr><th>Module</th><th>Elements</th><th>Comment</th><th>see</th></tr><tr><td>MOBJ</td><td>om:OMOBJ, m:math, legacy</td><td>mathematical Objects</td><td>p. 107</td></tr><tr><td>MTXT</td><td>phrase, term</td><td>phrase-level markup</td><td>below</td></tr><tr><td>DOC</td><td>ref, ignore</td><td>document structure</td><td>p. 89</td></tr><tr><td>RT</td><td>p, ol, ul, dl, table, link, note, idx</td><td>rich text structure</td><td>p. 128</td></tr><tr><td>EXT</td><td>omlet</td><td>for applets, images, ...</td><td>p. 205</td></tr></table>	Module	Elements	Comment	see	MOBJ	om:OMOBJ, m:math, legacy	mathematical Objects	p. 107	MTXT	phrase, term	phrase-level markup	below	DOC	ref, ignore	document structure	p. 89	RT	p, ol, ul, dl, table, link, note, idx	rich text structure	p. 128	EXT	omlet	for applets, images, ...	p. 205																
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Conf. MathPix confidence:

■ ≥0.9

■ 0.5-0.9

■ <0.5

— no value

Residual render vs scan (inkdrill):

● C component

● W weak

● S stable

● N noise

● K clean

● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																																								
kohlhase-omdoc_TAB_024	136	—	(no LaTeX source; 1110 × 564 px region)	—	14.6 Rich Text Structure129 <table><tr><th>Element</th><th>Optional Attributes</th><th>DC</th><th>Content</th></tr><tr><td>p</td><td>xml:id, style, class, index, verbalizes</td><td>+</td><td>⟨⟨math vernacular⟩⟩</td></tr><tr><td>ol</td><td>xml:id, style, class, index, verbalizes</td><td>+</td><td>li*</td></tr><tr><td>ul</td><td>xml:id, style, class, index, verbalizes</td><td>+</td><td>li*</td></tr><tr><td>li</td><td>xml:id, style, class, index, verbalizes</td><td>+</td><td>⟨⟨math vernacular⟩⟩</td></tr><tr><td>dl</td><td>xml:id, style, class, index, verbalizes</td><td>+</td><td>di*</td></tr><tr><td>di</td><td>xml:id, style, class, index, verbalizes</td><td>+</td><td>dt*,dd*</td></tr><tr><td>dt</td><td>xml:id, style, class, index, verbalizes</td><td>+</td><td>⟨⟨math vernacular⟩⟩</td></tr><tr><td>dd</td><td>xml:id, style, class, index, verbalizes</td><td>+</td><td>⟨⟨math vernacular⟩⟩</td></tr><tr><td>idx</td><td>(xml:id xref)</td><td>—</td><td>idt?, ide+</td></tr><tr><td>ide</td><td>index, sort-by, see, seealso, links</td><td>—</td><td>idp*</td></tr><tr><td>idt</td><td>style, class</td><td>—</td><td>⟨⟨math vernacular⟩⟩</td></tr><tr><td>idp</td><td>sort-by, see, seealso, links</td><td>—</td><td>⟨⟨math vernacular⟩⟩</td></tr><tr><td>table</td><td>xml:id, style, class, index, verbalizes</td><td>+</td><td>tr*</td></tr><tr><td>tr</td><td>xml:id, style, class, index, verbalizes</td><td>+</td><td>td*</td></tr><tr><td>td</td><td>xml:id, style, class, index, verbalizes</td><td>+</td><td>⟨⟨math vernacular⟩⟩</td></tr><tr><td>th</td><td>xml:id, style, class, index, verbalizes</td><td>+</td><td>⟨⟨math vernacular⟩⟩</td></tr><tr><td>link</td><td>xml:id, style, class, index, verbalizes</td><td>—</td><td>⟨⟨math vernacular⟩⟩</td></tr></table>	Element	Optional Attributes	DC	Content	p	xml:id, style, class, index, verbalizes	+	⟨⟨math vernacular⟩⟩	ol	xml:id, style, class, index, verbalizes	+	li*	ul	xml:id, style, class, index, verbalizes	+	li*	li	xml:id, style, class, index, verbalizes	+	⟨⟨math vernacular⟩⟩	dl	xml:id, style, class, index, verbalizes	+	di*	di	xml:id, style, class, index, verbalizes	+	dt*,dd*	dt	xml:id, style, class, index, verbalizes	+	⟨⟨math vernacular⟩⟩	dd	xml:id, style, class, index, verbalizes	+	⟨⟨math vernacular⟩⟩	idx	(xml:id xref)	—	idt?, ide+	ide	index, sort-by, see, seealso, links	—	idp*	idt	style, class	—	⟨⟨math vernacular⟩⟩	idp	sort-by, see, seealso, links	—	⟨⟨math vernacular⟩⟩	table	xml:id, style, class, index, verbalizes	+	tr*	tr	xml:id, style, class, index, verbalizes	+	td*	td	xml:id, style, class, index, verbalizes	+	⟨⟨math vernacular⟩⟩	th	xml:id, style, class, index, verbalizes	+	⟨⟨math vernacular⟩⟩	link	xml:id, style, class, index, verbalizes	—	⟨⟨math vernacular⟩⟩
Element	Optional Attributes	DC	Content																																																																										
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kohlhase-omdoc_TAB_025	141	—	(no LaTeX source; 1143 × 262 px region)	—	<table><tr><td>The exponential function e^x</td><td>The exponential function e^x is the solution to the differential equation $\partial f = f$ [where $f(0) = 1$].</td><td>implicit</td></tr><tr><td>The addition function $+$</td><td>Addition on the natural numbers is defined by the equations $x + 0 = x$ and $x + s(y) = s(x + y)$.</td><td>recursive</td></tr></table> Fig. 15.2. Some common definitions Definitions can have many forms, they can be <table><tr><th>Element</th><th colspan="2">Attributes</th><th>D</th><th>Content</th></tr><tr><td></td><th>Required</th><th>Optional</th><th>C</th><td></td></tr><tr><td>symbol</td><td>name</td><td>xml:id, role, scope, style, class</td><td>+</td><td>type*</td></tr><tr><td>type</td><td></td><td>xml:id, system, style, class</td><td>—</td><td>CMP*,⟨⟨mobj⟩⟩</td></tr><tr><td>axiom</td><td></td><td>xml:id, for, type, style, class</td><td>+</td><td>CMP*,FMP*</td></tr><tr><td>definition</td><td>for</td><td>xml:id, type, style, class, uniqueness, existence, consistency, exhaustivity</td><td>+</td><td>CMP*, (FMP* reequation+ ⟨⟨mobj⟩⟩)?, measure?, ordering?</td></tr><tr><td>requation</td><td></td><td>xml:id, style, class</td><td>—</td><td>⟨⟨mobj⟩⟩,⟨⟨mobj⟩⟩</td></tr><tr><td>measure</td><td></td><td>xml:id, style, class</td><td>—</td><td>⟨⟨mobj⟩⟩</td></tr><tr><td>ordering</td><td></td><td>xml:id, style, class</td><td>—</td><td>⟨⟨mobj⟩⟩</td></tr><tr><td colspan="5">where ⟨⟨mobj⟩⟩ is (OMOBJ m:math legacy)</td></tr></table> Fig. 15.3. Theory-constitutive elements in OMDoc	The exponential function e^x	The exponential function e^x is the solution to the differential equation $\partial f = f$ [where $f(0) = 1$].	implicit	The addition function $+$	Addition on the natural numbers is defined by the equations $x + 0 = x$ and $x + s(y) = s(x + y)$.	recursive	Element	Attributes		D	Content		Required	Optional	C		symbol	name	xml:id, role, scope, style, class	+	type*	type		xml:id, system, style, class	—	CMP*,⟨⟨mobj⟩⟩	axiom		xml:id, for, type, style, class	+	CMP*,FMP*	definition	for	xml:id, type, style, class, uniqueness, existence, consistency, exhaustivity	+	CMP*, (FMP* reequation+ ⟨⟨mobj⟩⟩)?, measure?, ordering?	requation		xml:id, style, class	—	⟨⟨mobj⟩⟩,⟨⟨mobj⟩⟩	measure		xml:id, style, class	—	⟨⟨mobj⟩⟩	ordering		xml:id, style, class	—	⟨⟨mobj⟩⟩	where ⟨⟨mobj⟩⟩ is (OMOBJ m:math legacy)																				
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kohlhase-omdoc_TAB_027	148	—	(no LaTeX source; 1137 × 469 px region)	—	14215 Mathematical Statements <table><tr><th>Element</th><th colspan="2">Attributes</th><th>D</th><th>Content</th></tr><tr><td></td><th>Required</th><th>Optional</th><th>C</th><td></td></tr><tr><td>assertion</td><td></td><td>xml:id, for, type, theory, class, style, status, just-by</td><td>+</td><td>CMP*, FMP*</td></tr><tr><td>type</td><td>system</td><td>xml:id, for, just-by, theory, class, style</td><td>—</td><td>CMP*, ⟨⟨mobj⟩⟩,⟨⟨mobj⟩⟩</td></tr><tr><td>example</td><td>for</td><td>xml:id, type, assertion, theory, class, style</td><td>+</td><td>CMP* ⟨⟨mobj⟩⟩*</td></tr><tr><td>alternative</td><td>for, theory, entailed-by, entails, entailed-by-thm</td><td>xml:id, type, theory, class, style</td><td>+</td><td>CMP*, (FMP* reequation* ⟨⟨mobj⟩⟩)</td></tr></table>	Element	Attributes		D	Content		Required	Optional	C		assertion		xml:id, for, type, theory, class, style, status, just-by	+	CMP*, FMP*	type	system	xml:id, for, just-by, theory, class, style	—	CMP*, ⟨⟨mobj⟩⟩,⟨⟨mobj⟩⟩	example	for	xml:id, type, assertion, theory, class, style	+	CMP* ⟨⟨mobj⟩⟩*	alternative	for, theory, entailed-by, entails, entailed-by-thm	xml:id, type, theory, class, style	+	CMP*, (FMP* reequation* ⟨⟨mobj⟩⟩)																																										
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Conf. MathPix confidence:

■

 ≥0.9

■

 0.5-0.9

■

 <0.5

— no value

Residual render vs scan (inkdrill):

●

 C component

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 W weak

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 S stable

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 N noise

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 K clean

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 not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																		
kohlhase-omdoc_TAB_028	149	—	(no LaTeX source; 1171 × 1052 px region)	—	15.3 The Unassuming Rest143 <table><tr><td>Value</td><td>Explanation</td></tr><tr><td>theorem, proposition</td><td>an important assertion with a proof</td></tr><tr><td colspan="2">Note that the meaning of the type (in this case the existence of a proof) is not enforced by OMDoc applications. It can be appropriate to give an assertion the type theorem, if the author knows of a proof (e.g. in the literature), but has not formalized it in OMDoc yet.</td></tr><tr><td>lemma</td><td>a less important assertion with a proof</td></tr><tr><td colspan="2">The difference of importance specified in this type is even softer than the other ones, since e.g. reusing a mathematical paper as a chapter in a larger monograph, may make it necessary to downgrade a theorem (e.g. the main theorem of the paper) and give it the status of a lemma in the overall work.</td></tr><tr><td>corollary</td><td>a simple consequence</td></tr><tr><td colspan="2">An assertion is sometimes marked as a corollary to some other statement, if the proof is considered simple. This is often the case for important theorems that are simple to get from technical lemmata.</td></tr><tr><td>postulate, conjecture</td><td>an assertion without proof or counter-example</td></tr><tr><td colspan="2">Conjectures are assertions, whose semantic value is not yet decided, but which the author considers likely to be true. In particular, there is no proof or counter-example (see Section 15.4).</td></tr><tr><td>false-conjecture</td><td>an assertion with a counter-example</td></tr><tr><td colspan="2">A conjecture that has proven to be false, i.e. it has a counter-example. Such assertions are often kept for illustration and historical purposes.</td></tr><tr><td>obligation, assumption</td><td>an assertion on which the proof of another depends</td></tr><tr><td colspan="2">These kinds of assertions are convenient during the exploration of a mathematical theory. They can be used and proven later (or assumed as an axiom).</td></tr><tr><td>formula</td><td>if everything else fails</td></tr><tr><td colspan="2">This type is the catch-all if none of the others applies.</td></tr></table> Fig. 15.10. Types of mathematical assertions	Value	Explanation	theorem, proposition	an important assertion with a proof	Note that the meaning of the type (in this case the existence of a proof) is not enforced by OMDoc applications. It can be appropriate to give an assertion the type theorem , if the author knows of a proof (e.g. in the literature), but has not formalized it in OMDoc yet.		lemma	a less important assertion with a proof	The difference of importance specified in this type is even softer than the other ones, since e.g. reusing a mathematical paper as a chapter in a larger monograph, may make it necessary to downgrade a theorem (e.g. the main theorem of the paper) and give it the status of a lemma in the overall work.		corollary	a simple consequence	An assertion is sometimes marked as a corollary to some other statement, if the proof is considered simple. This is often the case for important theorems that are simple to get from technical lemmata.		postulate, conjecture	an assertion without proof or counter-example	Conjectures are assertions, whose semantic value is not yet decided, but which the author considers likely to be true. In particular, there is no proof or counter-example (see Section 15.4).		false-conjecture	an assertion with a counter-example	A conjecture that has proven to be false, i.e. it has a counter-example. Such assertions are often kept for illustration and historical purposes.		obligation, assumption	an assertion on which the proof of another depends	These kinds of assertions are convenient during the exploration of a mathematical theory. They can be used and proven later (or assumed as an axiom).		formula	if everything else fails	This type is the catch-all if none of the others applies.																					
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This type is the catch-all if none of the others applies.																																																							
kohlhase-omdoc_TAB_029	150	—	(no LaTeX source; 1135 × 1126 px region)	—	14415 Mathematical Statements <table><tr><td>status</td><td>just-by points to</td></tr><tr><td>tautology</td><td>Proof of $\mathcal{F}$</td></tr><tr><td colspan="2"><i>All T-interpretations satisfy $\mathcal{A}$ and some C_i</i></td></tr><tr><td>tautologous-conclusion</td><td>Proof of $\mathcal{F}_c$.</td></tr><tr><td colspan="2"><i>All T-interpretations satisfy some C_j</i></td></tr><tr><td>equivalent</td><td>Proofs of $\mathcal{F}$ and $\mathcal{F}^{-1}$</td></tr><tr><td colspan="2"><i>$\mathcal{A}$ and $\mathcal{C}$ have the same T-models (and there are some)</i></td></tr><tr><td>theorem</td><td>Proof of $\mathcal{F}$</td></tr><tr><td colspan="2"><i>All T-models of $\mathcal{A}$ (and there are some) satisfy some C_i</i></td></tr><tr><td>satisfiable</td><td>Model of $\mathcal{A}$ and some C_i</td></tr><tr><td colspan="2"><i>Some T-models of $\mathcal{A}$ (and there are some) satisfy some C_i</i></td></tr><tr><td>contradictory-axioms</td><td>Refutation of $\mathcal{A}$</td></tr><tr><td colspan="2"><i>There are no T-models of $\mathcal{A}$</i></td></tr><tr><td>no-consequence</td><td>T-model of $\mathcal{A}$ and some C_i, T-model of $\mathcal{A} \cup \overline{\mathcal{C}}$.</td></tr><tr><td colspan="2"><i>Some T-models of $\mathcal{A}$ (and there are some) satisfy some C_i, some satisfy $\overline{\mathcal{C}}$</i></td></tr><tr><td>counter-satisfiable</td><td>Model of $\mathcal{A} \cup \overline{\mathcal{C}}$</td></tr><tr><td colspan="2"><i>Some T-models of $\mathcal{A}$ (and there are some) satisfy $\overline{\mathcal{C}}$</i></td></tr><tr><td>counter-theorem</td><td>Proof of $\overline{\mathcal{C}}$ from $\mathcal{A}$</td></tr><tr><td colspan="2"><i>All T-models of $\mathcal{A}$ (and there are some) satisfy $\overline{\mathcal{C}}$</i></td></tr><tr><td>counter-equivalent</td><td>Proof of $\overline{\mathcal{C}}$ from $\mathcal{A}$ and proof of $\mathcal{A}$ from $\overline{\mathcal{C}}$</td></tr><tr><td colspan="2"><i>$\mathcal{A}$ and $\overline{\mathcal{C}}$ have the same T-models (and there are some)</i></td></tr><tr><td>unsatisfiable-conclusion</td><td>Proof of $\overline{\mathcal{C}}$</td></tr><tr><td colspan="2"><i>All T-interpretations satisfy $\overline{\mathcal{C}}$</i></td></tr><tr><td>unsatisfiable</td><td>Proof of $\neg \mathcal{F}$</td></tr><tr><td colspan="2"><i>All T-interpretations satisfy $\mathcal{A}$ and $\overline{\mathcal{C}}$</i></td></tr></table> Where $\mathcal{F}$ is an assertion whose FMP has assumption elements $\mathcal{A}_1, \dots, \mathcal{A}_n$ and conclusion elements $C_1, \dots, C_m$. Furthermore, let $\mathcal{A} := \{\mathcal{A}_1, \dots, \mathcal{A}_n\}$ and $\mathcal{C} := \{C_1, \dots, C_m\}$, and $\mathcal{F}^{-1}$ be the sequent that has the C_i as assumptions and the $\mathcal{A}_i$ as conclusions. Finally, let $\overline{\mathcal{C}} := \{\overline{C}_1, \dots, \overline{C}_m\}$, where $\overline{C}_i$ is a negation of C_i. Fig. 15.12. Proof status for assertions in a theory T	status	just-by points to	tautology	Proof of $\mathcal{F}$	<i>All T-interpretations satisfy $\mathcal{A}$ and some C_i</i>		tautologous-conclusion	Proof of $\mathcal{F}_c$.	<i>All T-interpretations satisfy some C_j</i>		equivalent	Proofs of $\mathcal{F}$ and $\mathcal{F}^{-1}$	<i>$\mathcal{A}$ and $\mathcal{C}$ have the same T-models (and there are some)</i>		theorem	Proof of $\mathcal{F}$	<i>All T-models of $\mathcal{A}$ (and there are some) satisfy some C_i</i>		satisfiable	Model of $\mathcal{A}$ and some C_i	<i>Some T-models of $\mathcal{A}$ (and there are some) satisfy some C_i</i>		contradictory-axioms	Refutation of $\mathcal{A}$	<i>There are no T-models of $\mathcal{A}$</i>		no-consequence	T -model of $\mathcal{A}$ and some C_i , T -model of $\mathcal{A} \cup \overline{\mathcal{C}}$.	<i>Some T-models of $\mathcal{A}$ (and there are some) satisfy some C_i, some satisfy $\overline{\mathcal{C}}$</i>		counter-satisfiable	Model of $\mathcal{A} \cup \overline{\mathcal{C}}$	<i>Some T-models of $\mathcal{A}$ (and there are some) satisfy $\overline{\mathcal{C}}$</i>		counter-theorem	Proof of $\overline{\mathcal{C}}$ from $\mathcal{A}$	<i>All T-models of $\mathcal{A}$ (and there are some) satisfy $\overline{\mathcal{C}}$</i>		counter-equivalent	Proof of $\overline{\mathcal{C}}$ from $\mathcal{A}$ and proof of $\mathcal{A}$ from $\overline{\mathcal{C}}$	<i>$\mathcal{A}$ and $\overline{\mathcal{C}}$ have the same T-models (and there are some)</i>		unsatisfiable-conclusion	Proof of $\overline{\mathcal{C}}$	<i>All T-interpretations satisfy $\overline{\mathcal{C}}$</i>		unsatisfiable	Proof of $\neg \mathcal{F}$	<i>All T-interpretations satisfy $\mathcal{A}$ and $\overline{\mathcal{C}}$</i>	
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<i>$\mathcal{A}$ and $\mathcal{C}$ have the same T-models (and there are some)</i>																																																							
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contradictory-axioms	Refutation of $\mathcal{A}$																																																						
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no-consequence	T -model of $\mathcal{A}$ and some C_i , T -model of $\mathcal{A} \cup \overline{\mathcal{C}}$.																																																						
<i>Some T-models of $\mathcal{A}$ (and there are some) satisfy some C_i, some satisfy $\overline{\mathcal{C}}$</i>																																																							
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<i>All T-interpretations satisfy $\overline{\mathcal{C}}$</i>																																																							
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kohlhase-omdoc_TAB_030	156	—	(no LaTeX source; 1132 × 217 px region)	—	15015 Mathematical Statements <table><tr><th>Element</th><th>Attributes</th><th>D</th><th>Content</th></tr><tr><td></td><td>Req. T. Optional</td><td>C</td><td></td></tr></table>	Element	Attributes	D	Content		Req. T. Optional	C																																											
Element	Attributes	D	Content																																																				
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured																																																							

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																							
kohlhase-omdoc_TAB_031	162	—	(no LaTeX source; 1058 × 387 px region)	—	15616 Abstract Data Types <table><tr><th rowspan="2">Element</th><th colspan="2">Attributes</th><th>D</th><th>Content</th></tr><tr><th>Req.</th><th>Optional</th><th>C</th><th></th></tr><tr><td>adt</td><td></td><td>xml:id, class, style, parameters</td><td>+</td><td>sortdef+</td></tr><tr><td>sortdef</td><td>name</td><td>type, role, scope, class, style</td><td>+</td><td>(constructor insert)*, recognizer?</td></tr><tr><td>constructor</td><td>name</td><td>type, scope, class, style</td><td>+</td><td>argument*</td></tr><tr><td>argument</td><td></td><td></td><td>+</td><td>type, selector?</td></tr><tr><td>insert</td><td>for</td><td></td><td>—</td><td></td></tr></table>	Element	Attributes		D	Content	Req.	Optional	C		adt		xml:id, class, style, parameters	+	sortdef+	sortdef	name	type, role, scope, class, style	+	(constructor insert)*, recognizer?	constructor	name	type, scope, class, style	+	argument*	argument			+	type, selector?	insert	for		—						
Element	Attributes		D	Content																																								
	Req.	Optional	C																																									
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sortdef	name	type, role, scope, class, style	+	(constructor insert)*, recognizer?																																								
constructor	name	type, scope, class, style	+	argument*																																								
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insert	for		—																																									
kohlhase-omdoc_TAB_032	168	—	(no LaTeX source; 1107 × 385 px region)	—	16217 Representing Proofs <table><tr><th rowspan="2">Element</th><th colspan="2">Attributes</th><th>D</th><th>Content</th></tr><tr><th>Req.</th><th>Optional</th><th>C</th><th></th></tr><tr><td>proof</td><td>for</td><td>theory, xml:id, class, style</td><td>+</td><td>(context derive hypothesis symbol definition)*</td></tr><tr><td>proofobject</td><td></td><td>xml:id, for, class, style, theory</td><td>+</td><td>CMP*, (OMOBJ m:math legacy)</td></tr><tr><td>hypothesis</td><td></td><td>xml:id, class, style, inductive</td><td>—</td><td>CMP*, FMP*</td></tr><tr><td>derive</td><td></td><td>xml:id, class, style, method</td><td>—</td><td>CMP*, FMP*, method?</td></tr></table>	Element	Attributes		D	Content	Req.	Optional	C		proof	for	theory, xml:id, class, style	+	(context derive hypothesis symbol definition)*	proofobject		xml:id, for, class, style, theory	+	CMP*, (OMOBJ m:math legacy)	hypothesis		xml:id, class, style, inductive	—	CMP*, FMP*	derive		xml:id, class, style, method	—	CMP*, FMP*, method?										
Element	Attributes		D	Content																																								
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proof	for	theory, xml:id, class, style	+	(context derive hypothesis symbol definition)*																																								
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hypothesis		xml:id, class, style, inductive	—	CMP*, FMP*																																								
derive		xml:id, class, style, method	—	CMP*, FMP*, method?																																								
kohlhase-omdoc_TAB_033	178	—	(no LaTeX source; 1146 × 525 px region)	—	<table><tr><td>morphism</td><td></td><td>xml:id, base, class, style, type, hiding, consistency, exhaustivity</td><td>—</td><td>requation*, measure?, ordering?</td></tr><tr><td>inclusion</td><td>via</td><td>xml:id, conservativity, conservativity-just</td><td>—</td><td>EMPTY</td></tr><tr><td>theory-inclusion</td><td>from, to</td><td>xml:id, class, style, conservativity, conservativity-just</td><td>+</td><td>(CMP*,FMP*, morphism, obligation*)</td></tr><tr><td>axiom-inclusion</td><td>from, to</td><td>xml:id, class, style, conservativity, conservativity-just</td><td>+</td><td>morphism?, obligation*</td></tr></table> Fig. 18.1. Complex theories in OMDoc M. Kohlhase: OMDoc, LNAI 4180, pp. 173–186, 2006.	morphism		xml:id, base, class, style, type, hiding, consistency, exhaustivity	—	requation*, measure?, ordering?	inclusion	via	xml:id, conservativity, conservativity-just	—	EMPTY	theory-inclusion	from, to	xml:id, class, style, conservativity, conservativity-just	+	(CMP*,FMP*, morphism, obligation*)	axiom-inclusion	from, to	xml:id, class, style, conservativity, conservativity-just	+	morphism?, obligation*																			
morphism		xml:id, base, class, style, type, hiding, consistency, exhaustivity	—	requation*, measure?, ordering?																																								
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axiom-inclusion	from, to	xml:id, class, style, conservativity, conservativity-just	+	morphism?, obligation*																																								
kohlhase-omdoc_TAB_034	189	—	(no LaTeX source; 1135 × 272 px region)	—	<table><tr><td>axiom-inclusion</td><td>by</td><td>class, style</td><td>+</td><td>(decomposition* obligation* obligation*)</td></tr></table> Fig. 18.10. Development graphs in OMDoc Furthermore module DG provides path-just elements as children to the axiom-inclusion elements to justify that this relation holds, much like a proof element provides a justification for an assertion element for some property of mathematical objects.	axiom-inclusion	by	class, style	+	(decomposition* obligation* obligation*)																																		
axiom-inclusion	by	class, style	+	(decomposition* obligation* obligation*)																																								
kohlhase-omdoc_TAB_035	192	—	(no LaTeX source; 1150 × 317 px region)	—	<table><tr><td>presentation</td><td>via</td><td>xml:id, class, style, precedence, crossref-symbol</td><td></td><td>(use as style)*</td></tr><tr><td>xslt</td><td>format</td><td>xml:lang, requires, xref</td><td></td><td>XSLT fragment</td></tr><tr><td>use</td><td>format</td><td>xml:lang, requires, fixity, lbrack, rbrack, separator, element, attributes, crossref-symbol</td><td></td><td>(element text recurse map value-of)*</td></tr></table> Fig. 19.1. The OMDoc elements for notation information The normal (but of course not the only) way to generate presentation from XML documents is to use XSLT style sheets (see Chapter 25 for other approaches), but this does not work for the presentation of the elements of the namespace) of the elements for the presentation to work.²	presentation	via	xml:id, class, style, precedence, crossref-symbol		(use as style)*	xslt	format	xml:lang, requires, xref		XSLT fragment	use	format	xml:lang, requires, fixity, lbrack, rbrack, separator, element, attributes, crossref-symbol		(element text recurse map value-of)*																								
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xslt	format	xml:lang, requires, xref		XSLT fragment																																								
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kohlhase-omdoc_TAB_036	195	—	(no LaTeX source; 1141 × 380 px region)	—	<table><tr><th rowspan="2">Element</th><th colspan="2">Attributes</th><th>Content</th></tr><tr><th>Required</th><th>Optional</th><th></th></tr><tr><td>style</td><td>format</td><td>xml:lang, requires, xref</td><td>(element text recurse map value-of)*</td></tr><tr><td>element</td><td>name</td><td>cr:cr, cr, ns</td><td>(attribute element text value-of recurse map)*</td></tr><tr><td>attribute</td><td>name</td><td></td><td>(value-of text)*</td></tr><tr><td>text</td><td></td><td></td><td>(#PCDATA)</td></tr><tr><td>value-of</td><td>select</td><td></td><td>EMPTY</td></tr><tr><td>recurse</td><td></td><td>select</td><td>EMPTY</td></tr><tr><td>map</td><td></td><td>select</td><td>separator?, (element text recurse map)</td></tr><tr><td>separator</td><td></td><td></td><td>(element text recurse map)</td></tr></table>	Element	Attributes		Content	Required	Optional		style	format	xml:lang, requires, xref	(element text recurse map value-of)*	element	name	cr:cr, cr, ns	(attribute element text value-of recurse map)*	attribute	name		(value-of text)*	text			(#PCDATA)	value-of	select		EMPTY	recurse		select	EMPTY	map		select	separator?, (element text recurse map)	separator			(element text recurse map)
Element	Attributes		Content																																									
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separator			(element text recurse map)																																									
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured																																												

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																				
kohlhase-omdoc_TAB_037	201	—	(no LaTeX source; 717 × 258 px region)	—	ation and Presentation <table><tr><th>Precedence</th><th>Operators</th><th>Comment</th></tr><tr><td>200</td><td>+,-</td><td>unary</td></tr><tr><td>200</td><td>*</td><td>exponentiation</td></tr></table>	Precedence	Operators	Comment	200	+,-	unary	200	*	exponentiation											
Precedence	Operators	Comment																							
200	+,-	unary																							
200	*	exponentiation																							
kohlhase-omdoc_TAB_038	201	—	(no LaTeX source; 781 × 234 px region)	—	<table><tr><th>fixity</th><th>bracket-style</th><th>separator</th><th>yields</th></tr><tr><td>prefix</td><td>lisp</td><td>" "</td><td>$(f\ 1\ 2\ 3)$</td></tr><tr><td>postfix</td><td>lisp</td><td>" "</td><td>$(1\ 2\ 3)\ f$</td></tr><tr><td>prefix</td><td>math</td><td>" "</td><td>$f(1, 2, 3)$</td></tr><tr><td>postfix</td><td>math</td><td>" "</td><td>$(1, 2, 3)f$</td></tr></table> assuming lbrack="(" and rbrack=")" Attributes: combination and function style	fixity	bracket-style	separator	yields	prefix	lisp	" "	$(f\ 1\ 2\ 3)$	postfix	lisp	" "	$(1\ 2\ 3)\ f$	prefix	math	" "	$f(1, 2, 3)$	postfix	math	" "	$(1, 2, 3)f$
fixity	bracket-style	separator	yields																						
prefix	lisp	" "	$(f\ 1\ 2\ 3)$																						
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prefix	math	" "	$f(1, 2, 3)$																						
postfix	math	" "	$(1, 2, 3)f$																						
kohlhase-omdoc_TAB_039	202	—	(no LaTeX source; 1133 × 586 px region)	—	19.3 Notation of Symbols 197 <table><tr><th>Notation specification</th><th>Example</th></tr><tr><td><pre><presentation for="#forall" role="binding" separator=","> <use format="TeX">\forall</use> <use format="html">&#8704;</use> </presentation></pre></td><td><pre><OMBIND> <OMS cd="quant1" name="forall" /> <OMBVAR> <OMV name="X" /> </OMBVAR> <OMS cd="logic1" name="true" /> </OMBIND></pre></td></tr></table> Using XSLT templates induced from the presentation element on the OPEN-MATH expression yields $\forall X.\text{true}$, where the glyph $\forall$ carries a hyperlink⁶ to its definition, as the crossref-symbol on the presentation element has the default value yes. Internally, the hyperlinks are format-dependent, we have: $\text{\LaTeX:}$ <code>\href{../ocd/logic1.ps#true}{\forallforall}X.</code> <code>\href{../ocd/logic1.ps#true}{\sf true}}</code> HTML: <code>&#8704; X.</code> <code>true</code>	Notation specification	Example	<pre><presentation for="#forall" role="binding" separator=","> <use format="TeX">\forall</use> <use format="html">&#8704;</use> </presentation></pre>	<pre><OMBIND> <OMS cd="quant1" name="forall" /> <OMBVAR> <OMV name="X" /> </OMBVAR> <OMS cd="logic1" name="true" /> </OMBIND></pre>																
Notation specification	Example																								
<pre><presentation for="#forall" role="binding" separator=","> <use format="TeX">\forall</use> <use format="html">&#8704;</use> </presentation></pre>	<pre><OMBIND> <OMS cd="quant1" name="forall" /> <OMBVAR> <OMV name="X" /> </OMBVAR> <OMS cd="logic1" name="true" /> </OMBIND></pre>																								
kohlhase-omdoc_TAB_040	206	—	(no LaTeX source; 1133 × 329 px region)	—	<table><tr><td>output</td><td></td><td>xml:id, style, class</td><td>+</td><td>CMP*, FMP*</td></tr><tr><td>effect</td><td></td><td>xml:id, style, class</td><td>+</td><td>CMP*, FMP*</td></tr><tr><td>data</td><td></td><td>format, href, size, original, pto, pto-version</td><td>-</td><td><![CDATA[...]]></td></tr></table> Fig. 20.1. The OMDoc auxiliary elements for non-XML data ¹ Compare Chapter 9 in the OMDoc Primer.	output		xml:id, style, class	+	CMP*, FMP*	effect		xml:id, style, class	+	CMP*, FMP*	data		format, href, size, original, pto, pto-version	-	<![CDATA[...]]>					
output		xml:id, style, class	+	CMP*, FMP*																					
effect		xml:id, style, class	+	CMP*, FMP*																					
data		format, href, size, original, pto, pto-version	-	<![CDATA[...]]>																					
kohlhase-omdoc_TAB_041	210	—	(no LaTeX source; 1151 × 158 px region)	—	<table><tr><td>param</td><td>name</td><td>actuate, class, style value, valuetype</td><td>-</td><td>EMPTY</td></tr></table> Fig. 20.5. The OMDoc elements for external objects	param	name	actuate, class, style value, valuetype	-	EMPTY															
param	name	actuate, class, style value, valuetype	-	EMPTY																					
kohlhase-omdoc_TAB_042	214	—	(no LaTeX source; 1153 × 247 px region)	—	<table><tr><td>solution</td><td></td><td>xml:id, for, class, style</td><td>+</td><td>⟨⟨top-level element⟩⟩</td></tr><tr><td>mc</td><td></td><td>xml:id, for, class, style</td><td>-</td><td>choice, hint?, answer</td></tr><tr><td>choice</td><td></td><td>xml:id, class, style</td><td>+</td><td>CMP*, FMP*</td></tr><tr><td>answer</td><td>verdict</td><td>xml:id, class, style</td><td>+</td><td>CMP*, FMP*</td></tr></table> Fig. 21.1. The OMDoc auxiliary elements for exercises The QUIZ module provides the top-level elements exercise, hint, and solution. The first one is used for exercises and assessments. The ques-	solution		xml:id, for, class, style	+	⟨⟨top-level element⟩⟩	mc		xml:id, for, class, style	-	choice, hint?, answer	choice		xml:id, class, style	+	CMP*, FMP*	answer	verdict	xml:id, class, style	+	CMP*, FMP*
solution		xml:id, for, class, style	+	⟨⟨top-level element⟩⟩																					
mc		xml:id, for, class, style	-	choice, hint?, answer																					
choice		xml:id, class, style	+	CMP*, FMP*																					
answer	verdict	xml:id, class, style	+	CMP*, FMP*																					
kohlhase-omdoc_TAB_043	220	—	(no LaTeX source; 1157 × 457 px region)	—	<table><tr><td></td><td></td><td>tributes differ.</td><td>data dc:title solution</td></tr><tr><td>3</td><td>DAG encoding</td><td>Directed acyclic graphs built up using om:OMR elements are equal, iff their tree expansions are equal.</td><td>om:OMR ref</td></tr><tr><td>4</td><td>Dataset</td><td>If the content of the dc:type element is Dataset, then the order of the siblings of the parent metadata element is irrelevant.</td><td>dc:type</td></tr></table> Fig. 22.4. The OMDoc document model			tributes differ.	data dc:title solution	3	DAG encoding	Directed acyclic graphs built up using om:OMR elements are equal, iff their tree expansions are equal.	om:OMR ref	4	Dataset	If the content of the dc:type element is Dataset, then the order of the siblings of the parent metadata element is irrelevant.	dc:type								
		tributes differ.	data dc:title solution																						
3	DAG encoding	Directed acyclic graphs built up using om:OMR elements are equal, iff their tree expansions are equal.	om:OMR ref																						
4	Dataset	If the content of the dc:type element is Dataset, then the order of the siblings of the parent metadata element is irrelevant.	dc:type																						

Conf. MathPix confidence:

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 ≥0.9

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 <0.5

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Residual render vs scan (inkdrill):

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 C component

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 W weak

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 S stable

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 N noise

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 K clean

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Identifier	Page	Conf.	LaTeX source	Rendered	Scan image														
kohlhase-omdoc_TAB_044	227	—	(no LaTeX source; 1086 × 439 px region)	—	svn co https://svn.mathweb.org/repos/mathweb.org/branches/omdoc-1.2 This will create a directory <code>omdoc</code>, with the sub-directories <table><tr><th>directories</th><th>content</th></tr><tr><td><code>bin</code>, <code>lib</code>, <code>oz</code>, <code>thirdParty</code></td><td>programs and third-party software used in the administration and examples</td></tr><tr><td><code>css</code>, <code>xsl</code></td><td>style sheets for displaying OMDoc documents on the web, see Chapter 25 for a discussion.</td></tr><tr><td><code>doc</code></td><td>The OMDoc documentation, including the specification, papers about a the OMDoc format and tools.</td></tr><tr><td><code>dtd</code>, <code>rnc</code></td><td>The OMDoc document type definition and the RELAXNG schemata for OMDoc</td></tr><tr><td><code>examples</code></td><td>Various example documents in OMDoc format.</td></tr><tr><td><code>projects</code></td><td>various contributed developments for OMDoc. Documentation is usually in their <code>doc</code> sub-directory</td></tr></table>	directories	content	<code>bin</code> , <code>lib</code> , <code>oz</code> , <code>thirdParty</code>	programs and third-party software used in the administration and examples	<code>css</code> , <code>xsl</code>	style sheets for displaying OMDoc documents on the web, see Chapter 25 for a discussion.	<code>doc</code>	The OMDoc documentation, including the specification, papers about a the OMDoc format and tools.	<code>dtd</code> , <code>rnc</code>	The OMDoc document type definition and the RELAXNG schemata for OMDoc	<code>examples</code>	Various example documents in OMDoc format.	<code>projects</code>	various contributed developments for OMDoc. Documentation is usually in their <code>doc</code> sub-directory
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kohlhase-omdoc_TAB_045	233	—	(no LaTeX source; 1090 × 270 px region)	—															
kohlhase-omdoc_TAB_046	245	—	(no LaTeX source; 1089 × 134 px region)	—	24426 Applications and Projects 26.2 QMath: A Human-Oriented Language and Batch Formatter for OMDoc														
kohlhase-omdoc_TAB_047	248	—	(no LaTeX source; 1087 × 133 px region)	—	24426 Applications and Projects 26.3 Sentido Integrated Environment247 26.3 Sentido: An Integrated Environment for OMDoc														
kohlhase-omdoc_TAB_048	253	—	(no LaTeX source; 1090 × 216 px region)	—	25226 Applications and Projects 26.4 MBase, an Open Mathematical Knowledge Base <table><tr><td>Project Home</td><td>http://www.mathweb.org/mbase</td></tr><tr><td>Authors</td><td>Andreas Franke¹, Michael Kohlhase²</td></tr></table>	Project Home	http://www.mathweb.org/mbase	Authors	Andreas Franke ¹ , Michael Kohlhase ²										
Project Home	http://www.mathweb.org/mbase																		
Authors	Andreas Franke ¹ , Michael Kohlhase ²																		
kohlhase-omdoc_TAB_049	255	—	(no LaTeX source; 1090 × 136 px region)	—	25426 Applications and Projects 26.5 A Search Engine for Mathematical Formulae														
kohlhase-omdoc_TAB_050	259	—	(no LaTeX source; 1088 × 133 px region)	—	25826 Applications and Projects 26.6 Semantic Interrelation and Change Management														
kohlhase-omdoc_TAB_051	263	—	(no LaTeX source; 1088 × 178 px region)	—	26226 Applications and Projects 26.7 MathDox: Mathematical Documents on the Web Project Home: http://www.mathweb.org/mathdox/														
kohlhase-omdoc_TAB_052	267	—	(no LaTeX source; 1088 × 256 px region)	—	26626 Applications and Projects 26.8 OMDoc in ACTIVEMATH <table><tr><td>Project Home</td><td>http://www.activemath.org/</td></tr><tr><td>Authors</td><td>The ACTIVEMATH group: Erica Melis, Giorgi Goguadse, Alberto Gonzales-Palomo, Adrian</td></tr></table>	Project Home	http://www.activemath.org/	Authors	The ACTIVEMATH group: Erica Melis, Giorgi Goguadse, Alberto Gonzales-Palomo, Adrian										
Project Home	http://www.activemath.org/																		
Authors	The ACTIVEMATH group: Erica Melis, Giorgi Goguadse, Alberto Gonzales-Palomo, Adrian																		
kohlhase-omdoc_TAB_053	273	—	(no LaTeX source; 1088 × 135 px region)	—	27226 Applications and Projects 26.9 Authoring Tools for ACTIVEMATH														
kohlhase-omdoc_TAB_054	275	—	(no LaTeX source; 1088 × 133 px region)	—	27426 Applications and Projects 26.10 SWiM – An OMDoc-Based Semantic Wiki														

Conf. MathPix confidence: ■ ≥0.9■ 0.5–0.9■ <0.5— no value

Residual render vs scan (inkdrill): ● C component● W weak● S stable● N noise● K clean● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																												
kohlhase-omdoc_TAB_055	279	—	(no LaTeX source; 1088 × 301 px region)	—	27826 Applications and Projects 26.11 Induction Challenge OMDoc Manager (ICOM) <table><tr><td>Project Home</td><td>http://www.cs.nott.ac.uk/~lad/research/challenges/challenge_manager.html</td></tr><tr><td>Authors</td><td>Thomas D. Attfield, Monica C. Duarte, Lin Li, Ho-Ying Mak, Adam M. Neal, Lewis M. Toft, Zixuan Wang, Louise A. Dennis</td></tr></table>	Project Home	http://www.cs.nott.ac.uk/~lad/research/challenges/challenge_manager.html	Authors	Thomas D. Attfield, Monica C. Duarte, Lin Li, Ho-Ying Mak, Adam M. Neal, Lewis M. Toft, Zixuan Wang, Louise A. Dennis																																								
Project Home	http://www.cs.nott.ac.uk/~lad/research/challenges/challenge_manager.html																																																
Authors	Thomas D. Attfield, Monica C. Duarte, Lin Li, Ho-Ying Mak, Adam M. Neal, Lewis M. Toft, Zixuan Wang, Louise A. Dennis																																																
kohlhase-omdoc_TAB_056	282	—	(no LaTeX source; 1088 × 260 px region)	—	26.12 MAYA281 26.12 Maya: Maintaining Structured Developments <table><tr><td>Project Home</td><td>www.dfki.de/~inka/maya.html</td></tr><tr><td>Authors</td><td>Serge Autexier¹, Dieter Hutter¹, Till Mossakowski², Axel Schairer¹</td></tr></table>	Project Home	www.dfki.de/~inka/maya.html	Authors	Serge Autexier ¹ , Dieter Hutter ¹ , Till Mossakowski ² , Axel Schairer ¹																																								
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Authors	Serge Autexier ¹ , Dieter Hutter ¹ , Till Mossakowski ² , Axel Schairer ¹																																																
kohlhase-omdoc_TAB_057	287	—	(no LaTeX source; 1090 × 134 px region)	—	28626 Applications and Projects 26.13 Hets: The Heterogeneous Tool Set																																												
kohlhase-omdoc_TAB_058	291	—	(no LaTeX source; 1088 × 175 px region)	—	29026 Applications and Projects 26.14 CPOINT: An OMDoc Editor in MS PowerPoint <table><tr><td>Project Home</td><td>http://www.cs.cmu.edu/~ccans</td></tr></table>	Project Home	http://www.cs.cmu.edu/~ccans																																										
Project Home	http://www.cs.cmu.edu/~ccans																																																
kohlhase-omdoc_TAB_059	295	—	(no LaTeX source; 1090 × 136 px region)	—	29426 Applications and Projects 26.15 sTeX: A L^AT_EX-Based Workflow for OMDoc																																												
kohlhase-omdoc_TAB_060	299	—	(no LaTeX source; 1088 × 175 px region)	—	29826 Applications and Projects 26.16 An Emacs Mode for Editing OMDoc Documents <table><tr><td>Project Home</td><td>http://www.cs.cmu.edu/~ccans</td></tr></table>	Project Home	http://www.cs.cmu.edu/~ccans																																										
Project Home	http://www.cs.cmu.edu/~ccans																																																
kohlhase-omdoc_TAB_061	302	—	(no LaTeX source; 1090 × 175 px region)	—	26.17 Converting Mathematica Notebooks to OMDoc301 26.17 Converting Mathematica Notebooks to OMDoc <table><tr><td>Project Home</td><td>http://www.cs.cmu.edu/~ccans</td></tr></table>	Project Home	http://www.cs.cmu.edu/~ccans																																										
Project Home	http://www.cs.cmu.edu/~ccans																																																
kohlhase-omdoc_TAB_062	305	—	(no LaTeX source; 1090 × 133 px region)	—	26.18 Standardizing Context in System Interoperability																																												
kohlhase-omdoc_TAB_063	306	—	(no LaTeX source; 1099 × 421 px region)	—	26.18 Standardizing Context in System Interoperability305 <table><tr><td>PVS</td><td>ΩMEGA</td><td>PVS</td><td>ΩMEGA</td></tr><tr><td>set</td><td></td><td>subset?</td><td>subset</td></tr><tr><td>member</td><td>in</td><td></td><td>subset2</td></tr><tr><td>empty?</td><td>empty</td><td>strict_subset?</td><td>proper-subset</td></tr><tr><td>emptyset</td><td>emptyset</td><td></td><td>superset</td></tr><tr><td>nonempty?</td><td>not-empty</td><td>union</td><td>union</td></tr><tr><td>full?</td><td></td><td></td><td>union2</td></tr><tr><td>fullset</td><td></td><td></td><td>union-over-collection</td></tr><tr><td>singleton?</td><td>singleton</td><td>intersection</td><td>intersection</td></tr><tr><td>singleton</td><td></td><td></td><td>intersection-over-coll.</td></tr><tr><td>complement</td><td>set-complement</td><td>disjoint?</td><td>misses</td></tr></table>	PVS	ΩMEGA	PVS	ΩMEGA	set		subset?	subset	member	in		subset2	empty?	empty	strict_subset?	proper-subset	emptyset	emptyset		superset	nonempty?	not-empty	union	union	full?			union2	fullset			union-over-collection	singleton?	singleton	intersection	intersection	singleton			intersection-over-coll.	complement	set-complement	disjoint?	misses
PVS	ΩMEGA	PVS	ΩMEGA																																														
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member	in		subset2																																														
empty?	empty	strict_subset?	proper-subset																																														
emptyset	emptyset		superset																																														
nonempty?	not-empty	union	union																																														
full?			union2																																														
fullset			union-over-collection																																														
singleton?	singleton	intersection	intersection																																														
singleton			intersection-over-coll.																																														
complement	set-complement	disjoint?	misses																																														
kohlhase-omdoc_TAB_064	310	—	(no LaTeX source; 1088 × 258 px region)	—	26.19 Integrating Proof Assistants as Plugins in a Scientific Editor <table><tr><td>Project Home</td><td>http://www.ags.uni-sb.de/~omega/projects/verimathdoc</td></tr><tr><td>Authors</td><td>Serge Autexier, Christoph Benzmüller, Armin Fiedler, and Henri Lesourd</td></tr></table>	Project Home	http://www.ags.uni-sb.de/~omega/projects/verimathdoc	Authors	Serge Autexier, Christoph Benzmüller, Armin Fiedler, and Henri Lesourd																																								
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Conf. MathPix confidence:

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≥0.9

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0.5–0.9

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<0.5

— no value

Residual render vs scan (inkdrill):

●

C component

●

W weak

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S stable

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N noise

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K clean

●

not measured

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image				
kohlhase-omdoc_TAB_065	314	—	(no LaTeX source; 1088 × 173 px region)	—	26.20 VeriFun 313 26.20 OMDoc as a Data Format for veriFun				
kohlhase-omdoc_TAB_066	320	—	(no LaTeX source; 1152 × 479 px region)	—	<table><tr><td>assertion</td><td>aug</td><td>the <code>assertion</code> element now has an optional <code>for</code> attribute. Furthermore, an optional attribute <code>generated-via</code> has been added to allow generation via a theory morphism. Finally, two new attributes <code>status</code> and <code>just-by</code> have been added to mark up the deductive status of the assertion.</td><td>142</td></tr></table>	assertion	aug	the <code>assertion</code> element now has an optional <code>for</code> attribute. Furthermore, an optional attribute <code>generated-via</code> has been added to allow generation via a theory morphism. Finally, two new attributes <code>status</code> and <code>just-by</code> have been added to mark up the deductive status of the assertion.	142
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Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																								
kohlhase-omdoc_TAB_067	321	—	(no LaTeX source; 1150 × 1786 px region)	—	A.1 Changes from 1.1 to 1.2321 <table><tr><td>assumption</td><td>cha</td><td>This element can now have an attribute inductive for inductive assumptions. The natural language description in the optional CMP element is no longer allowed, use a phrase element in a CMP that is a sibling to the FMP instead.</td><td>145</td></tr><tr><td>adt</td><td>aug</td><td>the adt loses the CMP and commonname children, use the Dublin Core metadata elements dc:description and dc:subject instead. The type attribute is now on the sortdef element. Furthermore, an optional attribute generated-via has been added to allow generation via a theory morphism. Finally, an attribute parameters has been added to allow for parametric ADTs.</td><td>156</td></tr><tr><td>answer</td><td>cha</td><td>the answer element does not allow symbol children any more, if these are needed, the exercise should have its own theory.</td><td>210</td></tr><tr><td>attribute</td><td>aug</td><td>the attribute element now has a optional ns attribute for the namespace URI of the generated attribute node and an attribute select for an XPATH expression that specifies the value of the generated attribute.</td><td>191</td></tr><tr><td>axiom</td><td>aug</td><td>the axiom element now has an optional for attribute which can point to a list of symbols. Furthermore, an optional attribute generated-via has been added to allow generation via a theory morphism and an attribute type is now also allowed.</td><td>138</td></tr><tr><td>axiom-inclusion</td><td>lib</td><td>the axiom-inclusion element can now contain multiple path-just children to record multiple justifications. Furthermore, it can now have theory, generated-from, and generated-via attributes. New optional attributes conservativity and conservativity-just for stating and justifying conservativity.</td><td>180</td></tr><tr><td>catalogue</td><td>dep</td><td>the catalogue mechanism has been eliminated.</td><td></td></tr><tr><td>choice</td><td>cha</td><td>the choice element does not allow symbol children any more, if these are needed, the exercise should have its own theory</td><td>210</td></tr><tr><td>code</td><td>cha</td><td>Attributes classid and codebase are deprecated. The attributes pto and pto-version have moved to the data element. The attribute type has been removed and optional attributes theory, generated-from, and generated-via have been added.</td><td>202</td></tr><tr><td>commonname</td><td>dep</td><td>This element is deprecated in favor of a metadata/dc:subject element.</td><td></td></tr></table>	assumption	cha	This element can now have an attribute inductive for inductive assumptions. The natural language description in the optional CMP element is no longer allowed, use a phrase element in a CMP that is a sibling to the FMP instead.	145	adt	aug	the adt loses the CMP and commonname children, use the Dublin Core metadata elements dc:description and dc:subject instead. The type attribute is now on the sortdef element. Furthermore, an optional attribute generated-via has been added to allow generation via a theory morphism. Finally, an attribute parameters has been added to allow for parametric ADTs.	156	answer	cha	the answer element does not allow symbol children any more, if these are needed, the exercise should have its own theory.	210	attribute	aug	the attribute element now has a optional ns attribute for the namespace URI of the generated attribute node and an attribute select for an XPATH expression that specifies the value of the generated attribute.	191	axiom	aug	the axiom element now has an optional for attribute which can point to a list of symbols. Furthermore, an optional attribute generated-via has been added to allow generation via a theory morphism and an attribute type is now also allowed.	138	axiom-inclusion	lib	the axiom-inclusion element can now contain multiple path-just children to record multiple justifications. Furthermore, it can now have theory , generated-from , and generated-via attributes. New optional attributes conservativity and conservativity-just for stating and justifying conservativity.	180	catalogue	dep	the catalogue mechanism has been eliminated.		choice	cha	the choice element does not allow symbol children any more, if these are needed, the exercise should have its own theory	210	code	cha	Attributes classid and codebase are deprecated. The attributes pto and pto-version have moved to the data element. The attribute type has been removed and optional attributes theory , generated-from , and generated-via have been added.	202	commonname	dep	This element is deprecated in favor of a metadata/dc:subject element.	
assumption	cha	This element can now have an attribute inductive for inductive assumptions. The natural language description in the optional CMP element is no longer allowed, use a phrase element in a CMP that is a sibling to the FMP instead.	145																																										
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Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																												
kohlhase-omdoc_TAB_068	322	—	(no LaTeX source; 1150 × 1822 px region)	—	322 A Changes to the Specification <table><tr><td>conclusion</td><td>cha</td><td>The natural language description in the optional <code>CMP</code> element is no longer allowed, use a <code>phrase</code> element in a <code>CMP</code> that is a sibling to the <code>FMP</code> instead.</td><td>124</td></tr><tr><td>constructor</td><td>cha</td><td>The <code>role</code> attribute is now fixed to <code>object</code>. The <code>commonname</code> child has been replaced by an initial <code>metadata</code> element.</td><td>156</td></tr><tr><td>data</td><td>aug</td><td>new optional attributes <code>original</code> to specify whether the external resource referenced by the <code>href</code> attribute (value <code>external</code>) or the <code>data</code> content is the original (value <code>local</code>). The <code>data</code> element has acquired attributes <code>pto</code> and <code>pto-version</code> from the <code>code</code> and <code>private</code> elements.</td><td>203</td></tr><tr><td>dc:*</td><td>aug</td><td>All Dublin Core tags have been lowercased to synchronize with the tag syntax recommended by the Dublin Core Initiative. The tags were capitalized in OMDoc1.1. Furthermore, <code>dc:contributor</code>, <code>dc:creator</code>, <code>dc:publisher</code> have received an optional <code>xml:id</code> attribute, so that they can be cross-referenced by the new <code>who</code> of the <code>dc:date</code> element.</td><td>98</td></tr><tr><td>decomposition</td><td>aug</td><td>The <code>for</code> attribute is now optional, it need not be given, if the element is a child of a <code>theory-inclusion</code> element. Furthermore, it can now have a <code>theory</code>, <code>generated-from</code>, and <code>generated-via</code> attributes.</td><td>184</td></tr><tr><td>dc:description</td><td>aug</td><td>The <code>dc:description</code> can now have the optional <code>xml:id</code>, and CSS attributes</td><td>99</td></tr><tr><td>definition</td><td>aug</td><td>The <code>definition</code> element can now have the type <code>pattern</code> for pattern-defined functions. This is a degenerate case of the type <code>inductive</code>. Furthermore, an optional attribute <code>generated-via</code> has been added to allow generation via a theory morphism.</td><td>139</td></tr><tr><td>effect</td><td>aug</td><td>allows an optional <code>xml:id</code> attribute</td><td>204</td></tr><tr><td>example</td><td>aug</td><td>The <code>example</code> element now has the optional <code>theory</code> attribute that specifies the home theory. Furthermore, it can now have attributes <code>theory</code>, <code>generated-from</code>, and <code>generated-via</code>.</td><td>146</td></tr><tr><td>exercise</td><td>cha</td><td>the <code>exercise</code> element does not allow <code>symbol</code> children any more, if these are needed, the exercise should have its own theory. Furthermore, it can now have a <code>theory</code>, <code>generated-from</code>, and <code>generated-via</code> attributes.</td><td>209</td></tr><tr><td>extradata</td><td>cha</td><td>The content of the old <code>extradata</code> element can now be directly in the <code>metadata/dc:subject</code> element.</td><td></td></tr></table>	conclusion	cha	The natural language description in the optional <code>CMP</code> element is no longer allowed, use a <code>phrase</code> element in a <code>CMP</code> that is a sibling to the <code>FMP</code> instead.	124	constructor	cha	The <code>role</code> attribute is now fixed to <code>object</code> . The <code>commonname</code> child has been replaced by an initial <code>metadata</code> element.	156	data	aug	new optional attributes <code>original</code> to specify whether the external resource referenced by the <code>href</code> attribute (value <code>external</code>) or the <code>data</code> content is the original (value <code>local</code>). The <code>data</code> element has acquired attributes <code>pto</code> and <code>pto-version</code> from the <code>code</code> and <code>private</code> elements.	203	dc:*	aug	All Dublin Core tags have been lowercased to synchronize with the tag syntax recommended by the Dublin Core Initiative. The tags were capitalized in OMDoc1.1. Furthermore, <code>dc:contributor</code> , <code>dc:creator</code> , <code>dc:publisher</code> have received an optional <code>xml:id</code> attribute, so that they can be cross-referenced by the new <code>who</code> of the <code>dc:date</code> element.	98	decomposition	aug	The <code>for</code> attribute is now optional, it need not be given, if the element is a child of a <code>theory-inclusion</code> element. Furthermore, it can now have a <code>theory</code> , <code>generated-from</code> , and <code>generated-via</code> attributes.	184	dc:description	aug	The <code>dc:description</code> can now have the optional <code>xml:id</code> , and CSS attributes	99	definition	aug	The <code>definition</code> element can now have the type <code>pattern</code> for pattern-defined functions. This is a degenerate case of the type <code>inductive</code> . Furthermore, an optional attribute <code>generated-via</code> has been added to allow generation via a theory morphism.	139	effect	aug	allows an optional <code>xml:id</code> attribute	204	example	aug	The <code>example</code> element now has the optional <code>theory</code> attribute that specifies the home theory. Furthermore, it can now have attributes <code>theory</code> , <code>generated-from</code> , and <code>generated-via</code> .	146	exercise	cha	the <code>exercise</code> element does not allow <code>symbol</code> children any more, if these are needed, the exercise should have its own theory. Furthermore, it can now have a <code>theory</code> , <code>generated-from</code> , and <code>generated-via</code> attributes.	209	extradata	cha	The content of the old <code>extradata</code> element can now be directly in the <code>metadata/dc:subject</code> element.	
conclusion	cha	The natural language description in the optional <code>CMP</code> element is no longer allowed, use a <code>phrase</code> element in a <code>CMP</code> that is a sibling to the <code>FMP</code> instead.	124																																														
constructor	cha	The <code>role</code> attribute is now fixed to <code>object</code> . The <code>commonname</code> child has been replaced by an initial <code>metadata</code> element.	156																																														
data	aug	new optional attributes <code>original</code> to specify whether the external resource referenced by the <code>href</code> attribute (value <code>external</code>) or the <code>data</code> content is the original (value <code>local</code>). The <code>data</code> element has acquired attributes <code>pto</code> and <code>pto-version</code> from the <code>code</code> and <code>private</code> elements.	203																																														
dc:*	aug	All Dublin Core tags have been lowercased to synchronize with the tag syntax recommended by the Dublin Core Initiative. The tags were capitalized in OMDoc1.1. Furthermore, <code>dc:contributor</code> , <code>dc:creator</code> , <code>dc:publisher</code> have received an optional <code>xml:id</code> attribute, so that they can be cross-referenced by the new <code>who</code> of the <code>dc:date</code> element.	98																																														
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dc:description	aug	The <code>dc:description</code> can now have the optional <code>xml:id</code> , and CSS attributes	99																																														
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exercise	cha	the <code>exercise</code> element does not allow <code>symbol</code> children any more, if these are needed, the exercise should have its own theory. Furthermore, it can now have a <code>theory</code> , <code>generated-from</code> , and <code>generated-via</code> attributes.	209																																														
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured																																																	

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																				
					A.1 Changes from 1.1 to 1.2323 <table><tr><td>element</td><td>aug</td><td>The <code>element</code> element now allows the <code>map</code> and <code>separator</code> elements in the body. Furthermore, it carries the optional attributes <code>cr</code>id for parallel markup, <code>cr</code> for cross-references, and <code>ns</code> for specifying the namespace.</td><td>190</td></tr><tr><td>hint</td><td>aug</td><td>the <code>hint</code> element can now appear on top-level and has a <code>for</code> attribute. It does not allow <code>symbol</code> children any more, if these are needed, the exercise should have its own theory. Furthermore, the <code>exercise</code> can now have a <code>theory</code>, <code>generated-from</code>, and <code>generated-via</code> attributes.</td><td>209</td></tr><tr><td>hypothesis</td><td>cha</td><td>the <code>discharged-in</code> attribute has been eliminated. Scoping is now specified in terms of the enclosing <code>proof</code> element. Furthermore, the <code>symbol</code> child is no longer allowed inside the element. A sibling <code>symbol</code> should be used.</td><td>162</td></tr><tr><td>inclusion</td><td>aug</td><td>allows optional attributes <code>xml:id</code>, <code>conservativity</code>, and <code>conservativity-just</code> for stating and justifying conservativity.</td><td>179</td></tr><tr><td>imports</td><td>lib</td><td>the <code>xml:id</code> is now optional. New optional attributes <code>conservativity</code> and <code>conservativity-just</code> for stating and justifying conservativity.</td><td>150</td></tr><tr><td>input</td><td>aug</td><td>allows an optional <code>xml:id</code> attribute</td><td>204</td></tr><tr><td>legacy</td><td>new</td><td>An element for encapsulating legacy mathematics, can be used wherever <code>m:math</code> and <code>om:UMOBJ</code> are allowed.</td><td>120</td></tr><tr><td>loc</td><td>dep</td><td>The catalogue mechanism has been eliminated.</td><td></td></tr><tr><td>m:math</td><td>new</td><td>Content-MATHML is now allowed wherever OPENMATH objects were allowed before.</td><td>114</td></tr><tr><td>map</td><td>new</td><td>this element allows to map its style directives over a list of e.g. arguments</td><td>191</td></tr><tr><td>mc</td><td>aug</td><td>the <code>mc</code> element can now have a <code>for</code> attribute. It does not allow <code>symbol</code> children any more, if these are needed, the dominating <code>exercise</code> element should have its own theory. Furthermore, the <code>mc</code> element can now have a <code>theory</code>, <code>generated-from</code>, and <code>generated-via</code> attributes.</td><td>210</td></tr><tr><td>measure</td><td>aug</td><td>allows an optional <code>xml:id</code> attribute</td><td>141</td></tr><tr><td>metacomment</td><td>dep</td><td>This element is superseded by the <code>omtext</code> element.</td><td>124</td></tr></table>	element	aug	The <code>element</code> element now allows the <code>map</code> and <code>separator</code> elements in the body. Furthermore, it carries the optional attributes <code>cr</code> id for parallel markup, <code>cr</code> for cross-references, and <code>ns</code> for specifying the namespace.	190	hint	aug	the <code>hint</code> element can now appear on top-level and has a <code>for</code> attribute. It does not allow <code>symbol</code> children any more, if these are needed, the exercise should have its own theory. Furthermore, the <code>exercise</code> can now have a <code>theory</code> , <code>generated-from</code> , and <code>generated-via</code> attributes.	209	hypothesis	cha	the <code>discharged-in</code> attribute has been eliminated. Scoping is now specified in terms of the enclosing <code>proof</code> element. Furthermore, the <code>symbol</code> child is no longer allowed inside the element. A sibling <code>symbol</code> should be used.	162	inclusion	aug	allows optional attributes <code>xml:id</code> , <code>conservativity</code> , and <code>conservativity-just</code> for stating and justifying conservativity.	179	imports	lib	the <code>xml:id</code> is now optional. New optional attributes <code>conservativity</code> and <code>conservativity-just</code> for stating and justifying conservativity.	150	input	aug	allows an optional <code>xml:id</code> attribute	204	legacy	new	An element for encapsulating legacy mathematics, can be used wherever <code>m:math</code> and <code>om:UMOBJ</code> are allowed.	120	loc	dep	The catalogue mechanism has been eliminated.		m:math	new	Content-MATHML is now allowed wherever OPENMATH objects were allowed before.	114	map	new	this element allows to map its style directives over a list of e.g. arguments	191	mc	aug	the <code>mc</code> element can now have a <code>for</code> attribute. It does not allow <code>symbol</code> children any more, if these are needed, the dominating <code>exercise</code> element should have its own theory. Furthermore, the <code>mc</code> element can now have a <code>theory</code> , <code>generated-from</code> , and <code>generated-via</code> attributes.	210	measure	aug	allows an optional <code>xml:id</code> attribute	141	metacomment	dep	This element is superseded by the <code>omtext</code> element.	124
element	aug	The <code>element</code> element now allows the <code>map</code> and <code>separator</code> elements in the body. Furthermore, it carries the optional attributes <code>cr</code> id for parallel markup, <code>cr</code> for cross-references, and <code>ns</code> for specifying the namespace.	190																																																						
hint	aug	the <code>hint</code> element can now appear on top-level and has a <code>for</code> attribute. It does not allow <code>symbol</code> children any more, if these are needed, the exercise should have its own theory. Furthermore, the <code>exercise</code> can now have a <code>theory</code> , <code>generated-from</code> , and <code>generated-via</code> attributes.	209																																																						
hypothesis	cha	the <code>discharged-in</code> attribute has been eliminated. Scoping is now specified in terms of the enclosing <code>proof</code> element. Furthermore, the <code>symbol</code> child is no longer allowed inside the element. A sibling <code>symbol</code> should be used.	162																																																						
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kohlhase-omdoc_TAB_069	323	—	(no LaTeX source; 1148 × 1680 px region)	—																																																					

Conf. MathPix confidence:

■ ≥0.9

■ 0.5-0.9

■ <0.5

— no value

Residual render vs scan (inkdrill):

● C component

● W weak

● S stable

● N noise

● K clean

● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																												
kohlhase-omdoc_TAB_070	324	—	(no LaTeX source; 1150 × 1822 px region)	—	324 A Changes to the Specification <table><tr><td>morphism</td><td>aug</td><td>The <code>morphism</code> element now carries the optional attributes <code>consistency</code>, <code>exhaustivity</code>, <code>hiding</code>, and <code>type</code>. Furthermore the content model allows optional elements <code>measure</code> and <code>ordering</code> after the <code>reuation</code> children to specify termination information like in <code>definition</code>.</td><td>92</td></tr><tr><td>obligation</td><td>aug</td><td>allows an optional <code>xml:id</code> attribute</td><td>178</td></tr><tr><td>omdoc</td><td>aug</td><td>This element can now have a <code>theory</code>, <code>generated-from</code>, and <code>generated-via</code> attributes.</td><td>90</td></tr><tr><td>omgroup</td><td>cha</td><td>The values <code>dataset</code> and <code>labeled-dataset</code> are deprecated in Version 1.2 of OM-DOC, since we provide tables in module RT; see Section 14.6 for details. Furthermore, the element can now have the attributes, <code>modules</code>, <code>theory</code>, <code>generated-from</code>, and <code>generated-via</code>.</td><td>93</td></tr><tr><td>omlet</td><td>cha</td><td><code>omlet</code> can no longer occur at top-level (it just does not make sense). The data model for this element has been totally reworked, inspired by the <code>xhtml:object</code> element.</td><td>205</td></tr><tr><td>omstyle</td><td>aug</td><td>This element can now have <code>generated-from</code>, and <code>generated-via</code> attributes. New attribute <code>xref</code> that allows to inherit the information from another <code>omstyle</code> element.</td><td>188</td></tr><tr><td>om:*</td><td>aug</td><td>with OPENMATH2, the OPENMATH elements carry an optional <code>id</code> attribute for structure sharing via the <code>om:OMR</code> element. Furthermore, in OMDOC, they carry <code>cref</code> attributes for parallel markup with cross-references.</td><td>108</td></tr><tr><td>om:OMFOREIGN</td><td>new</td><td>The <code>om:OMFOREIGN</code> element can be used to encapsulate arbitrary XML data in OPENMATH attributions.</td><td>110</td></tr><tr><td>om:OMR</td><td>new</td><td>In the OPENMATH2 standard, this element is the main vehicle of the structure sharing representation.</td><td>112</td></tr><tr><td>omtext</td><td>aug</td><td>the <code>type</code> attribute can now also have the values <code>axiom</code>, <code>definition</code>, <code>theorem</code>, <code>proposition</code>, <code>lemma</code>, <code>corollary</code>, <code>postulate</code>, <code>conjecture</code>, <code>false-conjecture</code>, <code>obligation</code>, <code>assumption</code>, and <code>formula</code>. Furthermore, <code>omtext</code> can now have <code>theory</code>, <code>generated-from</code>, and <code>generated-via</code> and <code>verbalizes</code> attributes.</td><td>124</td></tr><tr><td>ordering</td><td>aug</td><td>Now allows the optional <code>xml:id</code> and <code>terminating</code> attributes. The latter points to a termination assertion.</td><td>141</td></tr></table>	morphism	aug	The <code>morphism</code> element now carries the optional attributes <code>consistency</code> , <code>exhaustivity</code> , <code>hiding</code> , and <code>type</code> . Furthermore the content model allows optional elements <code>measure</code> and <code>ordering</code> after the <code>reuation</code> children to specify termination information like in <code>definition</code> .	92	obligation	aug	allows an optional <code>xml:id</code> attribute	178	omdoc	aug	This element can now have a <code>theory</code> , <code>generated-from</code> , and <code>generated-via</code> attributes.	90	omgroup	cha	The values <code>dataset</code> and <code>labeled-dataset</code> are deprecated in Version 1.2 of OM-DOC, since we provide tables in module RT; see Section 14.6 for details. Furthermore, the element can now have the attributes, <code>modules</code> , <code>theory</code> , <code>generated-from</code> , and <code>generated-via</code> .	93	omlet	cha	<code>omlet</code> can no longer occur at top-level (it just does not make sense). The data model for this element has been totally reworked, inspired by the <code>xhtml:object</code> element.	205	omstyle	aug	This element can now have <code>generated-from</code> , and <code>generated-via</code> attributes. New attribute <code>xref</code> that allows to inherit the information from another <code>omstyle</code> element.	188	om:*	aug	with OPENMATH2, the OPENMATH elements carry an optional <code>id</code> attribute for structure sharing via the <code>om:OMR</code> element. Furthermore, in OMDOC, they carry <code>cref</code> attributes for parallel markup with cross-references.	108	om:OMFOREIGN	new	The <code>om:OMFOREIGN</code> element can be used to encapsulate arbitrary XML data in OPENMATH attributions.	110	om:OMR	new	In the OPENMATH2 standard, this element is the main vehicle of the structure sharing representation.	112	omtext	aug	the <code>type</code> attribute can now also have the values <code>axiom</code> , <code>definition</code> , <code>theorem</code> , <code>proposition</code> , <code>lemma</code> , <code>corollary</code> , <code>postulate</code> , <code>conjecture</code> , <code>false-conjecture</code> , <code>obligation</code> , <code>assumption</code> , and <code>formula</code> . Furthermore, <code>omtext</code> can now have <code>theory</code> , <code>generated-from</code> , and <code>generated-via</code> and <code>verbalizes</code> attributes.	124	ordering	aug	Now allows the optional <code>xml:id</code> and <code>terminating</code> attributes. The latter points to a termination assertion.	141
					morphism	aug	The <code>morphism</code> element now carries the optional attributes <code>consistency</code> , <code>exhaustivity</code> , <code>hiding</code> , and <code>type</code> . Furthermore the content model allows optional elements <code>measure</code> and <code>ordering</code> after the <code>reuation</code> children to specify termination information like in <code>definition</code> .	92																																									
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Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																
kohlhase-omdoc_TAB_071	325	—	(no LaTeX source; 1150 × 1750 px region)	—	A.1 Changes from 1.1 to 1.2325 <table><tr><td>output</td><td>aug</td><td>allows an optional <code>xml:id</code> attribute</td><td>204</td></tr><tr><td>pattern</td><td>aug</td><td>this element is no longer used, the pattern of a recursive equation is determined by the position as the first child.</td><td></td></tr><tr><td>path-just</td><td>aug</td><td>The element can now appear as a top-level element, if it does, the attribute <code>for</code> must point to the <code>axiom-inclusion</code> element it justifies. It also now allows an optional <code>xml:id</code> attribute</td><td>184</td></tr><tr><td>phrase</td><td>new</td><td>used to mark up phrases in <i>CMPs</i> and supply them with identifiers and links to context that can be used for presentation and referencing.</td><td>126</td></tr><tr><td>presentation</td><td>cha</td><td>The <code>theory</code> is not allowed any more, to refer to a symbol outside its theory use its <code>xml:id</code> attribute. The element now also allows a multilingual <i>CMP</i> group, so that it can be used as a notation definition element in mathematical vernacular.</td><td>192</td></tr><tr><td>private</td><td>cha</td><td>The <code>replaces</code> attribute is now called <code>reformulates</code>. The attributes <code>pto</code> and <code>pto-version</code> have moved to the <code>data</code> element. The attribute <code>type</code> has been removed and optional attributes <code>theory</code>, <code>generated-from</code>, and <code>generated-via</code> have been added.</td><td>202</td></tr><tr><td>proof</td><td>lib</td><td>The <code>for</code> attribute is now optional to allow for proofs as objects of mathematical discourse. Furthermore, it can now have <code>generated-from</code> and <code>generated-via</code> attributes.</td><td>161</td></tr><tr><td>proofobject</td><td>lib</td><td>The <code>for</code> attribute is now optional to allow for proofs as objects of mathematical discourse. Furthermore, it can now have <code>generated-from</code> and <code>generated-via</code> attributes.</td><td>169</td></tr><tr><td>recognizer</td><td>cha</td><td>The <code>role</code> attribute was fixed to <code>object</code>. The <code>commonname</code> child has been replaced by an initial <code>metadata</code> element.</td><td>157</td></tr><tr><td>ref</td><td>aug</td><td><code>ref</code> now has an optional <code>xml:id</code> attribute that identifies it.</td><td>94</td></tr><tr><td>selector</td><td>cha</td><td>The <code>role</code> attribute was fixed to <code>object</code>. The <code>commonname</code> child has been replaced by an initial <code>metadata</code> element.</td><td>157</td></tr><tr><td>solution</td><td>cha</td><td>the <code>solution</code> element now allows arbitrary OMDOC top-level elements as children. Furthermore, it can now have a <code>theory</code>, <code>generated-from</code>, and <code>generated-via</code> attributes.</td><td>209</td></tr></table>	output	aug	allows an optional <code>xml:id</code> attribute	204	pattern	aug	this element is no longer used, the pattern of a recursive equation is determined by the position as the first child.		path-just	aug	The element can now appear as a top-level element, if it does, the attribute <code>for</code> must point to the <code>axiom-inclusion</code> element it justifies. It also now allows an optional <code>xml:id</code> attribute	184	phrase	new	used to mark up phrases in <i>CMPs</i> and supply them with identifiers and links to context that can be used for presentation and referencing.	126	presentation	cha	The <code>theory</code> is not allowed any more, to refer to a symbol outside its theory use its <code>xml:id</code> attribute. The element now also allows a multilingual <i>CMP</i> group, so that it can be used as a notation definition element in mathematical vernacular.	192	private	cha	The <code>replaces</code> attribute is now called <code>reformulates</code> . The attributes <code>pto</code> and <code>pto-version</code> have moved to the <code>data</code> element. The attribute <code>type</code> has been removed and optional attributes <code>theory</code> , <code>generated-from</code> , and <code>generated-via</code> have been added.	202	proof	lib	The <code>for</code> attribute is now optional to allow for proofs as objects of mathematical discourse. Furthermore, it can now have <code>generated-from</code> and <code>generated-via</code> attributes.	161	proofobject	lib	The <code>for</code> attribute is now optional to allow for proofs as objects of mathematical discourse. Furthermore, it can now have <code>generated-from</code> and <code>generated-via</code> attributes.	169	recognizer	cha	The <code>role</code> attribute was fixed to <code>object</code> . The <code>commonname</code> child has been replaced by an initial <code>metadata</code> element.	157	ref	aug	<code>ref</code> now has an optional <code>xml:id</code> attribute that identifies it.	94	selector	cha	The <code>role</code> attribute was fixed to <code>object</code> . The <code>commonname</code> child has been replaced by an initial <code>metadata</code> element.	157	solution	cha	the <code>solution</code> element now allows arbitrary OMDOC top-level elements as children. Furthermore, it can now have a <code>theory</code> , <code>generated-from</code> , and <code>generated-via</code> attributes.	209
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kohlhase-omdoc_TAB_072	326	—	(no LaTeX source; 1150 × 1501 px region)	—	326 A Changes to the Specification <table><tr><td>sortdef</td><td>cha</td><td>The <code>role</code> attribute was fixed to <code>sort</code>. The <code>type</code> from the <code>adt</code> element is now on the <code>sortdef</code> element. The <code>commonname</code> child has been replaced by an initial <code>metadata</code> element.</td><td>156</td></tr><tr><td>dc:subject</td><td>aug</td><td>The <code>dc:subject</code> can now have the optional <code>dc:id</code>, and CSS attributes</td><td>99</td></tr><tr><td>style</td><td>aug</td><td>The <code>style</code> element now allows a <code>map</code> element in the body</td><td>189</td></tr><tr><td>symbol</td><td>cha</td><td>may no longer contain <code>selector</code>, since it only makes sense for constructors in data types. The <code>kind</code> attribute has been renamed to <code>role</code> for compatibility with OPENMATH2 and can have the additional values <code>binder</code>, <code>attribution</code>, <code>semantic-attribution</code>, and <code>error</code> corresponding to the OPENMATH 2 roles. Furthermore, an optional attribute <code>generated-via</code> has been added to allow generation via a theory morphism.</td><td>136</td></tr><tr><td>term</td><td>new</td><td>the <code>term</code> element can appear in mathematical text and contain it. It is used to link technical terms to symbols defined in content dictionaries via its <code>cd</code> and <code>name</code> attributes.</td><td>128</td></tr><tr><td>theory</td><td>cha</td><td>the <code>theory</code> element loses the <code>CMP</code> and <code>commonname</code> children, use the Dublin Core metadata elements <code>dc:description</code> and <code>dc:subject</code> instead. The <code>theory</code> element also gains the optional <code>cdbase</code> attribute to specify the disambiguating string prescribed for content dictionaries by the OPENMATH2 standard. The <code>xml:id</code> is now optional, it only needs to be specified, if the theory has constitutive elements. Finally, the element has gained the optional attributes <code>cdurl</code>, <code>cdbase</code>, <code>cdreviewdate</code>, <code>cdversion</code>, <code>cdrevision</code>, and <code>cdstatus</code> attributes for encoding the management metadata of OPENMATH content dictionaries.</td><td>149</td></tr><tr><td>dc:title</td><td>aug</td><td>The <code>dc:title</code> can now have the optional <code>dc:id</code>, and CSS attributes.</td><td>98</td></tr><tr><td>tgroup</td><td>new</td><td>The <code>tgroup</code> can be used to structure theories like documents.</td><td>149</td></tr></table>	sortdef	cha	The <code>role</code> attribute was fixed to <code>sort</code> . The <code>type</code> from the <code>adt</code> element is now on the <code>sortdef</code> element. The <code>commonname</code> child has been replaced by an initial <code>metadata</code> element.	156	dc:subject	aug	The <code>dc:subject</code> can now have the optional <code>dc:id</code> , and CSS attributes	99	style	aug	The <code>style</code> element now allows a <code>map</code> element in the body	189	symbol	cha	may no longer contain <code>selector</code> , since it only makes sense for constructors in data types. The <code>kind</code> attribute has been renamed to <code>role</code> for compatibility with OPENMATH2 and can have the additional values <code>binder</code> , <code>attribution</code> , <code>semantic-attribution</code> , and <code>error</code> corresponding to the OPENMATH 2 roles. Furthermore, an optional attribute <code>generated-via</code> has been added to allow generation via a theory morphism.	136	term	new	the <code>term</code> element can appear in mathematical text and contain it. It is used to link technical terms to symbols defined in content dictionaries via its <code>cd</code> and <code>name</code> attributes.	128	theory	cha	the <code>theory</code> element loses the <code>CMP</code> and <code>commonname</code> children, use the Dublin Core metadata elements <code>dc:description</code> and <code>dc:subject</code> instead. The <code>theory</code> element also gains the optional <code>cdbase</code> attribute to specify the disambiguating string prescribed for content dictionaries by the OPENMATH2 standard. The <code>xml:id</code> is now optional, it only needs to be specified, if the theory has constitutive elements. Finally, the element has gained the optional attributes <code>cdurl</code> , <code>cdbase</code> , <code>cdreviewdate</code> , <code>cdversion</code> , <code>cdrevision</code> , and <code>cdstatus</code> attributes for encoding the management metadata of OPENMATH content dictionaries.	149	dc:title	aug	The <code>dc:title</code> can now have the optional <code>dc:id</code> , and CSS attributes.	98	tgroup	new	The <code>tgroup</code> can be used to structure theories like documents.	149
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kohlhase-omdoc_TAB_073	327	—	(no LaTeX source; 1150 × 1284 px region)	—	A.2 Changes from 1.0 to 1.1327 <table><tr><td>type</td><td>aug</td><td>the type element now has the optional just-by and theory attribute. The first one points to an assertion or axiom that justifies the type judgment, the second specifies the home theory. The system attribute is now optional. Furthermore, the type element can have two math objects as children. If it does, then it is a term declaration, i.e. the first element is interpreted as a mathematical object and the second one is interpreted as its type. Finally, it can now have generated-from and generated-via attributes.</td><td>139</td></tr><tr><td>theory-inclusion</td><td>aug</td><td>the theory-inclusion element can now have obligation and decomposition children that justify it. Furthermore, it can now have a theory, generated-from, and generated-via attributes. New optional attributes conservativity and conservativity-just for stating and justifying conservativity.</td><td>178</td></tr><tr><td>theory</td><td>aug</td><td>the theory element can now be nested.</td><td>149</td></tr><tr><td>use</td><td>cha</td><td>can now contain element, text, recurse, map, and value-of to specify XML content. We have deprecated the larg-group and rarg-group attributes, since they were never used.</td><td>194</td></tr><tr><td>value</td><td>aug</td><td>this element is no longer used, the value of a recursive equation is determined by the position as the second child.</td><td></td></tr><tr><td>with</td><td>ren</td><td>the role of this element is now taken by the phrase element.</td><td>126</td></tr><tr><td>xslt</td><td>cha</td><td>the content of this element need not be escaped any more, it is now a valid XSLT fragment.</td><td>189</td></tr></table> A.2 Changes from 1.0 to 1.1	type	aug	the type element now has the optional just-by and theory attribute. The first one points to an assertion or axiom that justifies the type judgment, the second specifies the home theory. The system attribute is now optional. Furthermore, the type element can have two math objects as children. If it does, then it is a term declaration, i.e. the first element is interpreted as a mathematical object and the second one is interpreted as its type. Finally, it can now have generated-from and generated-via attributes.	139	theory-inclusion	aug	the theory-inclusion element can now have obligation and decomposition children that justify it. Furthermore, it can now have a theory , generated-from , and generated-via attributes. New optional attributes conservativity and conservativity-just for stating and justifying conservativity.	178	theory	aug	the theory element can now be nested.	149	use	cha	can now contain element , text , recurse , map , and value-of to specify XML content. We have deprecated the larg-group and rarg-group attributes, since they were never used.	194	value	aug	this element is no longer used, the value of a recursive equation is determined by the position as the second child.		with	ren	the role of this element is now taken by the phrase element.	126	xslt	cha	the content of this element need not be escaped any more, it is now a valid XSLT fragment.	189
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kohlhase-omdoc_TAB_074	328	—	(no LaTeX source; 1150 × 1763 px region)	—	328A Changes to the Specification <table><tr><th>element</th><th>state</th><th>comments</th><th>cf.</th></tr><tr><td>attribute</td><td>new</td><td>presentation of attributes for XML elements</td><td>191</td></tr><tr><td>alternative</td><td>cha</td><td>new form of the alternative-def element, it can now also used as an alternative to axiom. Compared to alternative-def it has a new optional attribute generated-by to show that an assertion is generated by expanding a some other element like adt.</td><td>145</td></tr><tr><td>alternative-def</td><td>dep</td><td>new form is alternative, since there can be alternative axioms too.</td><td></td></tr><tr><td>argument</td><td>cha</td><td>attribute sort is now of type IDREF, since it must be local in the definition.</td><td>156</td></tr><tr><td>assertion</td><td>aug</td><td>more values for the type attribute, new optional attribute generated-by to show that an assertion is generated by expanding a definition or an adt. New optional attribute proofs.</td><td>142</td></tr><tr><td>assertion-just</td><td>dep</td><td>this is now obligation</td><td></td></tr><tr><td>axiom</td><td>aug</td><td>new optional attribute generated-by to show that an axiom is generated by expanding a definition.</td><td>138</td></tr><tr><td>axiom-inclusion</td><td>cha</td><td>now allows a CMP group for descriptive text, includes a set of obligation elements instead of an assertion-just. The timestamp attribute is deprecated, use dc:date with appropriate action instead</td><td>180</td></tr><tr><td>CMP</td><td>cha</td><td>the attribute format is now deprecated, it makes no sense, since we are more strict and consistent about CMP content. CMP now allows an optional id attribute.</td><td>122</td></tr><tr><td>code</td><td>cha</td><td>Attributes width and height now in omlet, got attributes classid and codebase from private. Attribute format moved to data children. The multilingual group of CMP elements for description is deprecated, use metadata/dc:description instead. Child element data may appear multiple times (with different values of the format).</td><td>202</td></tr><tr><td>constructor</td><td>aug</td><td>new optional child recognizer for a recognizer predicate</td><td>156</td></tr><tr><td>Coverage</td><td>dep</td><td>this Dublin Core element specifies the place or time which the publication's contents addresses. This does not seem appropriate for the mathematical content of OMDoc.</td><td></td></tr><tr><td>data</td><td>aug</td><td>new optional attributes size to specify the size of the data file that is referenced by the href attribute and format for the format the data is in.</td><td>203</td></tr></table>	element	state	comments	cf.	attribute	new	presentation of attributes for XML elements	191	alternative	cha	new form of the alternative-def element, it can now also used as an alternative to axiom . Compared to alternative-def it has a new optional attribute generated-by to show that an assertion is generated by expanding a some other element like adt .	145	alternative-def	dep	new form is alternative , since there can be alternative axioms too.		argument	cha	attribute sort is now of type IDREF, since it must be local in the definition.	156	assertion	aug	more values for the type attribute, new optional attribute generated-by to show that an assertion is generated by expanding a definition or an adt . New optional attribute proofs .	142	assertion-just	dep	this is now obligation		axiom	aug	new optional attribute generated-by to show that an axiom is generated by expanding a definition .	138	axiom-inclusion	cha	now allows a CMP group for descriptive text, includes a set of obligation elements instead of an assertion-just . The timestamp attribute is deprecated, use dc:date with appropriate action instead	180	CMP	cha	the attribute format is now deprecated, it makes no sense, since we are more strict and consistent about CMP content. CMP now allows an optional id attribute.	122	code	cha	Attributes width and height now in omlet , got attributes classid and codebase from private . Attribute format moved to data children. The multilingual group of CMP elements for description is deprecated, use metadata/dc:description instead. Child element data may appear multiple times (with different values of the format).	202	constructor	aug	new optional child recognizer for a recognizer predicate	156	Coverage	dep	this Dublin Core element specifies the place or time which the publication's contents addresses. This does not seem appropriate for the mathematical content of OMDoc.		data	aug	new optional attributes size to specify the size of the data file that is referenced by the href attribute and format for the format the data is in.	203
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kohlhase-omdoc_TAB_075	329	—	(no LaTeX source; 1150 × 1825 px region)	—	A.2 Changes from 1.0 to 1.1329 <table><tr><td>dc:date</td><td>aug</td><td>new optional who attribute that can be used to specify who did the action on this date.</td><td>99</td></tr><tr><td>Translator</td><td>dep</td><td>this element is not part of Dublin Core, it got into OMDoc by mistake, we use dc:contributor with role=trl for this.</td><td>98</td></tr><tr><td>decomposition</td><td>aug</td><td>has a new required id attribute. It is no longer a child of theory-inclusion, but specifies which theory-inclusion it justifies by the new required attribute for.</td><td>184</td></tr><tr><td>definition</td><td>aug</td><td>new optional children measure and ordering to specify termination of recursive definitions. New optional attribute generated-by to show that it is generated by expanding a definition.</td><td>139</td></tr><tr><td>element</td><td>new</td><td>presentation of XML elements</td><td>190</td></tr><tr><td>FMP</td><td>aug</td><td>now allows multiple conclusion elements, to represent general Gentzen-type sequents (not only natural deduction.) FMP now allows an optional id attribute.</td><td>123</td></tr><tr><td>hypothesis</td><td>cha</td><td>new required attribute discharged-in to specify the derive element that discharges this hypothesis.</td><td>162</td></tr><tr><td>measure</td><td>new</td><td>specifies a measure function (as an OMOBJ)</td><td>141</td></tr><tr><td>metadata</td><td>aug</td><td>new optional attribute inherits that allows to inherit metadata from other declarations</td><td>92</td></tr><tr><td>method</td><td>cha</td><td>first child that used to be an om:OMSTR or ref element is now moved into a required xref attribute that holds an URI that points to the element that defines the method. The om:OMOBJ content of the other children (they were parameter elements) is now directly included in the method element.</td><td>164</td></tr><tr><td>obligation</td><td>new</td><td>takes over the role of assertion-just.</td><td></td></tr><tr><td>omgroup</td><td>aug</td><td>also allows the elements that can only appear in theory elements, so that omgroups can also be used for grouping inside theory elements. The type attribute is now restrained to one of narrative, sequence, alternative, contrast.</td><td>93</td></tr><tr><td>omlet</td><td>aug</td><td>obtained attributes width and height from private. New optional attributes action for the action to be taken when activated, and data a URIsref to data in a private element. 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New optional attribute generated-by to show that it is generated by expanding a definition.	139	element	new	presentation of XML elements	190	FMP	aug	now allows multiple conclusion elements, to represent general Gentzen-type sequents (not only natural deduction.) FMP now allows an optional id attribute.	123	hypothesis	cha	new required attribute discharged-in to specify the derive element that discharges this hypothesis.	162	measure	new	specifies a measure function (as an OMOBJ)	141	metadata	aug	new optional attribute inherits that allows to inherit metadata from other declarations	92	method	cha	first child that used to be an om:OMSTR or ref element is now moved into a required xref attribute that holds an URI that points to the element that defines the method. 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Conf. MathPix confidence:

≥0.9

0.5-0.9

<0.5

 — no value

Residual render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

not measured

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value
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kohlhase-omdoc_TAB_081	333	—	(no LaTeX source; 1142 × 568 px region)	—	334B Quick-Reference <table> <tr> <td>CMP</td><td>122</td><td>MTXT</td><td></td><td>xml:lang, xml:id</td><td>− (text OMOBJ m:math legacy with term omlet)*</td></tr> <tr> <td>code</td><td>202</td><td>EXT</td><td></td><td>xml:id, for, theory, generated-from, generated-via, requires, style, class</td><td>+ input?, output?, effect?, data*</td></tr> <tr> <td>conclusion</td><td>124</td><td>MTXT</td><td></td><td>xml:id, style, class</td><td>+ CMP*, (OMOBJ m:math legacy)?</td></tr> <tr> <td>constructor</td><td>156</td><td>ADT</td><td>name</td><td>type, scope, style, class, theory, generated-from, generated-via</td><td>+ argument*, recognizer?</td></tr> <tr> <td>dc:contributor</td><td>98</td><td>DC</td><td></td><td>xml:id, role, style, class</td><td>− ⟨⟨text⟩⟩</td></tr> <tr> <td>dc:creator</td><td>98</td><td>DC</td><td></td><td>xml:id, role, style, class</td><td>− ⟨⟨text⟩⟩</td></tr> </table>	CMP	122	MTXT		xml:lang, xml:id	− (text OMOBJ m:math legacy with term omlet)*	code	202	EXT		xml:id, for, theory, generated-from, generated-via, requires, style, class	+ input?, output?, effect?, data*	conclusion	124	MTXT		xml:id, style, class	+ CMP*, (OMOBJ m:math legacy)?	constructor	156	ADT	name	type, scope, style, class, theory, generated-from, generated-via	+ argument*, recognizer?	dc:contributor	98	DC		xml:id, role, style, class	− ⟨⟨text⟩⟩	dc:creator	98	DC		xml:id, role, style, class	− ⟨⟨text⟩⟩																																																
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kohlhase-omdoc_TAB_082	333	—	(no LaTeX source; 1141 × 852 px region)	—	<table> <tr> <td>m:csymbol</td><td>115</td><td>MML</td><td>definitionURL</td><td>id, xlink:href</td><td>− EMPTY</td></tr> <tr> <td>data</td><td>203</td><td>EXT</td><td></td><td>format, href, size, original</td><td>− <![CDATA[...]]></td></tr> <tr> <td>dc:date</td><td>99</td><td>DC</td><td></td><td>action, who</td><td>− ISO 8601 norm</td></tr> <tr> <td>dd</td><td>130</td><td>RT</td><td></td><td>xml:id, style, class, index, verbalizes</td><td>+ CMPcontent</td></tr> <tr> <td>di</td><td>130</td><td>RT</td><td></td><td>xml:id, style, class, index, verbalizes</td><td>+ dt+, dd*</td></tr> <tr> <td>dl</td><td>130</td><td>RT</td><td></td><td>xml:id, style, class, index, verbalizes</td><td>+ li*</td></tr> <tr> <td>dt</td><td>130</td><td>RT</td><td></td><td>xml:id, style, class, index, verbalizes</td><td>+ CMPcontent</td></tr> <tr> <td>decomposition</td><td>184</td><td>DG</td><td>links</td><td>theory, generated-from, generated-via</td><td>− EMPTY</td></tr> <tr> <td>definition</td><td>139</td><td>ST</td><td>xml:id, for</td><td>uniqueness, existence, consistency, exhaustivity, type, generated-from, generated-via, style, class</td><td>+ CMP*, (FMP equation OMOBJ m:math legacy)?, measure?, ordering?</td></tr> <tr> <td>dc:description</td><td>99</td><td>DC</td><td></td><td>xml:lang</td><td>− CMPcontent</td></tr> <tr> <td>derive</td><td>162</td><td>PF</td><td></td><td>xml:id, style, class</td><td>− CMP*, FMP?, method?</td></tr> <tr> <td>effect</td><td>204</td><td>EXT</td><td></td><td>xml:id, style, class</td><td>− CMP*, FMP*</td></tr> <tr> <td>element</td><td>190</td><td>PRES</td><td>name</td><td>xml:id, cr, ns</td><td>− (attribute element text recurse)*</td></tr> <tr> <td>example</td><td>146</td><td>ST</td><td>for</td><td>xml:id, type, assertion, proof, style, class, theory, generated-from, generated-via</td><td>+ CMP* (OMOBJ m:math legacy)?</td></tr> </table>	m:csymbol	115	MML	definitionURL	id, xlink:href	− EMPTY	data	203	EXT		format, href, size, original	− <![CDATA[...]]>	dc:date	99	DC		action, who	− ISO 8601 norm	dd	130	RT		xml:id, style, class, index, verbalizes	+ CMPcontent	di	130	RT		xml:id, style, class, index, verbalizes	+ dt+, dd*	dl	130	RT		xml:id, style, class, index, verbalizes	+ li*	dt	130	RT		xml:id, style, class, index, verbalizes	+ CMPcontent	decomposition	184	DG	links	theory, generated-from, generated-via	− EMPTY	definition	139	ST	xml:id, for	uniqueness, existence, consistency, exhaustivity, type, generated-from, generated-via, style, class	+ CMP*, (FMP equation OMOBJ m:math legacy)?, measure?, ordering?	dc:description	99	DC		xml:lang	− CMPcontent	derive	162	PF		xml:id, style, class	− CMP*, FMP?, method?	effect	204	EXT		xml:id, style, class	− CMP*, FMP*	element	190	PRES	name	xml:id, cr, ns	− (attribute element text recurse)*	example	146	ST	for	xml:id, type, assertion, proof, style, class, theory, generated-from, generated-via	+ CMP* (OMOBJ m:math legacy)?
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kohlhase-omdoc_TAB_086	334	—	(no LaTeX source; 1141 × 519 px region)	—	<table> <tr><td>hypothesis</td><td>162</td><td>PF</td><td></td><td>xml:id, style, class, inductive</td><td>— CMP*, FMP*</td></tr> <tr><td>dc:identifier</td><td>100</td><td>DC</td><td></td><td>scheme</td><td>— ANY</td></tr> <tr><td>ide</td><td>130</td><td>RT</td><td>index</td><td>xml:id,sort-by,see seealso, links, style, class</td><td>— idp*</td></tr> <tr><td>idp</td><td>130</td><td>RT</td><td></td><td>xml:id,sort-by,see seealso, links, style, class</td><td>— CMPcontent</td></tr> <tr><td>idt</td><td>130</td><td>RT</td><td></td><td>style, class</td><td>— CMPcontent</td></tr> <tr><td>idx</td><td>130</td><td>RT</td><td></td><td>xml:id,sort-by,see seealso, links, style, class</td><td>— idt?,idp*</td></tr> <tr><td>ignore</td><td>92</td><td>DOC</td><td></td><td>type, comment</td><td>— ANY</td></tr> <tr><td>imports</td><td>150</td><td>CTH</td><td>from</td><td>xml:id, type, style, class</td><td>+ morphism?</td></tr> <tr><td>inclusion</td><td>179</td><td>CTH</td><td>for</td><td>xml:id</td><td>—</td></tr> <tr><td>input</td><td>204</td><td>EXT</td><td></td><td>xml:id, style, class</td><td>— CMP*,FMP*</td></tr> <tr><td>insort</td><td>156</td><td>ADT</td><td>for</td><td></td><td>—</td></tr> <tr><td>dc:language</td><td>100</td><td>DC</td><td></td><td></td><td>— ISO 8601 norm</td></tr> <tr><td>li</td><td>130</td><td>RT</td><td></td><td>xml:id, style, class, index, verbalizes</td><td>— Math Vernacular</td></tr> </table>	hypothesis	162	PF		xml:id, style, class, inductive	— CMP*, FMP*	dc:identifier	100	DC		scheme	— ANY	ide	130	RT	index	xml:id,sort-by,see seealso, links, style, class	— idp*	idp	130	RT		xml:id,sort-by,see seealso, links, style, class	— CMPcontent	idt	130	RT		style, class	— CMPcontent	idx	130	RT		xml:id,sort-by,see seealso, links, style, class	— idt?,idp*	ignore	92	DOC		type, comment	— ANY	imports	150	CTH	from	xml:id, type, style, class	+ morphism?	inclusion	179	CTH	for	xml:id	—	input	204	EXT		xml:id, style, class	— CMP*,FMP*	insort	156	ADT	for		—	dc:language	100	DC			— ISO 8601 norm	li	130	RT		xml:id, style, class, index, verbalizes	— Math Vernacular
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kohlhase-omdoc_TAB_087	334	—	(no LaTeX source; 1141 × 120 px region)	—	<table> <tr><td>cc:license</td><td>102</td><td>CC</td><td></td><td>jurisdiction</td><td>— permissions, prohibitions, requirements</td></tr> <tr><td>link</td><td>130</td><td>RT</td><td></td><td>xml:id, style, class, index, verbalizes</td><td>— Math Vernacular</td></tr> </table>	cc:license	102	CC		jurisdiction	— permissions, prohibitions, requirements	link	130	RT		xml:id, style, class, index, verbalizes	— Math Vernacular																																																																		
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kohlhase-omdoc_TAB_088	334	—	(no LaTeX source; 1142 × 600 px region)	—	<table> <tr><td>m:math</td><td>114</td><td>MI4L</td><td></td><td>id, alternative</td><td>— choice, hint?, answer</td></tr> <tr><td>mc</td><td>210</td><td>QUIZ</td><td></td><td>xml:id, style, class, theory, generated-from, generated-via</td><td>—</td></tr> <tr><td>measure</td><td>141</td><td>ST</td><td></td><td>xml:id</td><td>— OMOBJ [m:math legacy</td></tr> <tr><td>metadata</td><td>92</td><td>DC</td><td></td><td>inherits</td><td>— (dc-element)*</td></tr> <tr><td>method</td><td>164</td><td>PF</td><td>xref</td><td></td><td>— (OMOBJ [m:math legacy premise proof proofobject])*</td></tr> <tr><td>morphism</td><td>175</td><td>CTH</td><td></td><td>xml:id, base, consistency, exhaustivity, type, hiding, style, class</td><td>— reequation*, measure?, ordering?</td></tr> <tr><td>note</td><td>131</td><td>RT</td><td></td><td>type,xml:id, style, class, index, verbalizes</td><td>— Math Vernacular</td></tr> </table>	m:math	114	MI4L		id, alternative	— choice, hint?, answer	mc	210	QUIZ		xml:id, style, class, theory, generated-from, generated-via	—	measure	141	ST		xml:id	— OMOBJ [m:math legacy	metadata	92	DC		inherits	— (dc-element)*	method	164	PF	xref		— (OMOBJ [m:math legacy premise proof proofobject])*	morphism	175	CTH		xml:id, base, consistency, exhaustivity, type, hiding, style, class	— reequation*, measure?, ordering?	note	131	RT		type,xml:id, style, class, index, verbalizes	— Math Vernacular																																				
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<tr><td>omLet</td><td>205</td><td>EXT</td><td></td><td>id, argstr, type, function, action, data, style, class</td><td>+ ANY</td></tr> <tr><td>om:OMOBJ</td><td>108</td><td>OM</td><td></td><td>id, cdbase, class, style</td><td>– $\langle\langle Omel\rangle\rangle?$</td></tr> <tr><td>omstyle</td><td>188</td><td>PRES</td><td>element</td><td>for, xml:id, xref, style, class</td><td>– (style xs t)*</td></tr> <tr><td>om:OMS</td><td>108</td><td>OM</td><td>cd, name</td><td>class, style</td><td>– EMPTY</td></tr> <tr><td>omtext</td><td>124</td><td>MTXT</td><td></td><td>xml:id, type, for, from, style, theory, generated-from, generated-via</td><td>+ CMP*, FMP?</td></tr> <tr><td>om:OMV</td><td>108</td><td>OM</td><td>name</td><td>class, style</td><td>– EMPTY</td></tr> <tr><td>ordering</td><td>141</td><td>ST</td><td></td><td>xml:id</td><td>– OMOBJ m:math legacy</td></tr> <tr><td>output</td><td>204</td><td>EXT</td><td></td><td>xml:id, style, class</td><td>– CMP*, FMP*</td></tr> <tr><td>p</td><td>130</td><td>RT</td><td></td><td>xml:id, style, class, index, verbalizes</td><td>– Math Vernacular</td></tr> <tr><td>param</td><td>207</td><td>EXT</td><td>name</td><td>value, valuetype</td><td>– EMPTY</td></tr> <tr><td>path-just</td><td>184</td><td>DG</td><td>local, globals</td><td>for, xml:id</td><td>– EMPTY</td></tr> <tr><td>cc:permissions</td><td>103</td><td>CC</td><td></td><td>reproduction, distribution, derivative, works</td><td>– EMPTY</td></tr> <tr><td>premise</td><td>164</td><td>PF</td><td>xref</td><td></td><td>– EMPTY</td></tr> </table>	obligation	178	CTH	induced-by, assertion	xml:id	– EMPTY	om:OMA	108	OM		id, cdbase	– $\langle\langle Omel\rangle\rangle^*$	om:OMATTR	110	OM		id, cdbase	– $\langle\langle Omel\rangle\rangle$	om:OMATP	110	OM		cdbase	– (ONS, $\langle\langle Omel\rangle\rangle$ om:OMFOREIGN))+	om:OMB	111	OM		id, class, style, class	– #PCDATA	om:OMBIND	109	OM		id, cdbase	– $\langle\langle Omel\rangle\rangle$, om:OMBVAR, $\langle\langle 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definition</td><td>OMDoc reference</td></tr><tr><td></td><td colspan="2">points to an assertion stating that the cases are consistent, i.e. that they give the same values, where they overlap</td></tr><tr><td>copyleft</td><td>cc:restrictions</td><td>required, not_required</td></tr><tr><td></td><td colspan="2">specifies whether derived works must be licensed with the same license as the current document.</td></tr><tr><td>cr</td><td>element</td><td>yes/no</td></tr><tr><td></td><td colspan="2">specifies whether an xlink:href cross-reference should be set on the result element.</td></tr><tr><td>cref</td><td>om:*</td><td>URI reference</td></tr><tr><td></td><td colspan="2">extra attribute for cross-references in parallel markup</td></tr><tr><td>crid</td><td>element</td><td>XPATH expression</td></tr><tr><td></td><td colspan="2">the path to the sub-element that corresponds to the result element.</td></tr><tr><td>crossref-symbol</td><td>presentation, use</td><td>no, yes, brackets, separator, 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<code>axiom-inclusions</code> or <code>theory-inclusions</code> that is the rest of the inclusion path.</td></tr><tr><td><code>hiding</code></td><td><code>morphism</code></td><td></td></tr><tr><td></td><td colspan="2">specifies the names of symbols that are in the domain of the morphism</td></tr><tr><td><code>href</code></td><td><code>data</code>, <code>link</code>, <code>on:OMR</code></td><td>URI reference</td></tr><tr><td></td><td colspan="2">a URI to an external file containing the data.</td></tr><tr><td><code>xml:id</code></td><td></td><td></td></tr><tr><td></td><td colspan="2">associates a unique identifier to an element, which can thus be referenced by an <code>for</code> or <code>xref</code> attribute.</td></tr><tr><td><code>xml:base</code></td><td></td><td></td></tr><tr><td></td><td colspan="2">specifies a base URL for a resource fragment</td></tr><tr><td><code>index</code></td><td><code>on RT elements</code></td><td></td></tr><tr><td></td><td colspan="2">A path identifier to 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The value should e.g. be a MIME type [FB96].		<code>for</code>	<code>*</code>			can be used to reference an element by its unique identifier given in its <code>xml:id</code> attribute.		<code>formalism</code>	<code>legacy</code>	URI reference		specifies the formalism in which the content is expressed		<code>format</code>	<code>legacy</code>	URI reference		specifies the encoding format of the content		<code>format</code>	<code>use</code>	<code>cmml</code> , <code>default</code> , <code>html</code> , <code>mathematica</code> , <code>pmml</code> , <code>TeX</code> , ...		specifies the output format for which the notation is specified		<code>from</code>	<code>imports</code> , <code>theory-inclusion</code> , <code>axiom-inclusion</code>	URI reference		pointer to source <code>theory</code> of a theory morphism		<code>from</code>	<code>omtext</code>	URI reference		points to the source of a relation given by a text type		<code>generated-from</code>	<code>top-level elements</code>	URI reference		points to a higher-level syntax element, that generates this statement.		<code>generated-via</code>	<code>top-level elements</code> , ...	URI reference		points to a theory-morphism, via which it is translated from the element pointed to by the <code>generated-from</code> attribute.		<code>globals</code>	<code>path-just</code>			points to the <code>axiom-inclusions</code> or <code>theory-inclusions</code> that is the rest of the inclusion path.		<code>hiding</code>	<code>morphism</code>			specifies the names of symbols that are in the domain of the morphism		<code>href</code>	<code>data</code> , <code>link</code> , <code>on:OMR</code>	URI reference		a URI to an external file containing the data.		<code>xml:id</code>				associates a unique identifier to an element, which can thus be referenced by an <code>for</code> or <code>xref</code> attribute.		<code>xml:base</code>				specifies a base URL for a resource fragment		<code>index</code>	<code>on RT elements</code>			A path identifier to establish multilingual correspondence		<code>induced-by</code>	<code>obligation</code>			points to the statement in the source theory that induces this proof obligation		<code>inductive</code>	<code>assumption</code> , <code>hypothesis</code>	<code>yes</code> , <code>no</code>		Marks an assumption or hypothesis inductive.		<code>inherits</code>	<code>metadata</code>	URI reference		points to a metadata element from which this one inherits.		<code>jurisdiction</code>	<code>cc:license</code>	IANA Top level Domain designator		specifies the country of jurisdiction for a Creative Commons license		<code>just-by</code>	<code>type</code>			points to an assertion that states the type property in question.		<code>role</code>	<code>symbol</code> , <code>constructor</code> , <code>recognizer</code> , <code>selector</code> , <code>sortdef</code>	<code>object</code> , <code>type</code> , <code>sort</code> , <code>binder</code> , <code>attribution</code> , <code>semantic-attribution</code> , <code>error</code>		specifies the role (possible syntactic roles) of the symbol in this declaration.		<code>role</code>	<code>dc:creator</code> , <code>dc:contributor</code>	MARC relators		specifies the role of a person who has contributed to the document		<code>role</code>	<code>presentation</code>	<code>applied</code> , <code>binding</code> , <code>key</code>		specifies which role of the symbol is annotated with notation information		<code>lbrack</code>	<code>presentation</code> , <code>use</code>			the left bracket to use in the notation of a function symbol		<code>links</code>	<code>decomposition</code>	
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Conf. MathPix confidence:

■

 ≥0.9

■

 0.5-0.9

■

 <0.5 — no value

Residual render vs scan (inkdrill):

●

 C component

●

 W weak

●

 S stable

●

 N noise

●

 K clean

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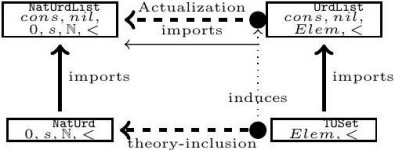
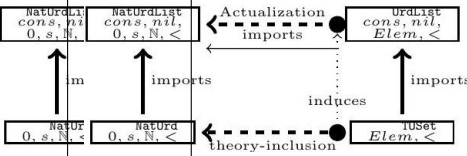
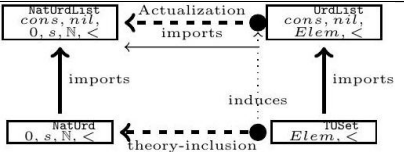
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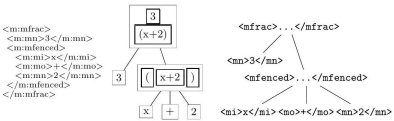
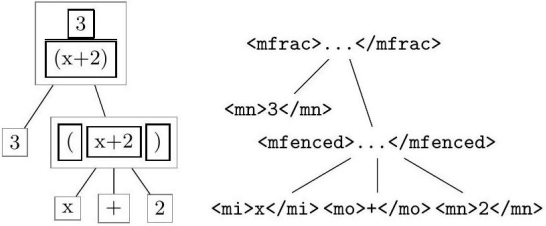
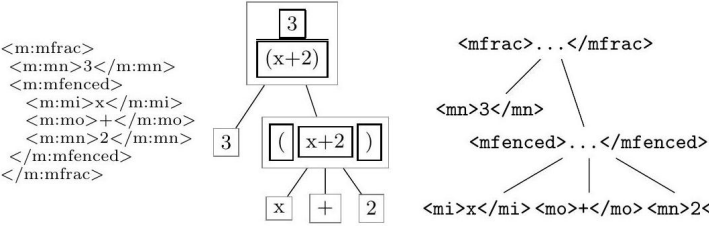
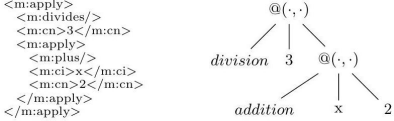
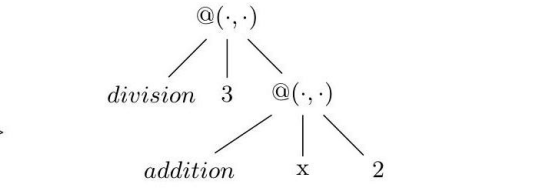
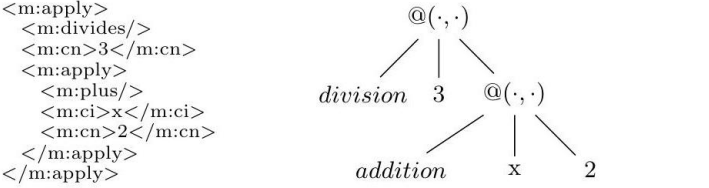
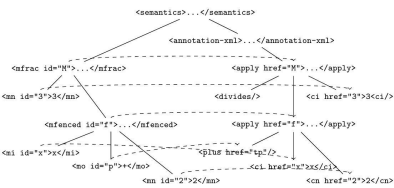
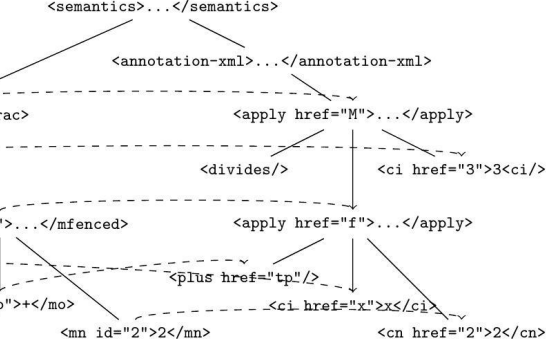
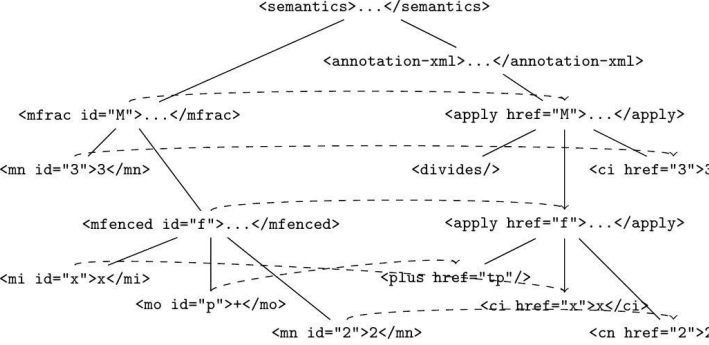
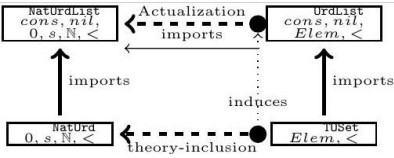
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The value should be number of kilobytes</td></tr><tr><td>sort</td><td colspan="2">argument</td></tr><tr><td></td><td colspan="2">specifies the argument sort of the constructor</td></tr><tr><td>style</td><td colspan="2">*</td></tr><tr><td></td><td colspan="2">specifies a token for a presentation style to be picked up in a presentation element.</td></tr><tr><td>system</td><td colspan="2">type</td></tr><tr><td></td><td colspan="2">A token that specifies the logical type system that governs the type specified in the type element.</td></tr><tr><td>theory</td><td colspan="2">*</td></tr><tr><td></td><td colspan="2">specifies the home theory of an OMDoc statement.</td></tr><tr><td>to</td><td>theory-inclusion, axiom-inclusion</td><td></td></tr><tr><td></td><td colspan="2">specifies the target theory</td></tr><tr><td>total</td><td>selector</td><td>no, yes</td></tr><tr><td></td><td colspan="2">specifies whether the symbol declared here is a total or partial function.</td></tr><tr><td>type</td><td>adt</td><td>free, generated, loose</td></tr><tr><td></td><td colspan="2">defines the semantics of an abstract data type free = no junk, no confusion, generated = no junk, loose is the general case.</td></tr><tr><td>type</td><td>assertion</td><td>theorem, lemma, corollary, conjecture, false-conjecture, obligation, postulate, formula, assumption, proposition</td></tr><tr><td></td><td colspan="2">tells you more about the intention of the assertion</td></tr><tr><td>type</td><td>definition</td><td>implicit, inductive, obj, recursive, simple</td></tr><tr><td></td><td colspan="2">specifies the definition principle</td></tr><tr><td>type</td><td>derive</td><td>conclusion, gap</td></tr><tr><td></td><td colspan="2">singles out special proof steps: conclusions and gaps (unjustified proof steps)</td></tr><tr><td>type</td><td>example</td><td>against, for</td></tr><tr><td></td><td colspan="2">specifies whether the objects in this example support or falsify some conjecture</td></tr><tr><td>type</td><td>ignore</td><td></td></tr><tr><td></td><td colspan="2">specifies the type of error, if ignore is used for in-place error markup</td></tr><tr><td>type</td><td>imports</td><td>global, local</td></tr><tr><td></td><td colspan="2">local imports only concern the assumptions directly stated in the theory. global imports also concern the ones the source theory inherits.</td></tr><tr><td>type</td><td>morphism</td><td></td></tr><tr><td></td><td colspan="2">specifies whether the morphism is recursive or merely pattern-defined</td></tr><tr><td>type</td><td>omgroup, omdoc</td><td>enumeration, sequence, itemize</td></tr><tr><td></td><td colspan="2">the first three give the text category, the second three are used for generalized tables</td></tr><tr><td>type</td><td>omtext</td><td>abstract, antithesis, comment, conclusion, elaboration, evidence, introduction, motivation, thesis</td></tr><tr><td></td><td colspan="2">a specification of the intention of the text fragment, in reference to context.</td></tr><tr><td>type</td><td>phrase</td><td></td></tr><tr><td></td><td colspan="2">the linguistic or mathematical type of the phrase</td></tr><tr><td>type</td><td>ref</td><td>include, cite</td></tr><tr><td></td><td colspan="2">specifies whether to replace the ref element by the fragment referenced by href attribute or to merely cite it.</td></tr><tr><td>uniqueness</td><td>definition</td><td>URI reference</td></tr><tr><td></td><td colspan="2">points to an assertion that states the uniqueness of the concept described in an implicit definition</td></tr><tr><td>value</td><td>param</td><td></td></tr></table>	separator	presentation, use			the separator for the arguments to use in the notation of a function symbol		show	omlet	new, replace, embed, other		specifies the desired presentation of the external object.		size	data			specifies the size the data specified by a data element. 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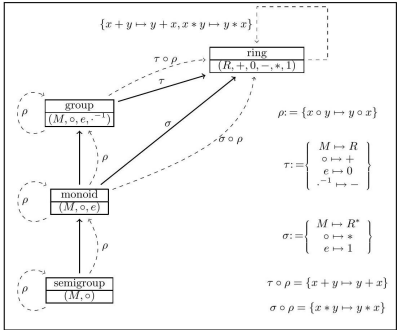
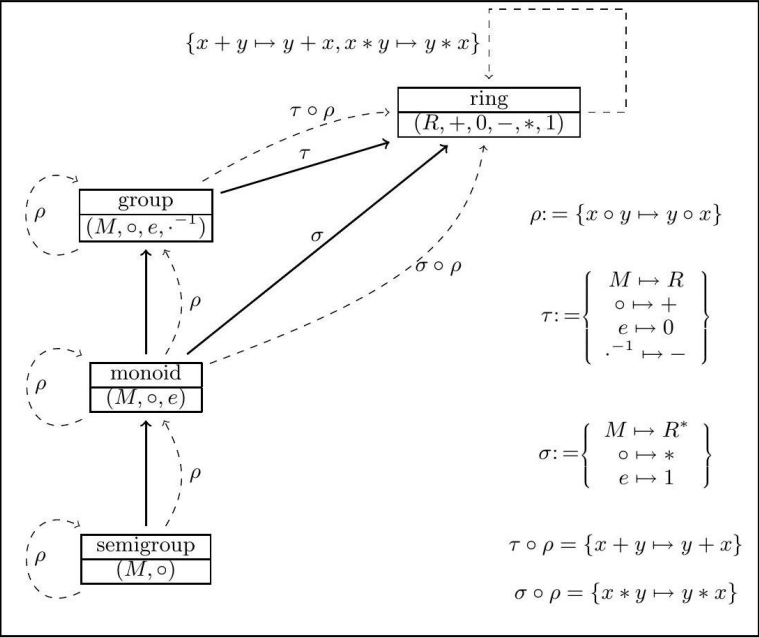
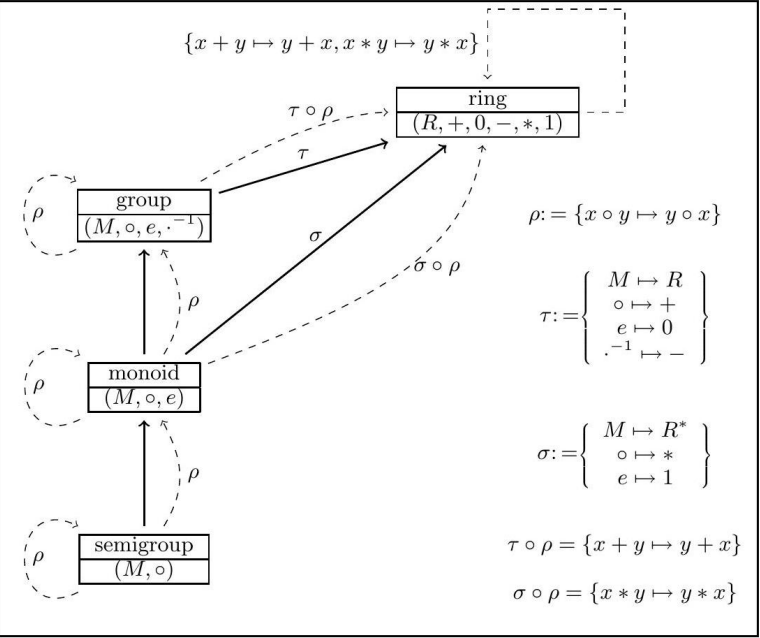
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Image regions — rendered against scan

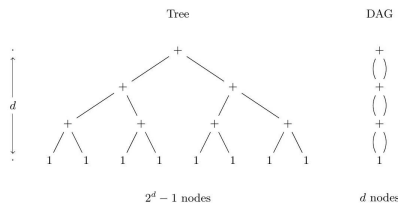
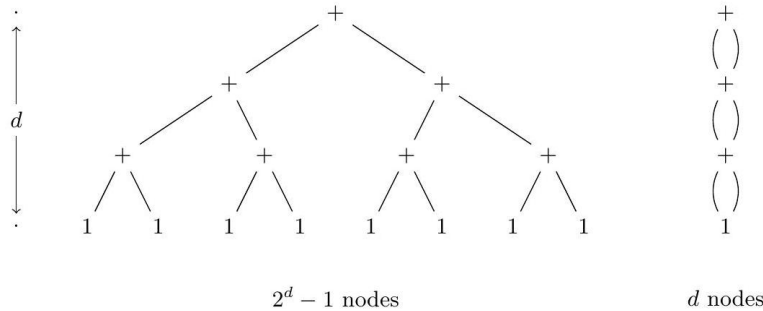
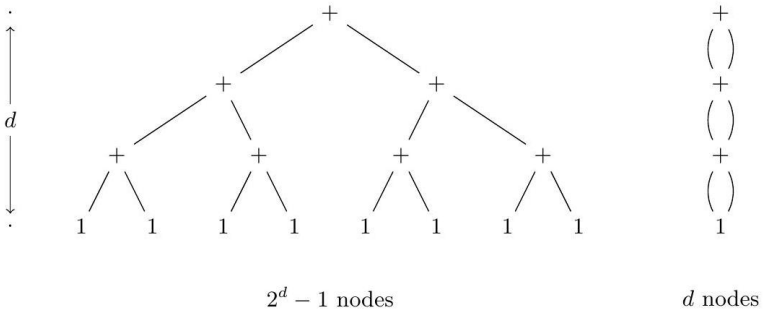
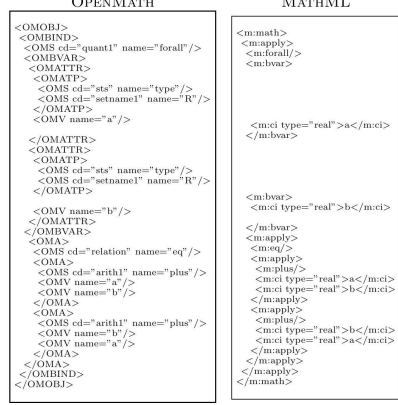
Two comparable cells per row. **Rendered:** the region’s own LaTeX compiled as its OWN document, or — where no LaTeX exists — the scan again, so every row has two cells and the ink difference reads as a floor rather than as missing-versus-present. A duplicated row is marked (*dup*) and its distance is a SELF-comparison, not agreement between two sources. **Scan:** the tex.zip image whose filename region 5-tuple matches this row, else the CDN crop. The Class column carries the residual class beside MathPix’s confidence, as the equation rows do. Setting a region’s LaTeX does NOT say it renders to what is on the page — that is what the two columns are for, and 281 is the open question they exist to let a reader answer. A region with no LaTeX can be reconstructed with pdfdrill vision; verify any LLM result against the real ink with inkdrill.

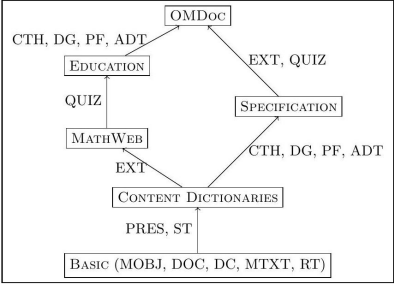
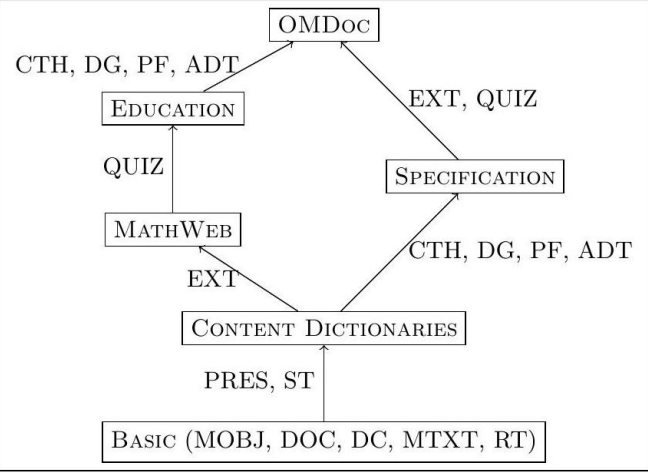
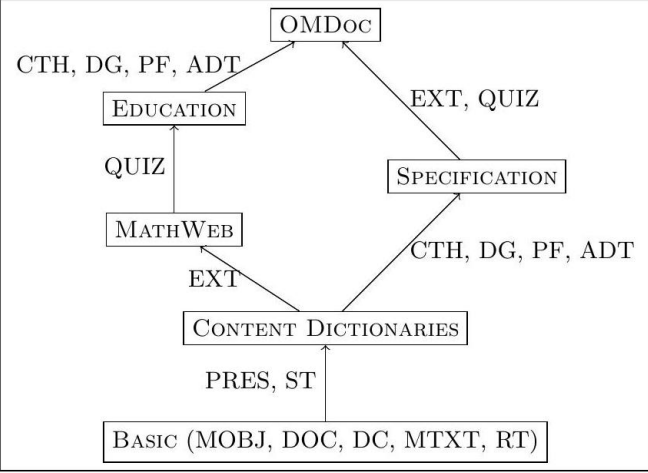
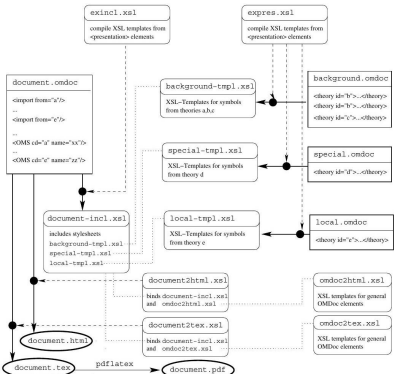
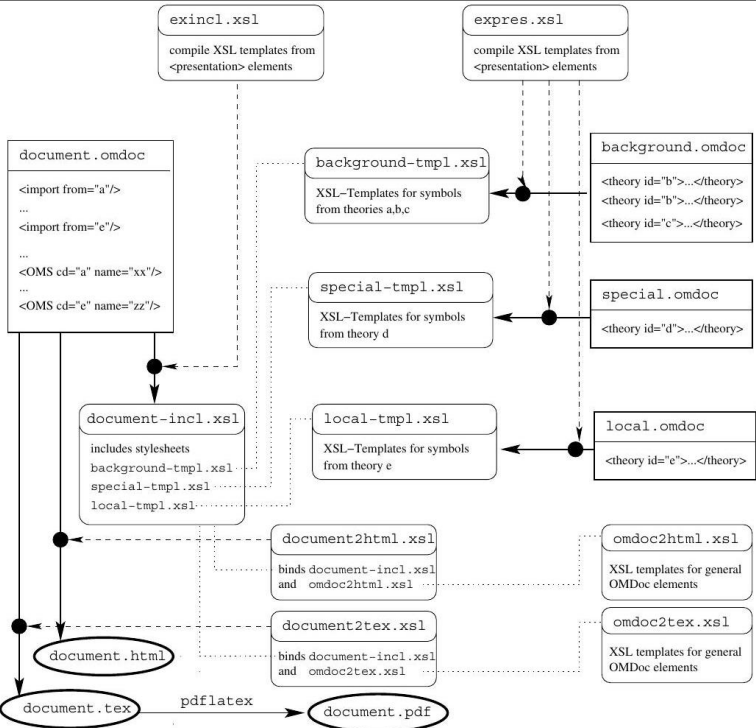
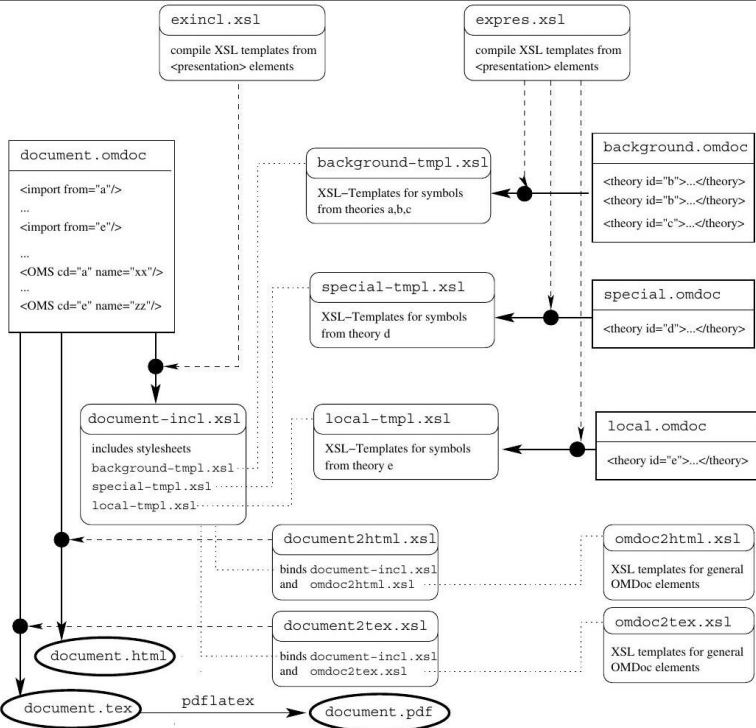
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kohlhase-omdoc_PIC_0005	072	— (dup)	1b348de5-ed54-4a9a-aba0-06a9a89f7dc6-072_29_1005_1436_193.jpg	<axiom xml:id="unit.ax"><FMP> $\exists x \in M. \forall y \in M. y \circ x = y$ </FMP></axiom>	<axiom xml:id="unit.ax"><FMP> $\exists x \in M. \forall y \in M. y \circ x = y$ </FMP></axiom>	<axiom xml:id="unit.ax"><FMP> $\exists x \in M. \forall y \in M. y \circ x = y$ </FMP></axiom>
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kohlhase-omdoc_PIC_0008	072	— (dup)	1b348de5-ed54-4a9a-aba0-06a9a89f7dc6-072_32_256_1791_215.jpg	$\forall S \subseteq M. S^* = S \setminus \{e\}$	$\forall S \subseteq M. S^* = S \setminus \{e\}$	$\forall S \subseteq M. S^* = S \setminus \{e\}$
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kohlhase-omdoc_PIC_0010	093	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
kohlhase-omdoc_PIC_0011	093	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
kohlhase-omdoc_PIC_0012	093	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
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kohlhase-omdoc_PIC_0014	131	— (dup)	1b348de5-ed54-4a9a-aba0-06a9a89f7dc6-131_29_582_1415_168.jpg	<CMP>If $a \in U$ and $a \in V$, then $a \in U \cap V$ or	<CMP>If $a \in U$ and $a \in V$, then $a \in U \cap V$ or	<CMP>If $a \in U$ and $a \in V$, then $a \in U \cap V$ or
kohlhase-omdoc_PIC_0015	148	— (dup)	1b348de5-ed54-4a9a-aba0-06a9a89f7dc6-148_29_856_1694_193.jpg	<FMP> $\forall S. \text{sgpr}(S) \rightarrow \forall x, y. \text{unit}(x, S) \wedge \text{unit}(y, S) \rightarrow x = y$ </FMP>	<FMP> $\forall S. \text{sgpr}(S) \rightarrow \forall x, y. \text{unit}(x, S) \wedge \text{unit}(y, S) \rightarrow x = y$ </FMP>	<FMP> $\forall S. \text{sgpr}(S) \rightarrow \forall x, y. \text{unit}(x, S) \wedge \text{unit}(y, S) \rightarrow x = y$ </FMP>
kohlhase-omdoc_PIC_0016	158	— (dup)	1b348de5-ed54-4a9a-aba0-06a9a89f7dc6-158_30_1039_965_218.jpg	<CMP> neut is a left unit for op .</CMP><FMP> $\forall x \in \text{set.op}(x, \text{neut}) = x$ </FMP>	<CMP> neut is a left unit for op .</CMP><FMP> $\forall x \in \text{set.op}(x, \text{neut}) = x$ </FMP>	<CMP> neut is a left unit for op .</CMP><FMP> $\forall x \in \text{set.op}(x, \text{neut}) = x$ </FMP>
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kohlhase-omdoc_PIC_0018	167	— (dup)	1b348de5-ed54-4a9a-aba0-06a9a89f7dc6-167_38_496_981_292.jpg	<CMP>Let $q \stackrel{\text{def}}{=} p_1 \cdots p_n + 1$ </CMP>	<CMP>Let $q \stackrel{\text{def}}{=} p_1 \cdots p_n + 1$ </CMP>	<CMP>Let $q \stackrel{\text{def}}{=} p_1 \cdots p_n + 1$ </CMP>
kohlhase-omdoc_PIC_0019	172	— (dup)	1b348de5-ed54-4a9a-aba0-06a9a89f7dc6-172_29_655_575_193.jpg	<CMP>If $a \in U$ or $a \in V$, then $a \in U \cup V$.</CMP>	<CMP>If $a \in U$ or $a \in V$, then $a \in U \cup V$.</CMP>	<CMP>If $a \in U$ or $a \in V$, then $a \in U \cup V$.</CMP>
kohlhase-omdoc_PIC_0020	172	— (dup)	1b348de5-ed54-4a9a-aba0-06a9a89f7dc6-172_29_539_881_218.jpg	<CMP>If $a \in U$, then $a \in U \cup V$.</CMP>	<CMP>If $a \in U$, then $a \in U \cup V$.</CMP>	<CMP>If $a \in U$, then $a \in U \cup V$.</CMP>
kohlhase-omdoc_PIC_0021	172	— (dup)	1b348de5-ed54-4a9a-aba0-06a9a89f7dc6-172_30_541_1101_218.jpg	<CMP>If $a \in V$, then $a \in U \cup V$.</CMP>	<CMP>If $a \in V$, then $a \in U \cup V$.</CMP>	<CMP>If $a \in V$, then $a \in U \cup V$.</CMP>
kohlhase-omdoc_PIC_0022	173	— (dup)	1b348de5-ed54-4a9a-aba0-06a9a89f7dc6-173_29_734_466_244.jpg	<assumption xml:id="t1_p1_l1.d1.a"> $a \in U$ </assumption>	<assumption xml:id="t1_p1_l1.d1.a"> $a \in U$ </assumption>	<assumption xml:id="t1_p1_l1.d1.a"> $a \in U$ </assumption>
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kohlhase-omdoc_PIC_0024	175	— (dup)	1b348de5-ed54-4a9a-aba0-06a9a89f7dc6-175_39_496_208_292.jpg	<CMP>Let $q \stackrel{\text{def}}{=} p_1 \cdots p_n + 1$ </CMP>	<CMP>Let $q \stackrel{\text{def}}{=} p_1 \cdots p_n + 1$ </CMP>	<CMP>Let $q \stackrel{\text{def}}{=} p_1 \cdots p_n + 1$ </CMP>
kohlhase-omdoc_PIC_0025	184	— (dup)	1b348de5-ed54-4a9a-aba0-06a9a89f7dc6-184_34_1024_213_168.jpg	<assertion xml:id="conv.inv" theory="group"> $\forall x, y \in M. x \circ x^{-1} = e$ </assertion>	<assertion xml:id="conv.inv" theory="group"> $\forall x, y \in M. x \circ x^{-1} = e$ </assertion>	<assertion xml:id="conv.inv" theory="group"> $\forall x, y \in M. x \circ x^{-1} = e$ </assertion>
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kohlhase-omdoc_DIA_0001	019	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)

kohlhase-omdoc_DIA_0002	019	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
kohlhase-omdoc_DIA_0003	022	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
kohlhase-omdoc_DIA_0004	024	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
kohlhase-omdoc_DIA_0005	030	— (dup)	1b348de5-ed54-4a9a-aba0-06a9a89f7dc6-030_343_1141_196_172.jpg			
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kohlhase-omdoc_DIA_0008	032	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
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share this crop						

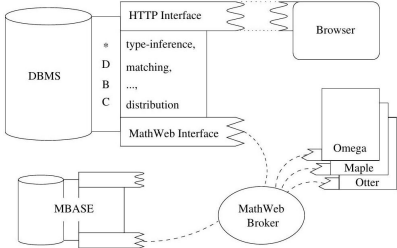
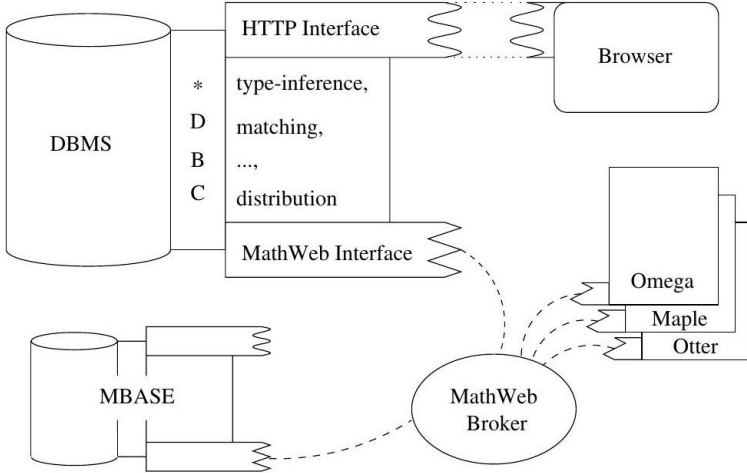
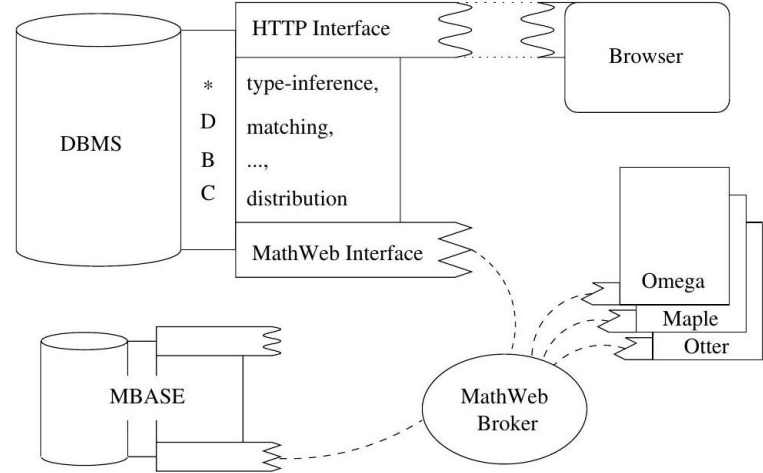
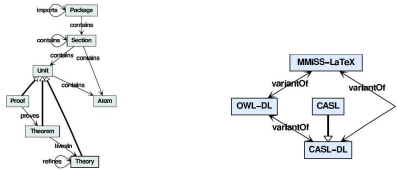
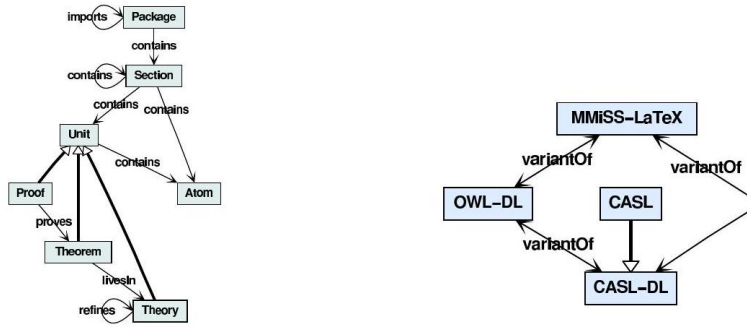
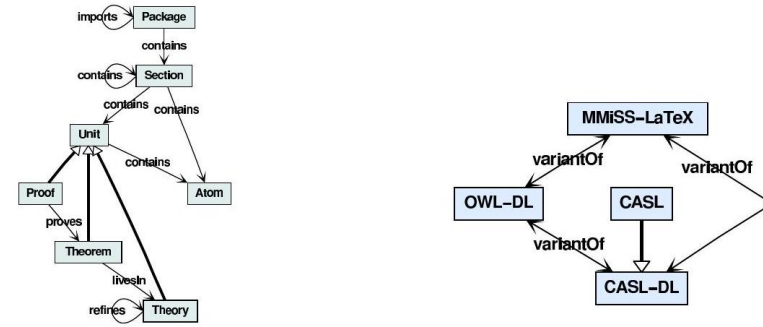
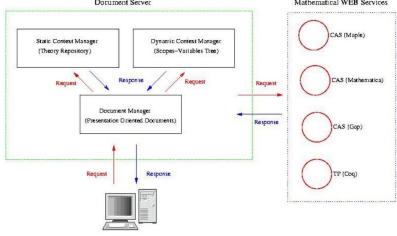
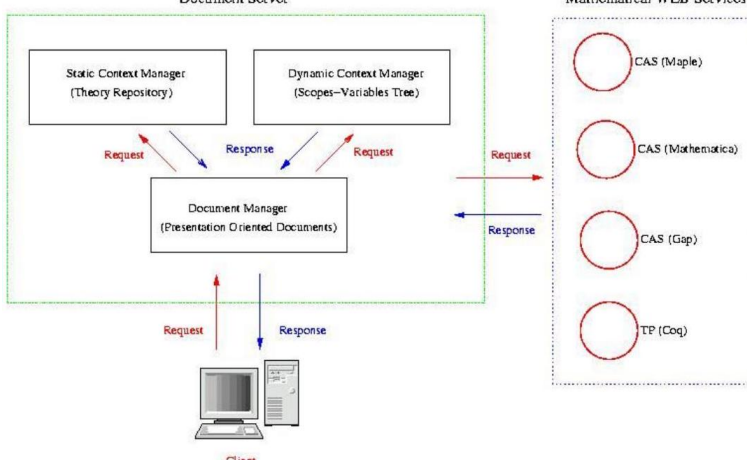
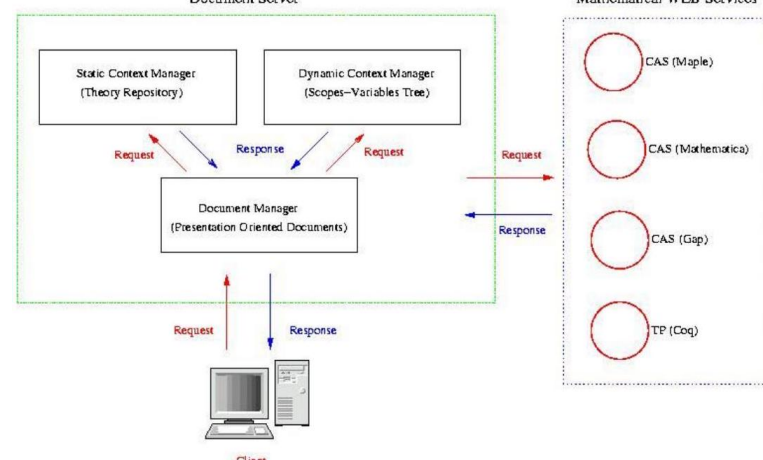
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kohlhase-omdoc_DIA_0031	069	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
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kohlhase-omdoc_DIA_0037	072	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
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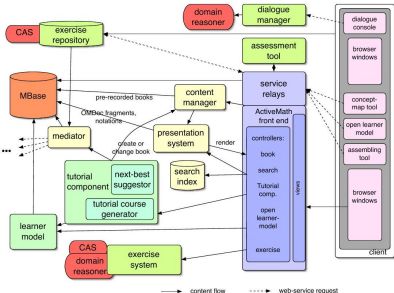
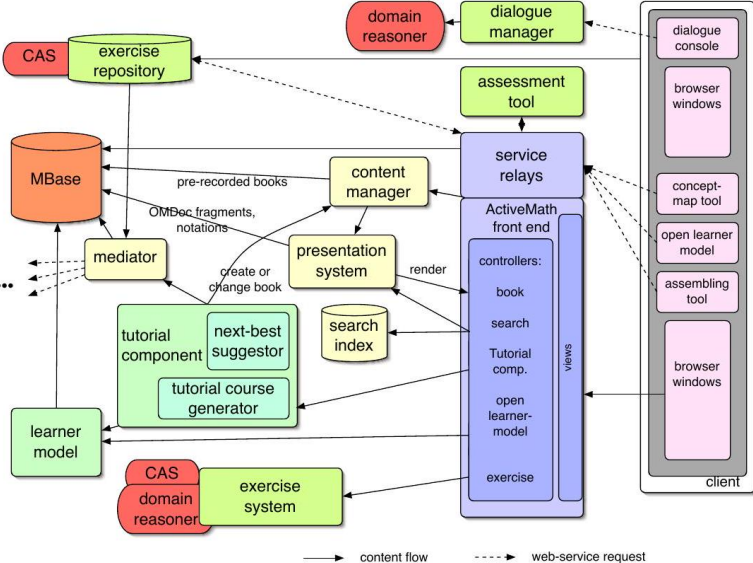
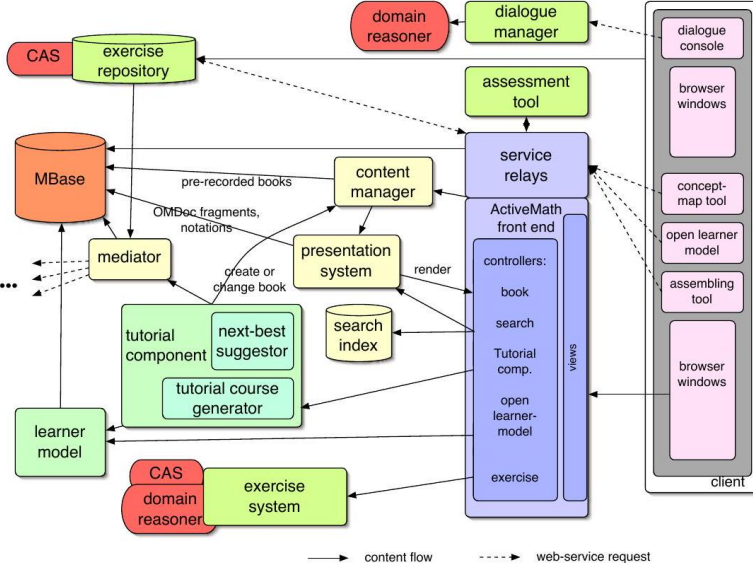
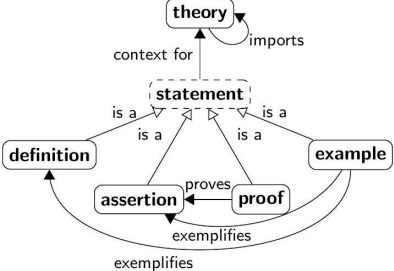
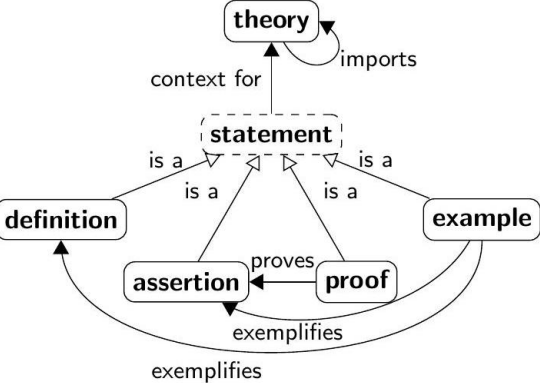
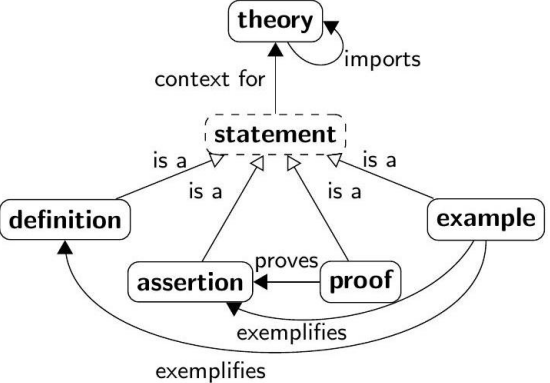
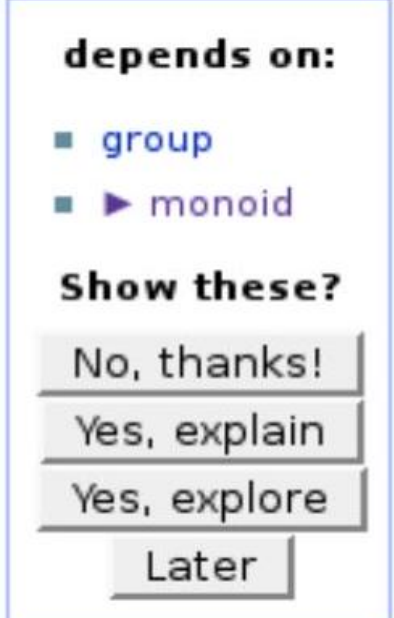
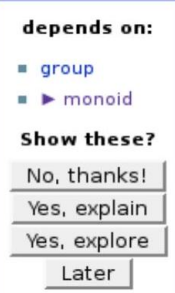
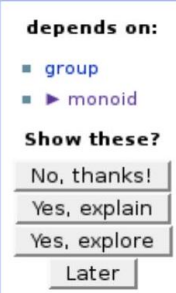
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kohlhase-omdoc_DIA_0061	101	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
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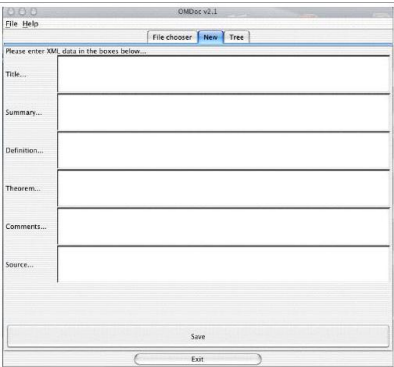
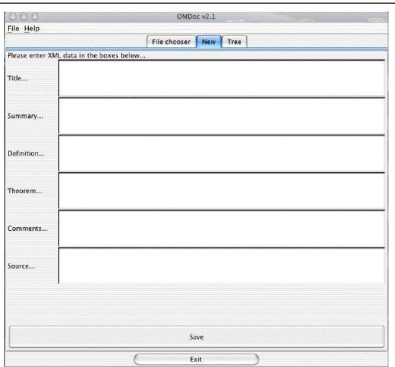
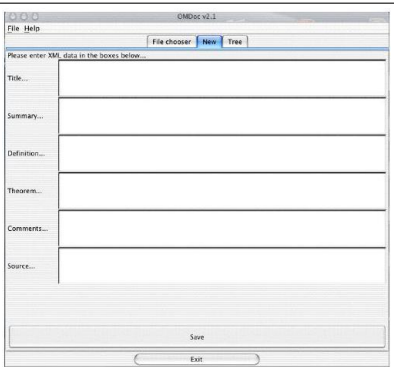
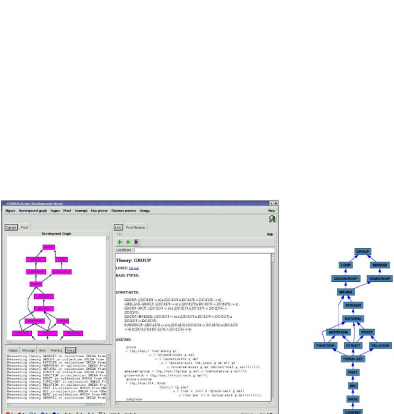
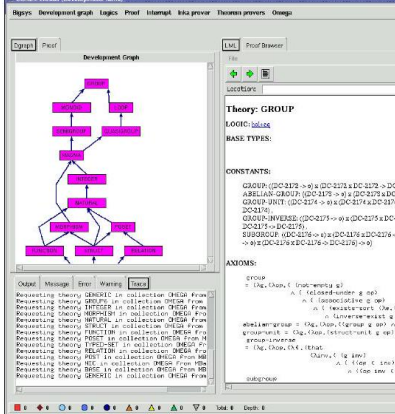
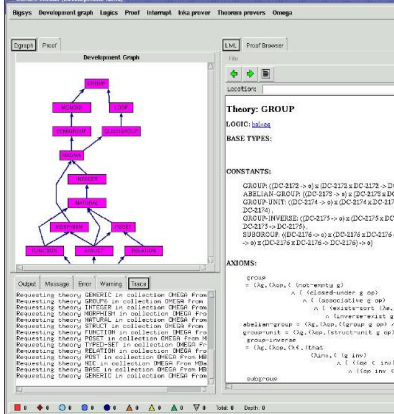
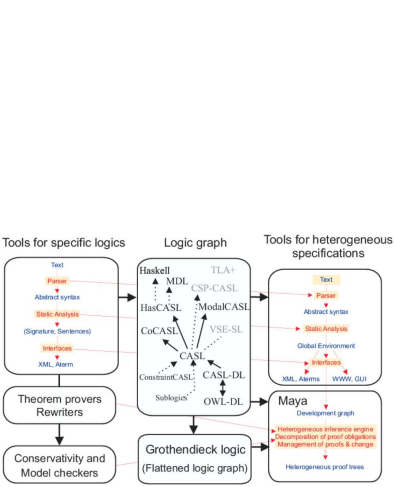
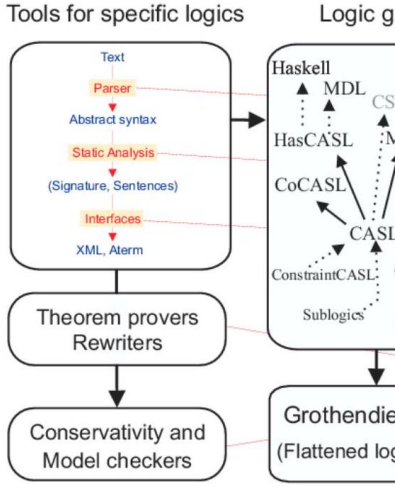
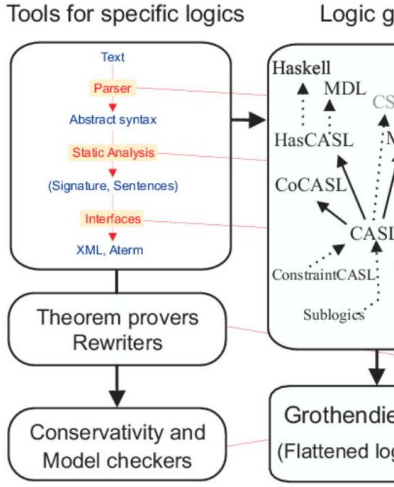
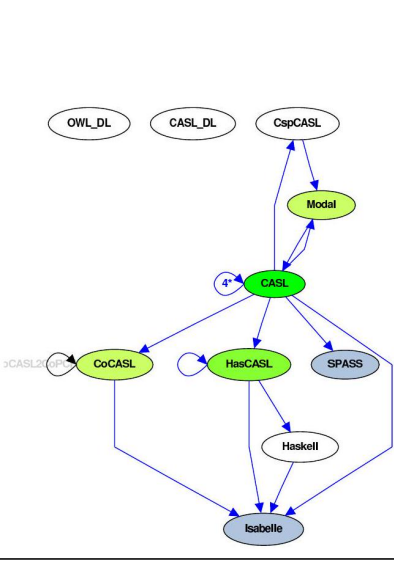
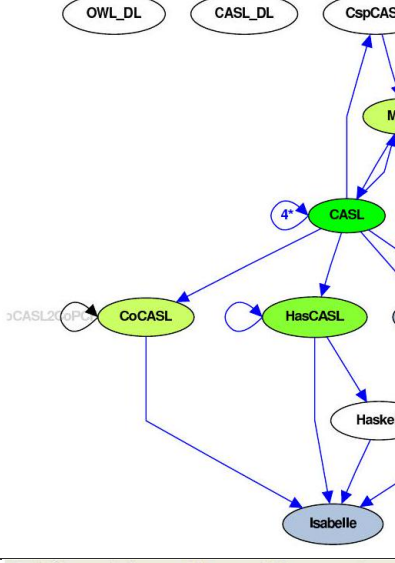
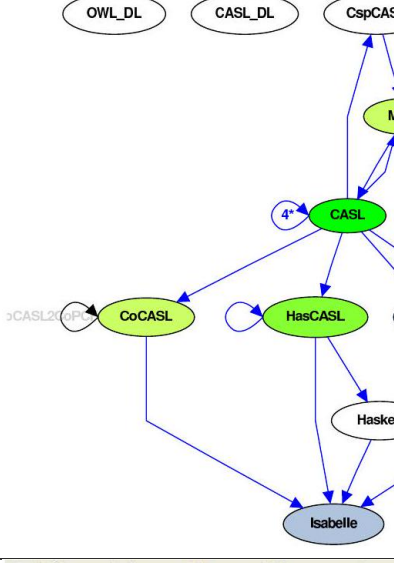
						
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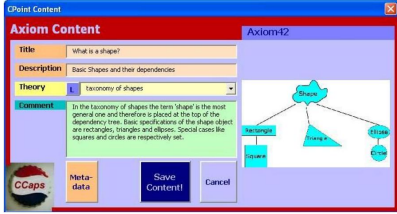
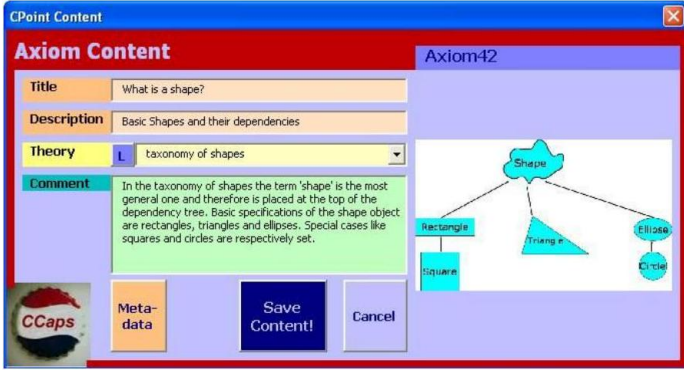
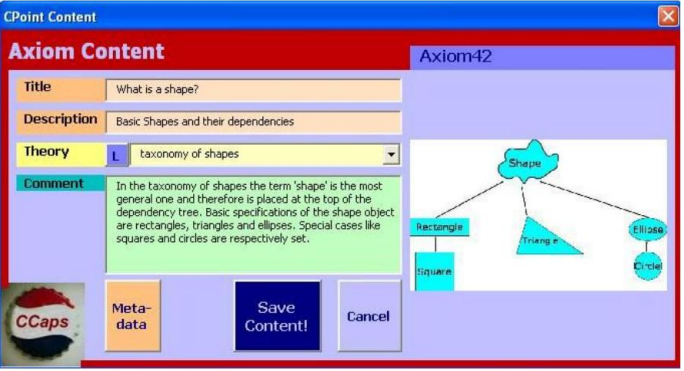
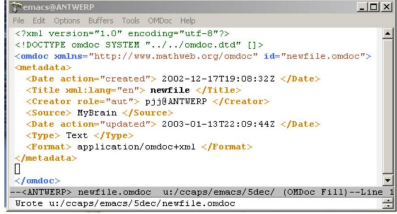
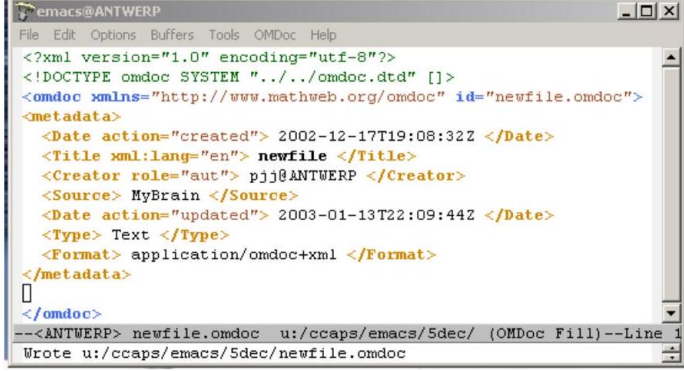
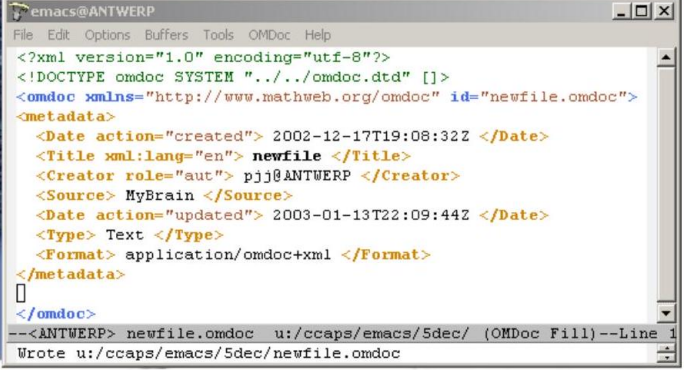
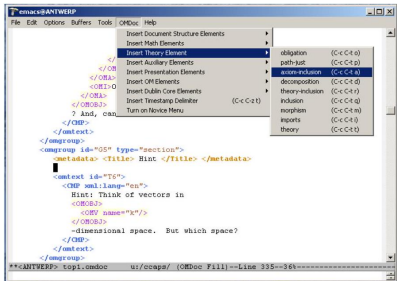
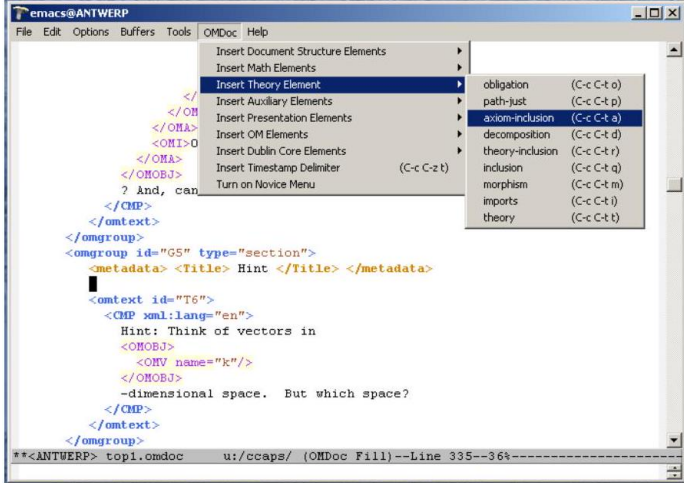
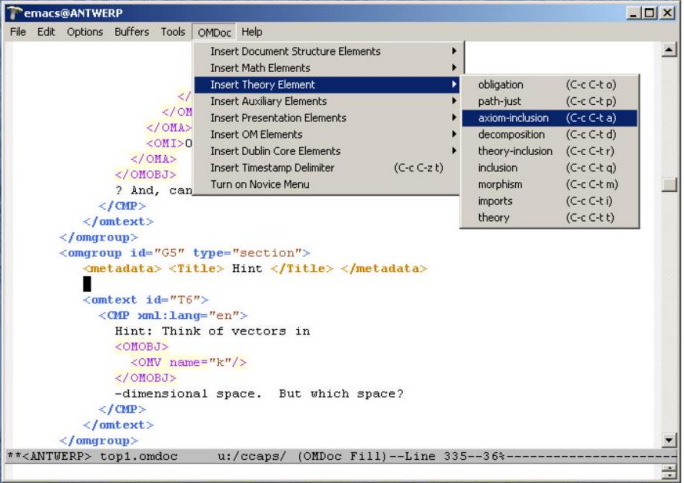
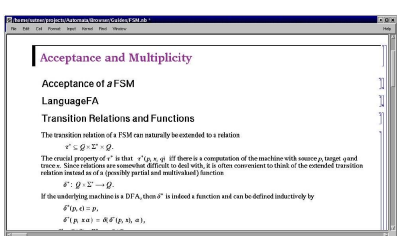
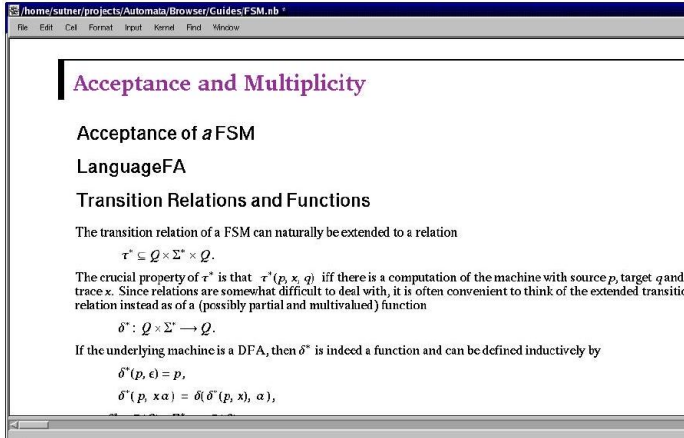
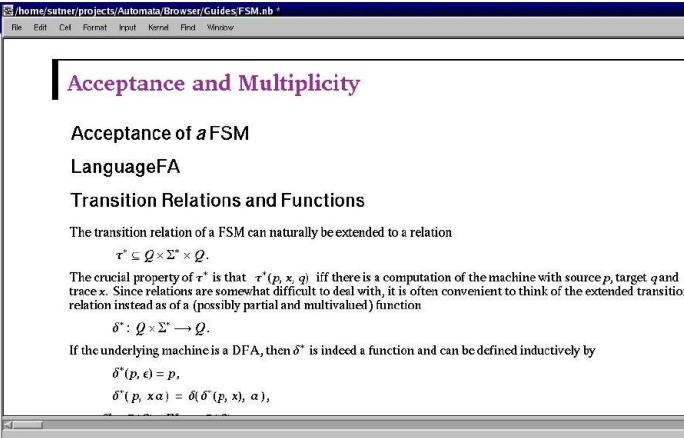
						
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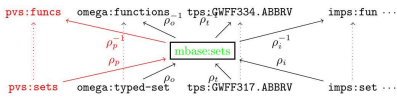
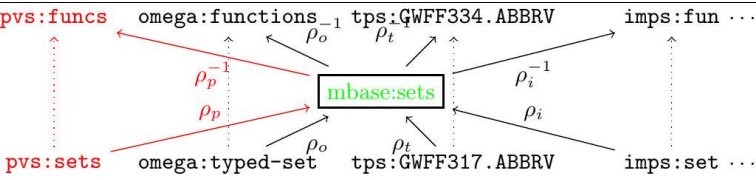
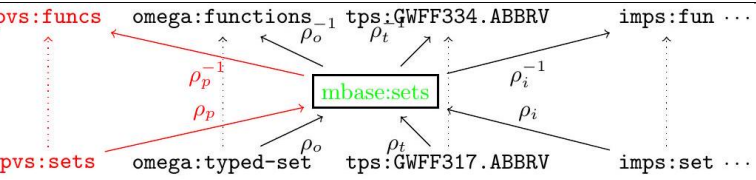
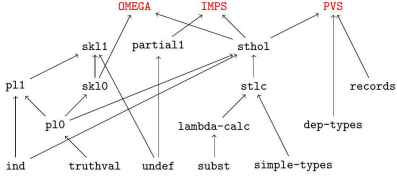
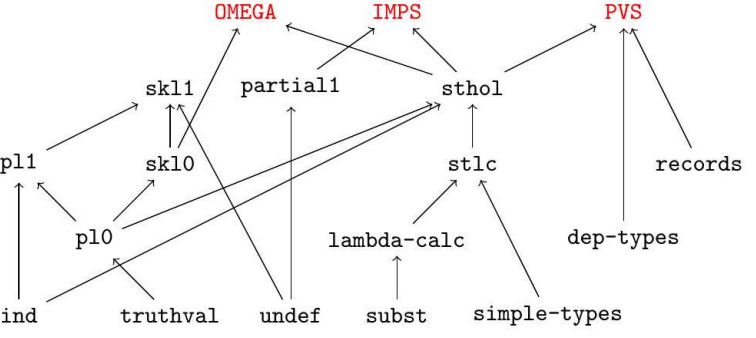
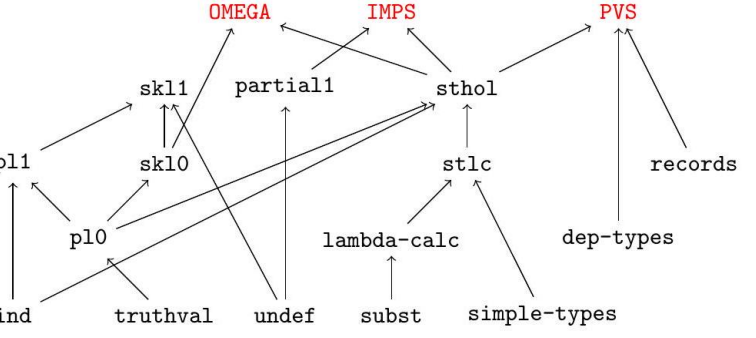
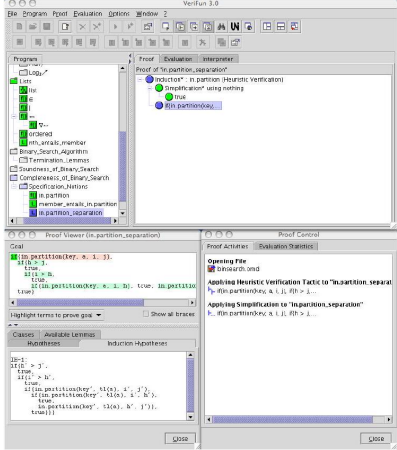
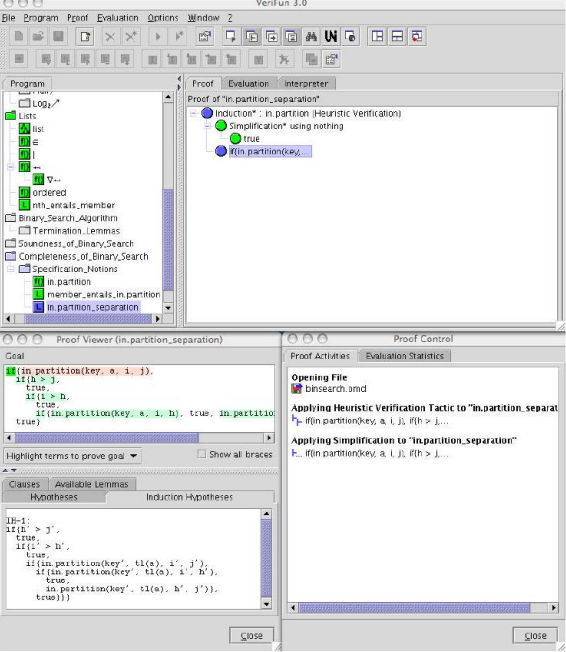
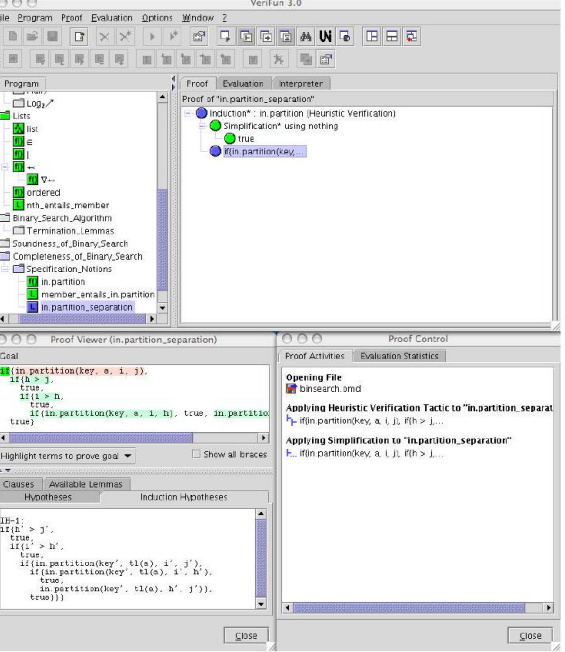
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kohlhase-omdoc_DIA_0194	304	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
kohlhase-omdoc_DIA_0195	304	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
kohlhase-omdoc_DIA_0196	307	— (dup)	1b348de5-ed54-4a9a-aba0-06a9a89f7dc6-307_238_1035_207_224.jpg			
kohlhase-omdoc_DIA_0197	308	— (dup)	1b348de5-ed54-4a9a-aba0-06a9a89f7dc6-308_432_979_560_255.jpg			
kohlhase-omdoc_DIA_0198	314	— (dup)	1b348de5-ed54-4a9a-aba0-06a9a89f7dc6-314_858_745_1120_579.jpg			
kohlhase-omdoc_DIA_0199	316	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
kohlhase-omdoc_DIA_0200	316	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
kohlhase-omdoc_DIA_0201	317	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
kohlhase-omdoc_DIA_0202	344	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
kohlhase-omdoc_DIA_0203	345	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
kohlhase-omdoc_DIA_0204	345	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
kohlhase-omdoc_DIA_0205	345	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
kohlhase-omdoc_DIA_0206	345	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)

kohlhase-omdoc_DIA_0237	364	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
kohlhase-omdoc_DIA_0238	365	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
kohlhase-omdoc_DIA_0239	366	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
kohlhase-omdoc_DIA_0240	367	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
kohlhase-omdoc_DIA_0241	368	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
kohlhase-omdoc_DIA_0242	369	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
kohlhase-omdoc_DIA_0243	370	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
kohlhase-omdoc_DIA_0244	371	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)
kohlhase-omdoc_DIA_0245	387	— (dup)	no image for this region (68 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)